

Appendix A

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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.045C3
		Application Number	
Title of Invention	Circuit Providing Load Isolation and Noise Reduction		
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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.045C3
		Application Number	
Title of Invention	Circuit Providing Load Isolation and Noise Reduction		

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Application Information:

Title of the Invention	Circuit Providing Load Isolation and Noise Reduction		
Attorney Docket Number	NETL.045C3	Small Entity Status Claimed	☒
Application Type	Nonprovisional		
Subject Matter	Utility		
Suggested Class (if any)		Sub Class (if any)	
Suggested Technology Center (if any)			
Total Number of Drawing Sheets (if any)	9	Suggested Figure for Publication (if any)	

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Prior Application Status	Pending	Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)
	Continuation of	14324990	2014-07-07
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Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)	Patent Number	Issue Date (YYYY-MM-DD)
14324990	Continuation of	13412243	2012-03-05	8782350	2014-07-15
Prior Application Status		Patented	Remove		
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13412243	non provisional of	12422853	2009-04-13	8154901	2012-04-10
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12422853	non provisional of	61044839	2008-04-14		
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Title of Invention	Circuit Providing Load Isolation and Noise Reduction		

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MEMORY MODULE WITH LOAD ISOLATION AND NOISE REDUCTION

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application is a continuation of U.S. Pat. Appl. No. 14/324,990, filed July 7, 2014, which is a continuation of U.S. Pat. Appl. No. 13/412,243, filed March 5, 2012, now U.S. Patent No. 8,782,350, which is a continuation of U.S. Pat. Appl. No. 12/422,853, filed April 13, 2009, now U.S. Patent No. 8,154,901, which claims the benefit of priority from U.S. Provisional Appl. No. 61/1044,839, filed April 14, 2008, from U.S. Provisional Appl. No. 61/044,825, filed April 14, 2008, and from U.S. Provisional Appl. No. 61/044,801, filed April 14, 2008, each of which are incorporated in its entirety by reference herein. This application is also related to U.S. Pat. Appl. No. 12/422,912, filed on April 13, 2009, now abandoned, and U.S. Pat. Appl. No. 12/422,925, filed on April 13, 2009, now U.S. Pat. No. 8,001,434, both of which are incorporated in their entirety by reference herein. This application is further related to U.S. Pat. Appl. No. 13/183,253, filed July 14, 2011, now U.S. Pat. No. 8,359,501, U.S. Pat. Appl. No. 13/745,790, filed January 19, 2013, now U.S. Pat. Appl. No. 8,689,064, and U.S. Pat. Appl. No. 14/229,844, filed March 29, 2014. This application is further related to U.S. Pat. Appl. No. 12/422,912, filed on April 13, 2009, now abandoned, U.S. Pat. Appl. No. 12/422,925, filed on April 13, 2009, now U.S. Pat. No. 8,001,434, U.S. Pat. Appl. No. 13/183,253, filed July 14, 2011, now U.S. Pat. No. 8,359,501, U.S. Pat. Appl. No. 13/745,790, filed January 19, 2013, now U.S. Pat. No. 8,689,064, U.S. Pat. Appl. No. 14/299,844, filed March 29, 2014, U.S. Pat. Appl. No. 12/504,131, filed July 16, 2009, now U.S. Pat. No. 8,417,870, U.S. Pat. Appl. No. 12/761,179, filed April 15, 2010, now U.S. Pat. No. 8,516,185, U.S. Pat. Appl. No. 13/970,606, filed August 20, 2013, U.S. Pat. Appl. No. 13/287,042, filed November 1, 2011, now, U.S. Pat. No. 8,756,364, U.S. Pat. Appl. No. 13/971,231, filed August 20, 2013, and U.S. Pat. Appl. No. 13/288,850, filed November 3, 2011.

BACKGROUND

[0002] The present application relates generally to the field of electrical signal integrity and more specifically to circuits for improving electrical signal integrity. Description of the Related Art

[0003] As the demand on the performance of electronic systems increase, the systems include higher speed processors and higher memory densities with relatively heavy fan-outs. However, higher speed processors and higher memory densities translate to higher power dissipation and increased memory access time, impairing the performance of the system.

[0004] Traditional systems sometimes incorporate multiplexers and/or demultiplexers to help quiet inactive signal paths, minimizing the overall dynamic power dissipation, and isolating unused subsystems. Figure 1 schematically illustrates an example of such a system 10 which incorporates a multiplexer-demultiplexer 12 to isolate unused subsystems of a plurality of subsystems 16, 18. This arrangement can reduce the load on each driver 14 when the original signal path includes multiple destinations by inserting a multiplexer on each signal path between the driver 14, the receiver 20 and the multiple destinations. However, the multiplexer and/or demultiplexer 12 can present transmission line discontinuity regions in electronic systems (e.g., in electronic systems with signal paths with electrical lengths longer than 1/4 of the wavelength of the operational frequency or the signal transition rate). In such systems, signal reflections may occur, changing wave characteristics and degrading system performance. These signal reflections may cause signal distortions and signal integrity issues which can contribute to reduced system performance and eventual failure of the system, thereby limiting the effectiveness of the multiplexer and/or demultiplexer in addressing the signal integrity and power dissipation issues.

[0005] In memory applications, some systems control "chip select" or device select signals, along with the memory address and control signals, based on the device select signals, in order to disable memory devices when they are not being accessed. Figure 2 schematically illustrates an example of such a system 20 which isolates unused subsystems of a plurality of subsystems 16, 18 using device select signals 22, 24 (e.g., chip select signals). The devices are disabled by not activating (asserting) the address, control, and/or device select signals, thus logically isolating the memory devices which are not accessed. This arrangement generally requires a device select signal for each physical memory rank and the address and device select signals are generally controllable based on whether the corresponding memory device or group of memory devices is being accessed. Generating controllable address and device select signals can introduce timing delays on the device select, address, and control paths. In addition, this arrangement does not address the power dissipation and may degrade

performance degradation by, for example, requiring that secondary control signals be generated for the memory devices.

SUMMARY

[0006] Certain embodiments described herein include a memory module comprising a printed circuit board comprising at least one connector configured to be operatively coupled to a memory controller of a computer system. The memory module comprises a plurality of memory devices on the printed circuit board and a circuit. The circuit comprises a first set of ports comprising a plurality of bi-directional ports, each port of the first set of ports operatively coupled to at least one memory device of the plurality of memory devices. The memory module further comprises a second set of ports comprising one or more bi-directional ports, each port of the second set of ports operatively coupled to the at least one connector. The memory module of certain embodiments comprises a switching circuit configured to selectively operatively couple one or more ports of the second set of ports to one or more ports of the first set of ports. The one or more ports of the first set of ports and the one or more ports of the second set of ports each comprise a correction circuit in certain embodiments. The correction circuit reduces noise in one or more signals transmitted between the one or more ports of the first set of ports and the one or more ports of the second set of ports.

[0007] A method of using a memory module with a computer system is described in certain embodiments. The method includes providing a memory module comprising a printed circuit board having at least one connector configured to be operatively coupled to a memory controller of a computer system and to a plurality of memory devices. The memory module further including a circuit which comprises a first set of ports comprising a plurality of bi-directional ports, each port of the first set of ports operatively coupled to at least one memory device of the plurality of memory devices. The circuit further comprises a second set of ports comprising one or more bi-directional ports. Each port of the second set of ports is operatively coupled to the at least one connector. The circuit is configured to selectively operatively couple one or more ports of the second set of ports to one or more ports of the first set of ports in some embodiments. The one or more ports of the first set of ports and the one or more ports of the second set of ports each comprise a correction circuit. The correction

circuit reduces noise in one or more signals transmitted between the one or more ports of the first set of ports and the one or more ports of the second set of ports. The method includes activating the circuit to selectively operatively couple at least one of the one or more ports of the second set of ports to at least one of the one or more ports of the first set of ports. The method also includes activating the circuit to reduce noise in the one or more signals.

[0008] Certain embodiments are described herein provide a circuit comprising a first set of ports comprising a plurality of bi-directional ports. Each port of the first set of ports may be configured to be operatively coupled to at least one memory device of a plurality of memory devices on a printed circuit board. The circuit further comprises a second set of ports comprising one or more bi-directional ports, each port of the second set of ports configured to be operatively coupled to at least one connector on the printed circuit board.

The at least one connector is configured to be operatively coupled to a memory controller of a computer system. The circuit further comprises a switching sub-circuit configured to selectively operatively couple one or more ports of the second set of ports to one or more ports of the first set of ports. The one or more ports of the first set of ports and the one or more ports of the second set of ports each comprise a correction circuit in certain embodiments. The correction circuit can reduce noise in one or more signals transmitted between the one or more ports of the first set of ports and the one or more ports of the second set of ports.

[0009] In certain embodiments, a circuit comprises a first set of ports comprising a plurality of bi-directional ports. Each port of the first set of ports can be configured to be operatively coupled to at least one subsystem of a plurality of subsystems. The circuit may comprise a second set of ports comprising one or more bi-directional ports, each port of the second set of ports configured to be operatively coupled to at least one subsystem of one or more subsystems. In certain embodiments, the circuit comprises a switching sub-circuit configured to selectively operatively couple one or more ports of the second set of ports to one or more ports of the first set of ports. The one or more ports of the first set of ports and the one or more ports of the second set of ports each comprise a correction circuit in certain embodiments. The correction circuit can reduces noise in one or more signals transmitted between the one or more ports of the first set of ports and the one or more ports of the second set of ports.

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] Figure 1 schematically illustrates an example system which isolates unused subsystems and incorporates a multiplexer-demultiplexer.

[0011] Figure 2 schematically illustrates an example system which isolates unused subsystems using device select signals.

[0012] Figure 3 schematically illustrates an example memory system incorporating a memory module in accordance with certain embodiments described herein.

[0013] Figure 4 schematically illustrates an example correction circuit in accordance with certain embodiments described herein.

[0014] Figure 5 schematically illustrates an example programmable impedance matching circuit in accordance with certain embodiments described herein.

[0015] Figure 6 schematically illustrates an example self-adjusting damper circuit in accordance with certain embodiments described herein.

[0016] Figure 7 schematically illustrates example configuration including an example filter element connected to an example repeater/amplifier in accordance with certain embodiments described herein.

[0017] Figure 8 schematically illustrates an example system having an example circuit connecting the system with one or more other systems in accordance with certain embodiments described herein.

[0018] Figure 9 schematically illustrates an example system including an example circuit in accordance with certain embodiments described herein.

[0019] Figure 10 is a flowchart of an example method 400 of using a memory module with a computer system in accordance with certain embodiments described herein.

DETAILED DESCRIPTION

[0020] Electronic systems incorporating devices and methods in accordance with

embodiments described herein may include memory subsystems that contain multiple memory boards or modules that are driven by a single memory controller which synchronizes memory access and manages memory coherency. However, having a single memory controller may gradually degrade system performance and increase power consumption as the physical and the logical size of the memory subsystem increases. Increases in memory speed may further increase the power consumption. Additionally, impedance mismatches between the system board, the memory boards, and the memory devices may also substantially degrade the performance and power efficiency of traditional memory systems. Such memory systems may be used with a power supply which helps to stabilize the voltage level, and/or a thermo-mechanical component to moderate memory subsystem temperature, and/or software for throttling the processor to lower the memory access rate.

[0021] Embodiments described herein address these speed and power issues in memory systems. For example, a circuit of certain embodiments can comprise an isolation switch and can logically isolate inactive subsystems and reduce signal reflection and/or other noise at, for example, the interface between the system board and the memory board and at the interface between the memory board and the memory devices.

[0022] Certain embodiments described herein present a circuit for isolating inactive subsystems and maintaining and improving signal integrity in a variety of systems. Certain embodiments described herein are used to increase the performance of electronic systems without increasing the power dissipation or the memory access time. In certain embodiments, a plurality of circuits are configured so as to improve the interconnection (e.g., timing, power consumption, and/or signal integrity) between each of the plurality of circuits and a plurality of corresponding devices (e.g., memory devices) of a system or subsystem (e.g., a memory module of a computer system). For example, the data width of each of the plurality of circuits may be selected to optimize the interconnection (e.g., timing, power consumption, and/or signal integrity). In various embodiments, the data width of each of the plurality of devices may be 4, 8, or 16 bits, for example.

[0023] Certain embodiments described herein describe a circuit for use in high performance electronic systems which may contain transmission lines and relatively heavy fan-outs. A circuit is described herein which, in certain embodiments, includes at least some of the following components: a signal router, a programmable coarse impedance matching circuit, a

self-adjusting damper circuit, a filter and/or a repeater/amplifier.

[0024] Figure 3 schematically illustrates an example system incorporating a memory module 100 in accordance with certain embodiments described herein. The memory module 100 comprises a printed circuit board (PCB) 102. The PCB 102 includes at least one connector 104 configured to be operatively coupled to a memory controller 106 of a computer system 108. The memory module 100 includes a plurality of memory devices 110 on the PCB 102.

[0025] The memory module 100 of certain embodiments further includes a circuit 112 comprising a first set of ports 122 comprising a plurality of bi-directional ports 124, each port 124 of the first set of ports 122 operatively coupled to at least one memory device 110 of the plurality of memory devices 110. The circuit 112 further comprises a second set of ports 126 comprising one or more bi-directional ports 128, each port 128 of the second set of ports 126 operatively coupled to the at least one connector 104. The circuit 112 further comprises a switching sub-circuit 140 configured to selectively operatively couple one or more ports 128 of the second set of ports 126 to one or more ports 124 of the first set of ports 122. The one or more ports 124 of the first set of ports 122 and the one or more ports 128 of the second set of ports 126 each comprises a correction circuit 150 (see, e.g., Figure 4) which reduces noise in one or more signals transmitted between the one or more ports 124 of the first set of ports 122 and the one or more ports 128 of the second set of ports 126.

[0026] In certain embodiments, the memory module 100 has a memory capacity of 512-MB, 1-GB, 2-GB, 4-GB, 8-GB, or higher or lower. Other memory capacities are also compatible with certain embodiments described herein. In addition, memory modules 100 having widths of 4 bits, 8 bits, 16 bits, 32 bits, as well as other widths (in bytes or in bits), are compatible with embodiments described herein. In certain embodiments, the PCB 102 has an industry-standard form factor. For example, the PCB 102 can have a low profile (LP) form factor with a height of 30 millimeters and a width of 133.35 millimeters. In certain other embodiments, the PCB 102 has a very high profile (VHP) form factor with a height of 50 millimeters or more. In certain other embodiments, the PCB 102 has a very low profile (VLP) form factor with a height of 18.3 millimeters. Other form factors including, but not limited to, small-outline (SO-DIMM), unbuffered (UDIMM), registered (RDIMM), fully-buffered (FBDIMM), mini-DIMM, mini-RDIMM, VLP mini-DIMM, micro-DIMM, and SRAM DIMM are also compatible with certain embodiments described herein. For example, in other

embodiments, certain non-DIMM form factors are possible such as, for example, single inline memory module (SIMM), multi-media card (MMC), and small computer system interface (SCSI).

[0027] In certain embodiments, the plurality of memory devices 110 of the memory module 100 may be arranged as one or more ranks, each rank of memory generally having a bit width. In certain embodiments, each rank may comprise an independent set of memory devices 111 of the plurality of memory devices 110 that can be accessed by the memory controller 106 to access the full bit-width of the memory bus of the memory module 100. For example, a memory module 100 in which each rank of the memory module is 64 bits wide is described as having an "x 64" organization. Similarly, a memory module 100 having 72-bit-wide ranks is described as having an "x 72" organization. The number of memory devices 111 of a memory module 100 can be increased by increasing the number of memory devices 111 per rank or by increasing the number of ranks of the memory module 100. For example, a memory module with four ranks with each rank having N 512-MB memory devices 111 has double the memory capacity of a memory module with two ranks with each rank having N 512-MB memory devices 111 and four times the memory capacity of a memory module with one rank with each rank having N 512-MB memory devices.

[0028] During operation, one or more ranks of a memory module 100 may be selected or activated from the plurality of ranks by control signals that are received from a component of the system (e.g., a system memory controller 106 or a local memory controller of the memory module 100). Examples of such control signals include, but are not limited to, rank-select signals, also called chip-select signals. In certain other embodiments, the memory module 100 comprises only one rank of memory devices 111.

[0029] As discussed, the PCB 102 includes at least one connector 104 configured to be operatively coupled to a memory controller 106 of a computer system 108, such as a host computer system 108. For example, the connector 104 is electrically coupled, logically coupled, or both, with the memory controller 106 in some other manner. In certain embodiments, the connector 104 comprises a wireless radio frequency (RF) interface or a portion thereof and is wirelessly connected (e.g., via one or more RF antennas) to the memory controller 106. In some embodiments, some other connection mechanism is used besides the connector 104. Examples of host computer systems 108 include, but are not limited to, blade servers, 1U servers, personal

computers (PCs), data storage systems and other applications in which space is constrained or limited. The memory controller 106 may comprise a disk controller of the computer system 108, for example. The memory controller 106 of certain embodiments may be mounted on a system board 109 of the host computer 108. The connector 104 can comprise a plurality of edge connections distributed along one or more edges of the PCB 102, which fit into a corresponding slot connector of the host system. The connector 104 of certain embodiments provides a conduit for power voltage as well as data, address, and control signals between the memory module 100 and the host system 108. For example, the connector 104 can comprise a standard 240-pin DDR2 edge connector. Additionally, in certain embodiments, more than one memory module 100 is coupled to the host system 108.

[0030] The plurality of memory devices 110 on the PCB 102 may include one or more volatile memory components. For example, the plurality of memory devices 110 of certain embodiments comprises two or more dynamic random-access memory (DRAM) elements 111. For example, in the block diagram of Figure 3, the plurality of memory devices 110 comprises eight DRAM devices 111. Types of DRAM devices 111 compatible with certain embodiments described herein include, but are not limited to, DDR, DDR2, DDR3, and synchronous DRAM (SDRAM). The memory devices 111 may comprise other types of memory elements such as static random-access memory (SRAM). In addition, volatile memory devices 111 having bit widths of 4, 8, 16, 32, as well as other bit widths, are compatible with certain embodiments described herein. Memory devices 111 compatible with certain embodiments described herein have packaging which include, but are not limited to, thin small-outline package (TSOP), ball-grid-array (BGA), fine-pitch BGA (FBGA), micro, BGA (μ BGA), mini-BGA (mBGA), and chip-scale packaging (CSP).

[0031] The first set of ports 122 of the circuit 112 comprises a plurality of bidirectional ports 124 where each port of the first set of ports 122 is operatively coupled to at least one memory device 111 of the plurality of memory devices 110. In addition, the second set of ports 126 of the connector 104 comprises one or more bi-directional ports 128 where each port of the second set of ports 126 is operatively coupled to the at least one connector 104. The switching sub-circuit 140 is configured to selectively operatively couple one or more ports 128 of the second set of ports 126 to one or more ports 124 of the first set of ports 122. The operative coupling of the first set of ports 122 to the memory devices 110, the second set of ports 126 to the connector 104, and the selective operative coupling between the first and

second set of ports 122, 126 may comprise electrical coupling, logical coupling, or both. In other embodiments, optical coupling or some other manner of coupling may be employed.

[0032] In certain embodiments, each bi-directional port 128 of the second set of ports 126 may correspond to a signal of a memory bus 130 which operatively couples the memory controller 106 and the memory module 100 via the connector 104. For example, the memory controller 106 may generally read data from and/or write data to the plurality of memory devices 110 on the memory module 100 through the memory bus 130 and through the bi-directional ports 124, 128. In certain embodiments, the number of bi-directional ports 128 of the second set of ports 126 may correspond to the number of data bits in the memory bus 130 (i.e., the bit-width of the memory bus 130).

[0033] In certain embodiments, each of the bi-directional ports 124 in the first set of ports 122 may correspond to and be operatively coupled to a signal port of one of the memory devices 111 of the plurality of memory devices 110. The bi-directional ports 124 correspond to the data signal ports, for example, of the plurality of memory devices 110. Because each memory device 111 includes associated data signal ports, the number of bidirectional ports 124 in the first set of ports of certain embodiments may be larger than the number of bi-directional ports 128 in the second set of ports 122 (e.g., the number of bits in the memory bus 130). In general, the overall number of ports of the plurality of memory devices 110 and corresponding bi-directional ports 124 in the first set of ports 126 may depend on the organization of the memory module. For example, the number of ports may depend on the number of memory devices and/or the number of ranks in the memory module. In certain embodiments, the memory module 100 includes a plurality of circuits 112. The circuits 112 may be configured so as to improve the interconnection (e.g., timing, power consumption, and/or signal integrity) between each of the plurality of circuits 112 and a plurality of corresponding memory devices 111 of memory module 100. For example, the data width of each of the plurality of circuits 112 may be selected to optimize the interconnection (e.g., timing, power consumption, and/or signal integrity). In various embodiments, the data width of each of the plurality of devices may be 4, 8, or 16 bits, for example. Other data widths are possible in other embodiments. In some embodiments, the data width of each circuit 112 corresponds to the data width of each of the memory devices 111 such that each circuit 112 is operatively coupled to one of the plurality of memory devices 110. In other embodiments, the

data width of each circuit 112 corresponds to a multiple of the data width of each memory device 111 such that each circuit 112 may be operatively coupled to the data bits of more than one memory device 111. In various embodiments, the data width of each circuit 112 corresponds to a segment of the data width of each memory device 111 or to segments of multiple memory devices 111. As such, in some embodiments, each circuit 112 may be operatively coupled to a segment of the data bits of one memory device 111 or to segments of more than one memory device 111. Other configurations are possible which optimize the interconnection (e.g., timing, power consumption, and/or signal integrity). In addition, in certain embodiments, the location of each of the plurality of circuits 112 on the PCB 102 is selected to optimize the interconnection, timing, power consumption, and signal integrity.

[0034] In other embodiments, the bi-directional ports 124 correspond to one or more other bits of the memory bus 130 instead of, or in addition to, the data bits. For example, one or more of the ports 124 may correspond to address, chip select, control (e.g., read/write) signals. Moreover, in some embodiments, one or more ports of the first set of ports 122 is not bi-directional. For example, in one embodiment, signals corresponding to address, chip select and/or control signals from the first set of ports 122 to the plurality of memory devices 110 and not vice versa. For example, one or more of the address, chip select, and control signals may be used to control the routing of the bi-directional ports 124 which correspond to data bits.

[0035] As discussed, the switching sub-circuit 140 is configured to selectively operatively couple one or more ports of the second set of ports 126 to one or more ports of the first set of ports 122. In certain embodiments, the switching sub-circuit 112 may be capable of interconnecting any number of the second set of ports 126 to any number of the first set of ports 122 simultaneously. In certain embodiments, the switching sub-circuit 140 can pass signals (e.g., information such as the data signals described above) from one port 128 of the second set of ports 126 to an operatively coupled port 124 of the first set of ports 122 and vice versa. Additionally, the switching sub-circuit 140 of some embodiments can pass signals from one port 128 of the second set of ports 126 to multiple operatively coupled ports 124 of the first set of ports 122 and vice versa. In certain embodiments, the switching sub-circuit 140 isolates and provides a known static value to all un-driven ports, thereby reducing any dynamic power consumption associated with floating or toggling electrical nodes.

[0036] The switching sub-circuit 140 of certain embodiments can function as an isolation switch which isolates and/or inactivates individual subsystems not actively transferring information. For example, in an example embodiment, the plurality of memory devices 110 comprises multiple independently accessible ranks. When the memory controller accesses one of the ranks of the memory module 100, the switching sub-circuit 140 isolates and/or inactivates at least the one or more other, un-accessed ranks. For example, the switching sub-circuit 140 does not operatively couple ports from the second of set of ports 126 to the first set of ports 122 which are coupled to the memory devices 111 of the unaccessed one or more ranks of memory devices 110. In certain embodiments, the switching sub-circuit 140 may function as or be characterized as a scalable isolation switch router which routes the information from one subsystem to another (e.g., host computer 108 to the memory module 100 via the memory controller 106), establishing a one-to-one connection for transferring data from point-to-point, and/or establishing a one-to-many connection for broadcasting data. The switching sub-circuit 140 comprises or is characterized as a bidirectional multiplexer-demultiplexer in certain embodiments. In other embodiments, the switching sub-circuit 140 comprises or is characterized as a router.

[0037] In certain embodiments, the memory module 100 further includes circuitry (not shown) for controlling the operation of the circuit 112 and the components therein (e.g., the switching sub-circuit 140 and the correction circuit 150, as described more fully below). For example, in one embodiment, the memory module 100 comprises a local memory controller which is operatively coupled to one or more control inputs (not shown) of the circuit 112 to control the selective coupling of the second set of ports 126 to the first set of ports 122. In certain other embodiments, some other component of the memory module 100 controls the operation of the circuit 112 or the circuit 112 comprises internal control circuitry. In various embodiments, the circuitry for controlling the circuit 112 includes discrete logic, one or more microprocessors, a field-programmable gate array (FPGA), or a computer programmable logic device (CPLD). In yet other configurations, an external component controls the operation of the circuit 112.

[0038] The circuit 112 of certain embodiments comprises one physically integral component (e.g., one integrated circuit package mounted on the memory module 100).

Various alternative configurations for the physical organization of the circuit 112 are

possible, however. In certain other embodiments, the circuit 112 comprises multiple, physically separate components. For example, in one embodiment, the circuit 112 comprises one physically separate component per memory device 111 (e.g., integrate circuit packages mounted on the memory module 100). In another embodiment, the circuit 112 comprises one physically separate component per memory rank. In some embodiments, the circuit 112 comprises one physically separate component corresponding to multiple memory devices 111.

[0039] Figure 4 schematically illustrates an example correction circuit 150 in accordance with certain embodiments described herein. As shown, in certain embodiments, the correction circuit 150 comprises one or more correction elements (e.g., at least one coarse correction element 152 to provide coarse correction of the noise and at least one fine correction fine correction element 154 to provide fine correction of the noise). The correction circuit 150 of certain embodiments further comprises at least one filter element 156. In certain embodiments, the correction circuit 150 further comprises at least one repeater/amplifier element 158, which improves the signal quality of the one or more signals.

[0040] Referring to Figures 3 and 4, the correction circuit 150 reduces noise in one or more signals transmitted between the one or more ports of the first set of ports 122 and the one or more ports of the second set of ports 126. For example, the correction circuit 150 may reduce noise due to impedance mismatches (e.g., due to discontinuity regions) at nodes between the memory controller 106 and the plurality of memory devices 110. In certain embodiments, the noise may comprise signal reflections. For example, in certain embodiments, the signal paths between the host computer 108 and the plurality of memory devices 110 or portions thereof may be characterized as transmission lines if they are greater than or about equal to a particular fraction of the wavelength of the signal, such as, for example, greater than or equal to about 1/4 of the wavelength of the signal. In other configurations, the signal paths may be characterized as transmission lines when the signal path is greater than or equal to some other fraction of the wavelength of the signal such as, for example, 1/2, 1/8, or 1/10 of the wavelength. In certain embodiments, the system (e.g., memory module 100) incorporating the circuit 112 has signal delays which are related to signal characteristics such as the signal transition rate and/or the rise/fall time of the signals. For example, the system incorporating the circuit 112 may include signal delays which are more than some fraction (e.g., 1/2) of the

signal transition rate and/or the rise/fall time of the signals. In other embodiments, the signal delays are related in some other way to the signal transition rate and/or the rise/fall time of the signals (e.g., are more than 1/4, 1/8 of those values) or are related to some other signal property.

[0041] The reflections may be caused by one or more discontinuity regions along the signal paths. For example, discontinuity regions comprise points or regions at which the characteristic impedance of the corresponding signal path changes. For example, the interface between the system board 109 of the host computer 108 and the memory module 100 (e.g., the connector 104) may comprise a discontinuity region along each of the signal paths. The interfaces between the connector 104 of the memory module 100 and each of the memory devices 111 mounted on the PCB 102 may comprise one or more discontinuity regions along each of the signal paths.

[0042] The noise may comprise cross-coupling between the signal paths in certain embodiments. For example, a cross-coupling impedance may be present between one or more of the signal paths or portions thereof. In certain embodiments, the cross-coupling impedance may include cross-coupling due to cross-coupling capacitance, inductance (e.g., mutual inductance), transconductance, or a combination thereof. Additionally, a variety of other noise are possible such as, for example, electromagnetic interference or other types of noise.

[0043] Referring to Figure 4, in certain embodiments, the correction circuit 150 includes one or more elements to provide for coarse correction of signal noise due to impedance mismatches (e.g., discontinuity regions) in the system. The at least one coarse correction element 152 comprises at least one programmable impedance matching circuit in certain embodiments. Figure 5 schematically illustrates an example programmable impedance matching circuit 160 in accordance with certain embodiments described herein. For example, the programmable impedance matching circuit 160 may comprises at least one programmable resistor 162. The resistance of the programmable resistor 162 may be controllable by a control signal 164. In certain embodiments, the coarse correction element 152 may be integrated with the switching sub-circuit 140, the fine correction element 154, the filter element 156, and/or the repeater/amplifier element 158. Alternatively, in other embodiments, the coarse correction element 152 may be implemented with discrete components.

[0044] In other embodiments, the coarse correction element 152 comprises a network of resistors 162 which can be configured to provide programmable impedance matching. In some embodiments, the coarse correction element 152 may comprise a set of programmable coarse impedance matching circuits 160 that are connected to each of the ports 124, 128 of the first set of ports 122 and the ports 128 of the second set of ports 126. Advantageously, the impedance values of the coarse correction element 152 may be dynamically controllable, configurable, and/or programmable, so as to match the characteristic impedances of the transmission lines (e.g., signal paths) in the system. The values of these impedances may differ (e.g., may be programmed differently) based on which particular signal path is activated (e.g., by the switching sub-circuit 140). Where the coarse correction element 152 comprises at least one impedance matching circuit, the granularities of the configurable impedance values may be configured based on the application of the particular system. In some embodiments, the impedance matching circuit of the coarse correction element 152 may include a real time controllable impedance control input for preprogramming and/or dynamically controlling the impedance matching circuit. In this manner, the impedance matching circuit 152 may, in certain embodiments, provide dynamically controllable variable impedance which can be pre-set for matching the impedance of the subsystem that drives the circuit 112.

[0045] In certain embodiments, the at least one fine correction element 154 of the correction circuit 150 further reduces signal reflections due to impedance mismatches and/or reduces other noise. In certain embodiments, for example, the coarse correction element 152 may be characterized as a first refining stage and the fine correction element 154 may be referred to as a second refining stage. The fine correction element 154 may smooth out the signal after it has undergone coarse correction (e.g., impedance matching) and reduces or removes transmission line characteristics from the received signal. Referring again to Figure 4, the fine correction element 154 can be connected, for example, in parallel with the programmable coarse correction element 152. In some embodiments, the fine correction element 154 operates to reduce the impedance mismatches by introducing a current source that damps to the slew rate of the input signal, reducing signal reflection. The fine correction element 154 actively counteracts the noise in certain embodiments. In addition, the fine correction element 154 of certain embodiments is dynamically responsive to one or more changes in the noise, such as, for example, changes in the frequency of the noise component.

[0046] The fine correction element 154 may include at least one damper circuit. The damper circuit includes at least one self-adjusting damper circuit 170 in certain embodiments, such as the example self-adjusting damper circuit 170 schematically illustrated by Figure 6. In certain embodiments, the self-adjusting damper circuit 170 comprises an input terminal 172 adapted to receive an input signal 174 comprising an input noise component and an input signal component. The damper circuit 170 further comprises an output terminal 176. In certain embodiments, the damper circuit 170 includes a first sub-circuit 184 adapted to generate a first signal 187 responsive to at least a portion of the noise component. For example, the first sub-circuit 184 may generate the first signal 187 in response to the portion of the noise component not reduced by the coarse correction element 152 in certain embodiments. The damper circuit 170 of certain embodiments further comprises a second sub-circuit 188 adapted to generate a second signal 190 corresponding to a delayed version of at least a portion of the input signal. The damper circuit 170 combines the first signal 187 with the second signal 190 to generate an output signal 108 on the output terminal 176 wherein the output signal comprises an output noise component smaller than the input noise component. For example, the first signal 187 and the second signal 190 may be combined at the node 182 of the damper circuit 170. The output signal 176 of certain embodiments is substantially representative of the signal component of the input signal 174. The first subcircuit 184 of certain embodiments comprises one or more current sources 185, 186. For example, the current sources 185, 186 may each comprise a controlled current source such as a voltage-controlled current source 185, 186 in certain embodiments. In some embodiments, the first sub-circuit 184 may comprise or be characterized as a negative feedback current source. In various embodiments, the self-adjusting damper circuit 170 includes an enable control input (not shown) for enabling the self-adjusting damper circuit 170. Examples of self-adjusting damper circuits 170 in accordance with certain embodiments described herein are described in U.S. Provisional Application No. 61/044,825, filed April 14, 2008, and copending U.S. Application No. 12/422,912, filed by the assignee of the present application on April 13, 2009. Both applications are incorporated in their entirety by reference herein.

[0047] In certain embodiments, the at least one filter element 156 and the at least one repeater/amplifier element 158 may also reduce noise from other sources such as, for example, signal coupling (e.g., cross-coupling). The filter element 156 may reduce noise from the one or more signals where the noise includes a predetermined range of frequencies.

For example, the filter element 156 may include at least one band-pass filter and may reduce any low and/or high frequency noise components. In other embodiments, the filter element 156 comprises at least one other type of a filter such as a notch filter, a low-pass filter, or a high-pass filter.

[0048] One or more of the filter element 156 and the repeater/amplifier element 158 of certain embodiments enhance signal slew rate to reduce the transient power dissipation at target devices or components (e.g., one or more of the memory devices 110). Figure 7 schematically illustrates example configuration including an example filter element 156 connected to an example repeater/amplifier element 158 in accordance with certain embodiments described herein. The repeater/amplifier element 158 includes an output enable control input 159 for controlling the bi-directional signal path. In various embodiments one or more of the filter element 156 and repeater/amplifier element 158 are programmable and/or dynamically controllable. The repeater/amplifier element may further include a reference input 157. In certain embodiments, the repeater/amplifier element 158 may include one or more operational amplifiers, for example. Referring again to Figure 4, in certain embodiments, only filter element 156, only the repeater, only the amplifier, or any combination of the three elements may be included in the circuit.

[0049] The filter element 156 and/or the repeater/amplifier element 158 of certain embodiments provide wave reshaping functions and generally reduces signal noise. For example, the filter element 156 and/or the repeater/amplifier element 158 may reduce signal ringing due to reflection (e.g., reflections at one or more interfaces along the signal paths such as interfaces between the circuit 112, the connector 104, the memory devices 110 and/or the system board 109 of Figure 3). In certain embodiments, the repeater/amplifier element 158 reshapes the signal, which may be deteriorated due to transmission over lossy transmission lines having discontinuity regions. The repeater/amplifier element 158 may be configured to repeat the input signal, generating an output signal with earlier transition trigger points and faster slew rates in order to regain speed and to reduce the transient power.

[0050] In certain embodiments, the circuit 112 or portions thereof (e.g., the switching sub-circuit 140, the correction circuit 150 or elements thereof, etc.) is selected from a group consisting of: a programmable-logic device (PLD), an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA), a custom-designed semiconductor device,

and a complex programmable-logic device (CPLD). For example, in certain embodiments, the circuit 112 or portions of thereof is implemented in one or more of a PLD, ASIC, FPGA, or CPLD. In other embodiments, the circuit 112 or portions thereof is implemented using some other technology, such as, for example, using discrete components.

[0051] In various embodiments, the elements of the circuit 112 such as the switching sub-circuit 140, the coarse correction element 152, the fine correction element 154, the filter element 156, and the repeater/amplifier element 158, comprise or may be characterized as independent, fully controllable functional blocks. Additionally, in certain embodiments, any subset of one or more of these functional blocks may be implemented in order to enhance performance, reduce power dissipation, or provide other advantages.

[0052] Figure 8 schematically illustrates an example system 200 having an example circuit 112 connecting a system 200 with one or more other systems 220 in accordance with certain embodiments described herein. The circuit 112 may be any of the circuits 112 discussed herein, such as the circuit 112 discussed above with respect to Figure 3, for example. The system 200 of Figure 8 includes a driver 202, a receiver 201, and two subsystems 203, 204. For illustration purposes, the system 200 of Figure 8 is described as an open system that communicates with other systems 220 via the driver 202 and the receiver 201. Furthermore, in some embodiments, a connector (not shown) connects the system 200 with other systems by means of transmission lines. For example, the system 200 may be a memory system and the receiver 201 and the driver 202 of Figure 8 may represent memory signals (e.g., data, address and/or control signals) transmitted between the system 220 (e.g., a memory controller and the subsystems 203, 204 (e.g., two sets of independently addressable memory devices such as DRAM devices). In certain embodiments, for example, the circuit 112 may be disposed on a memory board that is populated with sets of memory devices 203, 204.

[0053] In various embodiments, the receiver 201 may or may not contain any active circuits. For example, the receiver 201 may be a connector such as a socket or DIMM interface with a system board (e.g., the connector 104 described above in regards to Figure 3). The output signal from the receiver 201 may exhibit the characteristics of the signal on a transmission line such as, for example, wave dispersion, attenuation, and reflection. In certain embodiments, the output signal from the receiver 201 goes through a number of refining, filtering and/or amplification stages in circuit 112 as described herein to improve the

performance of the overall system. For example, the signal quality is enhanced by introducing at least one circuit 112 including at least one correction circuit 150 comprising one or more coarse correction elements 152 (e.g., impedance matching circuits), fine correction elements 154 (e.g., self-adjusting dampers), filter elements 156 (e.g., bandpass filters) and/or repeater/amplifier elements 158 as described herein on the signal paths (e.g., from receiver 101 to subsystem 103). For example, the circuit 112 may reduce power consumption by selectively isolating the inactive (or un-accessed) subsystems and may enhance the access rate of a targeted subsystem. For example, in one embodiment, the system 200 comprises a memory module 100 such as one of the memory modules described herein, the subsystems 203 and 204 comprise memory devices such as the memory devices 111 described herein, and the other system 220 comprises a host computer 108 including a memory controller 106 as described herein.

[0054] In certain embodiments, the performance is enhanced and power dissipation is reduced by removing the load associated with other (e.g., unused or inactive) subsystems (e.g., the subsystem 204) and reshaping the signal waveform using a filter element 156 and/or the repeater/amplifier element 158. In some embodiments, there are two refining stages for reducing signal reflections due to impedance mismatches. In some embodiments, for example, the first refining stage is the coarse correction element 152 (e.g., an impedance matching circuit) that matches the impedance of the transmission line (e.g., wire) prior to a receiver 201 with the characteristic impedance of the transmission line (e.g., wire) in the system 100. The second stage may be the fine correction element 154 (e.g., a self adjusting damper for further reducing signal reflections due to impedance mismatches and/or reducing other noise.

[0055] Figure 9 schematically illustrates an example system 300 including an example circuit 112 in accordance with certain embodiments described herein. The circuit 112 may be any of the circuits 112 discussed herein, such as the circuit 112 discussed above with respect to Figure 3, for example. The circuit 112 comprises a first set of ports 122 comprising a plurality of bi-directional ports, each port of the first set of ports 122 configured to be operatively coupled to at least one subsystem 311 of a plurality of subsystems 310. The circuit 112 further comprises a second set of ports 126 comprising one or more bi-directional ports, each port of the second set of ports 126 configured to be operatively coupled to at least

one subsystem 321 of one or more subsystems 320. In certain embodiments, the circuit 112 is configured to selectively operatively couple one or more ports of the second set of ports 126 to one or more ports of the first set of ports 122. In certain embodiments, the one or more ports of the first set of ports 122 and the one or more ports of the second set of ports 126 each comprises a correction circuit 150 which reduces noise in one or more signals transmitted between the one or more ports of the first set of ports 122 and the one or more ports of the second set of ports 126. The correction circuit 150 of certain embodiments may comprise one of the correction circuits 150 described herein such as the correction circuits 150 described above with respect to Figures 4-7 and may include/ at least one coarse correction element 152, at least one fine correction element 154, at least one filter element 156, and at least one repeater/amplifier element 158, or any combination or sub-combination thereof. In other embodiments, the fine correction element 154 may be placed after the switching sub-circuit 140 in order to achieve better results. For example, the fine correction element 154 may be connected after the filter element 156 (e.g., between the filter 156 and the repeater/amplifier element 158) in certain embodiments.

[0056] In certain embodiments, the system 300 includes a first set of interfaces 330 positioned between (e.g., logically and/or physically between) and operatively coupled to the first set of ports 122 and to the plurality of subsystems 310. In addition, the system 300 of certain embodiments may include a second set of interfaces 340 positioned between (e.g., logically and/or physically between) and operatively coupled to the second set of ports 126 and the one or more subsystems 320. For example, the first and/or second sets of interfaces 330, 340 may comprise port connections.

[0057] In an example configuration, the plurality of subsystems 310 include a plurality of memory devices 311 (e.g., DRAM devices) mounted on a memory module and the first set of interfaces 330 comprise interfaces between the circuit 112 and the plurality of memory devices 311. For example, each of the first set of ports 122 may correspond to a memory signal (e.g., a data, address, and/or control signal) and each of the first set of interfaces 330 is operatively coupled to a corresponding memory signal port of a corresponding memory device 311. The one or more subsystems 320 of the example configuration comprise a host computer system and the second set of interfaces 340 comprises memory board (e.g., a PCB of a memory module) interfaces to the host system. For example, the second set of interfaces

340 form part of a connector. Each of the second set of ports 126 and corresponding interfaces of the second set of interfaces 340 corresponds to a memory signal (e.g., a data, address, and/or control signal) of at least one memory bus between the host system and the memory module.

[0058] The coarse correction elements 152 (e.g., impedance matching circuits) and the fine correction elements (e.g., self-adjusting dampers) of the second set of ports 126 of the example configuration are configured to reduce the signal reflection due to the impedance mismatching between the system board and the memory board. The coarse correction elements 152 (e.g., impedance matching circuits) and the fine correction elements (e.g., self-adjusting dampers) of the second set of ports 122 of the example configuration are configured to reduce the signal reflection due to the impedance mismatching between the circuit 112, the memory board and/or the plurality of memory devices. The filter elements 156 (e.g., a bandpass filter) reduce or filter out cross-coupling between signals, for example. For example, the filter elements 158 reduce or filter out cross-coupling between data, address, and/or control signals (e.g., wires).

[0059] Figure 10 is a flowchart of an example method 400 of using a memory module 100 with a computer system in accordance with certain embodiments described herein. While the method 400 is described herein by reference to the memory module 100 schematically illustrated by Figure 3, other memory modules, electronic systems or subsystems, and/or circuits are also compatible with the embodiments described herein. In certain embodiments, at operational block 402, the method 400 comprises providing a memory module 100 comprising a printed circuit board 102 having at least one connector 104 configured to be operatively coupled to a memory controller 106 of a computer system 108 and to a plurality of memory devices 110. The memory module 100 further comprises a circuit 112 including a first set of ports 122 comprising a plurality of bi-directional ports 124. In certain embodiments, each port 124 of the first set of ports 122 is operatively coupled to at least one memory device 111 of the plurality of memory devices 110. The circuit of certain embodiments further includes a second set of ports 126 comprising one or more bidirectional ports 128' where each port 128 of the second set of ports 126 is operatively coupled to the at least one connector 104.

[0060] The circuit 112 of certain embodiments is configured to selectively operatively couple

one or more ports 128 of the second set of ports 126 to one or more ports 124 of the first set of ports 122. The one or more ports 124 of the first set of ports 122 and the one or more ports 128 of the second set of ports 126 each comprises at least one correction circuit 150 which reduces noise in one or more signals transmitted between the one or more ports 124 of the first set of ports 122 and the one or more ports 128 of the second set of ports 126.

[0061] The method 400 of certain embodiments further includes activating the circuit 112 at operational block 404 to selectively operatively couple at least one of the one or more ports 128 of the second set of ports 126 to at least one of the one or more ports 124 of the first set of ports 122. For example, in certain embodiments, activating the circuit 112 to perform the selective operative coupling may include sending the appropriate control signal or combination of control signals to a switching sub-circuit 140 of the circuit 112. In certain embodiments, the method 400 further includes activating the circuit 112 to reduce noise in the one or more signals at operational block 406. For example, in one embodiment, activating the circuit 112 to reduce noise comprises activating one or more of the elements of the correction circuit 150 (e.g., at least one coarse correction element 152, at least one fine correction element 154, at least one filter element 156, and at least one repeater/amplifier element 158) of ports 124, 128 of the first and second sets of ports 122, 126 corresponding to active signal paths between the memory controller 106 and the memory devices 110. Activating one or more of the elements 152, 154, 156, 158 of the correction circuit 150 may include sending control signals (e.g., enable signals) to the appropriate elements 152, 154, 156, 158 and/or receiving the control signals (e.g., enable signals) at the appropriate elements 152, 154, 156, 158.

[0062] The operational blocks 402, 404, 406 of the method 400 may be performed in a different order than provided or one or more of the operational blocks may be performed substantially at the same time or during overlapping time periods. For example, in certain embodiments, at least the operational block 404 and the operational block 406 are performed substantially simultaneously. In addition, in some embodiments, one or more of the operational blocks 402, 404, 406 may not be included, an additional operational block may be included, or both. The devices and methods described herein may be incorporated in any system (e.g. a board level system) or device that contains one or more connectors or wires.

[0063] Although certain preferred embodiments and examples are discussed above, it is

understood that the inventive subject matter extends beyond the specifically disclosed embodiments to other alternative embodiments and/or uses of the invention and obvious modifications and equivalents thereof. It is intended that the scope of the inventions disclosed herein should not be limited by the particular disclosed embodiments. Thus, for example, in any method or process disclosed herein, the acts or operations making up the method/process may be performed in any suitable sequence and are not necessarily limited to any particular disclosed sequence. Various aspects and advantages of the embodiments have been described where appropriate. It is to be understood that not necessarily all such aspects or advantages may be achieved in accordance with any particular embodiment. Thus, for example, it should be recognized that the various embodiments may be carried out in a manner that achieves or optimizes one advantage or group of advantages as taught herein without necessarily achieving other aspects or advantages as may be taught or suggested herein.

WHAT IS CLAIMED IS:

1. A memory module operable to communicate data with a system memory controller via a memory bus in response to address and control signals from the system memory controller, comprising:

a printed circuit board comprising at least one connector configured to be operatively coupled to the memory bus, the at least one connector including a plurality of edge connections distributed along one or more edges of the printed circuit board;

a plurality of memory devices on the printed circuit board, the plurality of memory devices being arranged in multiple ranks, including a first rank and a second rank, each rank comprising an independent set of memory devices that can be accessed by the system memory controller to access a full bit-width of the memory bus;

a circuit coupled between the at least one connector and the plurality of memory devices, the circuit comprising a first set of ports coupled to respective ones of the plurality of edge connections, the first set of ports including a set of bidirectional ports for conducting data signals, each port of the set of bidirectional ports in the first set of ports including a signal waveform reshaping circuit, the circuit further comprising a second set of ports coupled to the plurality of memory devices, the second set of ports including a first subset of ports coupled to the first rank and a second subset of ports coupled to the second rank, the circuit further comprising a switching sub-circuit configured to operatively couple the first subset of ports to respective ones of the set of bidirectional ports and to isolate the second subset of ports from the set of bidirectional ports in response to a set of address and control signals from the system memory controller, the set of address and control signals causing the memory module to select the first rank and not the second rank for memory access by the system memory controller; and

control circuitry configured to control the switching sub-circuit and the programmable impedance matching circuits in response to one or more signals in the address and control signals.

2. The memory module of claim 1, wherein each port in the second set of ports includes an impedance matching circuit that is pre-set for matching impedance associated with a respective one of the plurality of memory devices.

3. The memory module of claim 2, wherein the first set of ports includes a set of uni-directional ports in addition to the set of bidirectional ports.
4. The memory module of claim 2, wherein each port of the set of bidirectional ports in the first set of ports further includes a programmable impedance matching circuit having at least one variable impedance that can be programmed to match a characteristic impedance of a corresponding signal path in the system.
5. The memory module of claim 1, wherein each of the programmable impedance matching circuits comprises at least one network of programmable resistors that is dynamically controllable.
6. The memory module of claim 1, wherein the circuit further comprises self-adjusting damper circuits corresponding to respective ones of the plurality of edge connections.
7. The memory module of claim 1, wherein the circuit further comprises signal waveform reshaping circuits corresponding to respective ones of the plurality of edge connections.
8. The memory module of claim 1, wherein the circuit is further capable of reshaping data signals between the system memory controller and memory devices in the first rank; and
wherein the control circuitry is further configured to control reshaping of the data signals in response to one or more signals in the address and control signals.
9. The memory module of claim 8, wherein the circuit comprises first dynamically programmable repeater/amplifier elements corresponding to respective ones of the plurality of edge connections.
10. The memory module of claim 8, wherein the circuit comprises second dynamically programmable repeater/amplifier elements corresponding to respective ones of the second set of ports.

11. The memory module of claim 10, wherein a number of ports in the second set of ports is larger than a number of edge connections in the edge connector.

12. The memory module of claim 1, wherein the circuit is further configured to provide a known static value to each port in the second subset of ports while the second subset of ports are being isolated from the set of bidirectional ports.

13. The memory module of claim 1, wherein the number of bi-directional ports in the first set of ports corresponds to a bit width of the memory bus.

14. The memory module of claim 1, wherein the circuit comprises multiple physically separate integrated circuit packages each having a data width corresponding to a data width of one of the plurality of memory devices.

15. The memory module of claim 1, wherein the circuit comprises multiple physically separate integrated circuit packages each having a data width corresponding to a data width of multiple memory devices of the plurality of memory devices.

16. A circuit mountable on a memory module, the memory module operable to communicate data with a system memory controller via a memory bus in response to address and control signals from the system memory controller, the memory module including a plurality of memory devices on a printed circuit board, the plurality of memory devices being arranged in multiple ranks, including a first rank and a second rank, each rank comprising an independent set of memory devices that can be accessed by the system memory controller to access a full bit-width of the memory bus, the printed circuit board comprising at least one connector configured to be operatively coupled to the memory bus, the at least one connector including a plurality of edge connections distributed along one or more edges of the printed circuit board, the circuit comprising:

a first set of ports coupled to respective ones of the plurality of edge connections, the first set of ports including a set of bidirectional ports for conducting data signals, each port of the set of bidirectional ports in the first set of ports including a signal waveform reshaping circuit;

a second set of ports coupled to the plurality of memory devices, the second set of ports including a first subset of ports coupled to the first rank and a second subset of ports

coupled to the second rank; and

a switching sub-circuit configured to operatively couple the first subset of ports to respective ones of the set of bidirectional ports and to isolate the second subset of ports from the set of bidirectional ports in response to a set of address and control signals from the system memory controller, the set of address and control signals causing the memory module to select the first rank and not the second rank for memory access by the system memory controller.

17. The circuit of claim 16, wherein each port in the second set of ports includes an impedance matching circuit that is pre-set for matching impedance associated with a respective one of the plurality of memory devices.

18. The circuit of claim 16, wherein the circuit is further configured to provide a known static value to each port in the second subset of ports while the second subset of ports are being isolated from the set of bidirectional ports.

19. The circuit of claim 6, wherein the circuit comprises multiple physically separate integrated circuit packages each having a data width corresponding to a data width of one of the plurality of memory devices.

20. The circuit of claim 16, wherein the circuit comprises multiple physically separate integrated circuit packages each having a data width corresponding to a data width of multiple memory devices of the plurality of memory devices.

MEMORY MODULE WITH LOAD ISOLATION AND NOISE REDUCTION

ABSTRACT

Certain embodiments described herein include a memory module having a printed circuit board including at least one connector configured to be operatively coupled to a memory controller of a computer system. The memory module further includes a plurality of memory devices on the printed circuit board and a circuit including a first set of ports operatively coupled to at least one memory device. The circuit further includes a second set of ports operatively coupled to the at least one connector. The circuit includes a switching circuit configured to selectively operatively couple one or more ports of the second set of ports to one or more ports of the first set of ports. Each port of the first set and the second set comprises a correction circuit which reduces noise in one or more signals transmitted between the first set of ports and the second set of ports.

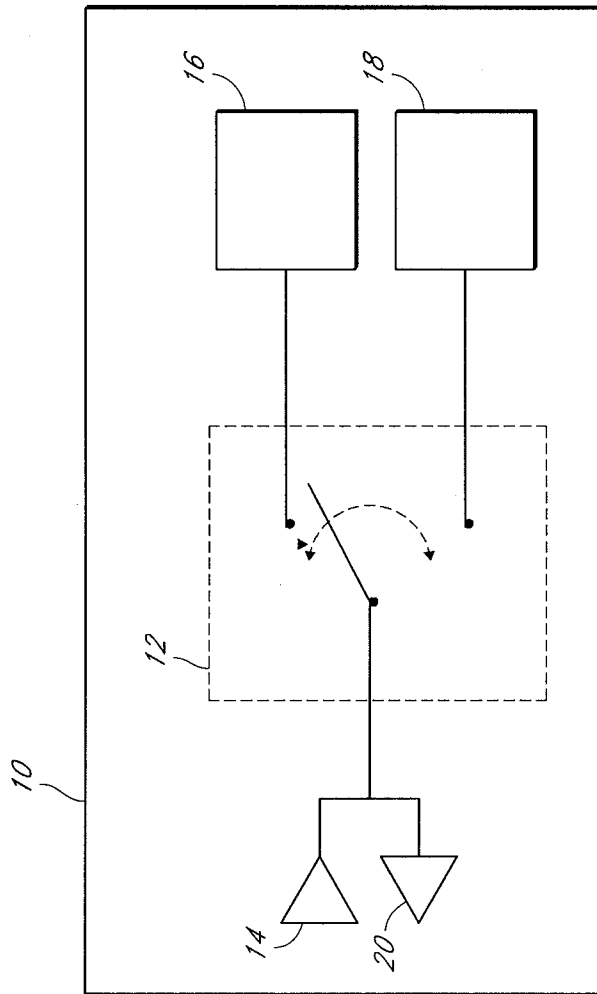


FIG. 1

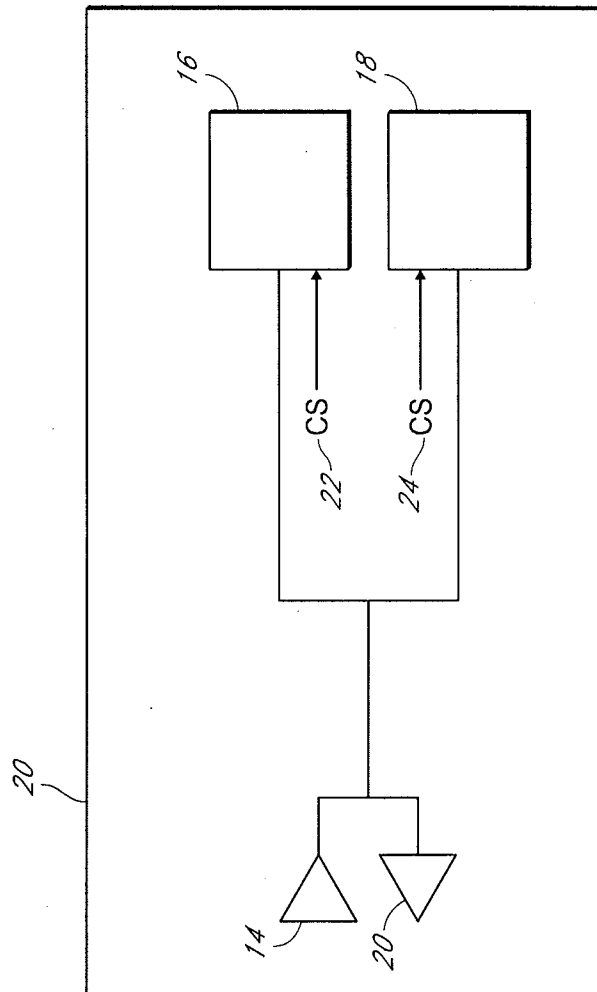


FIG. 2

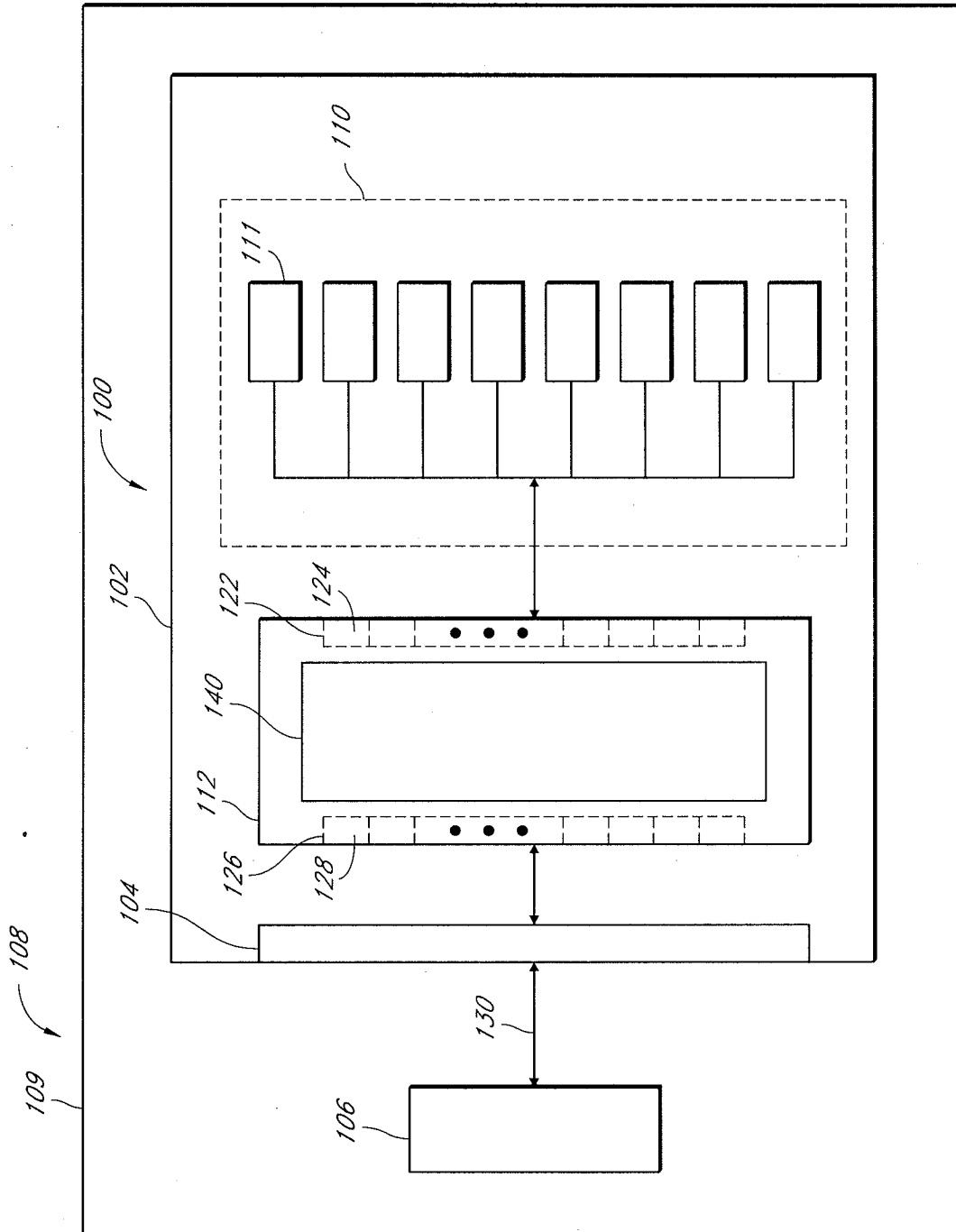


FIG. 3

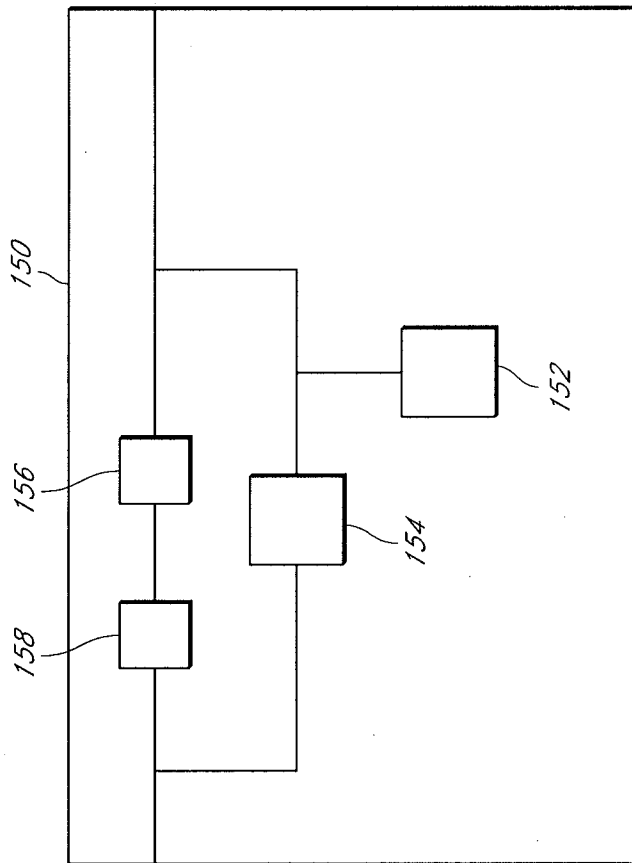


FIG. 4

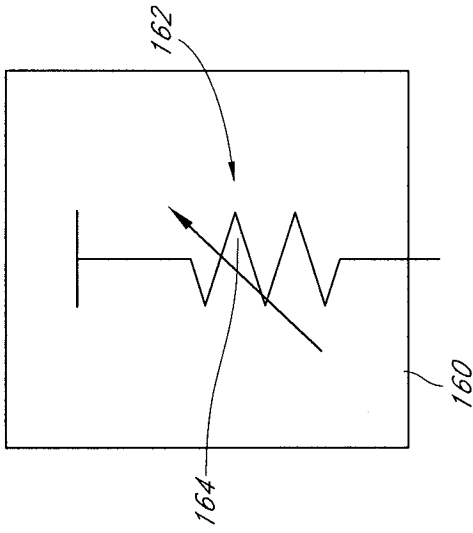


FIG. 5

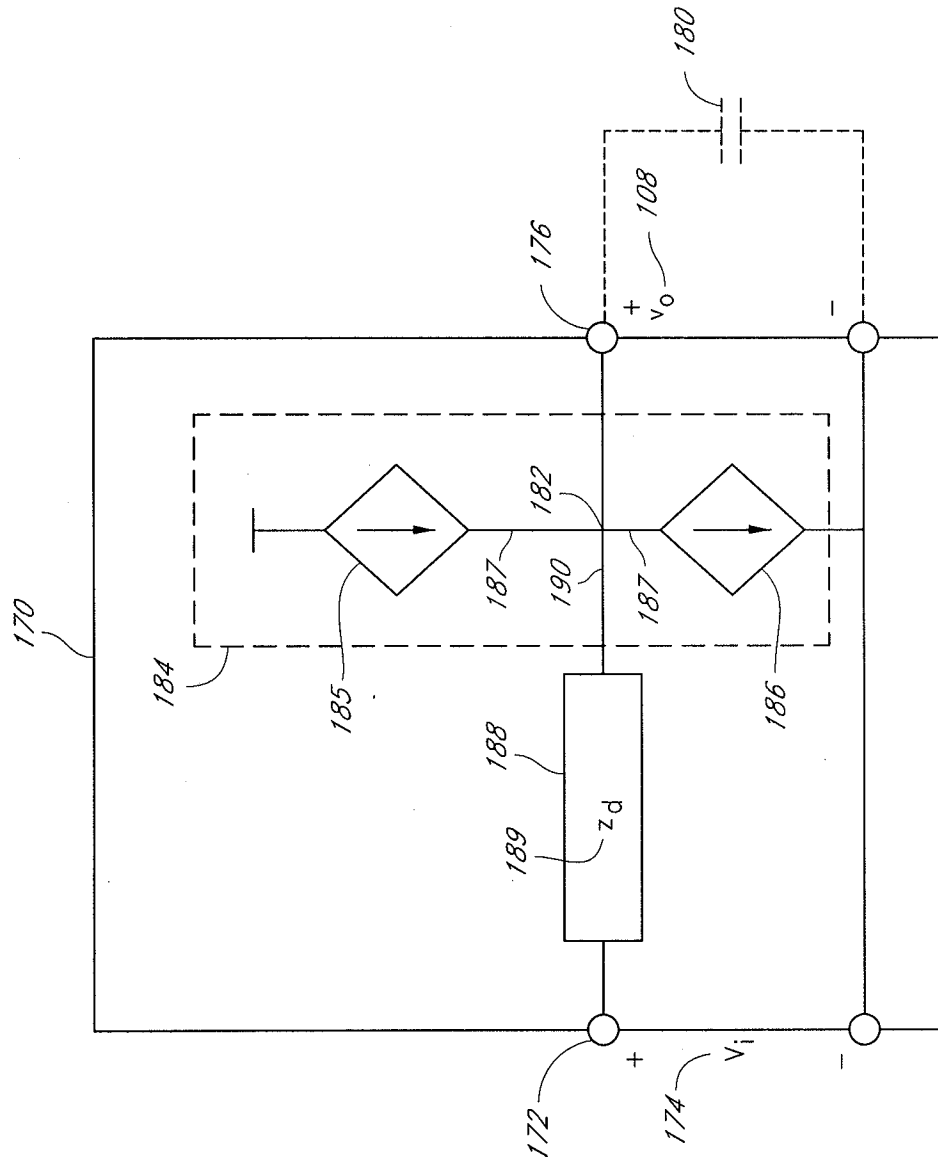


FIG. 6

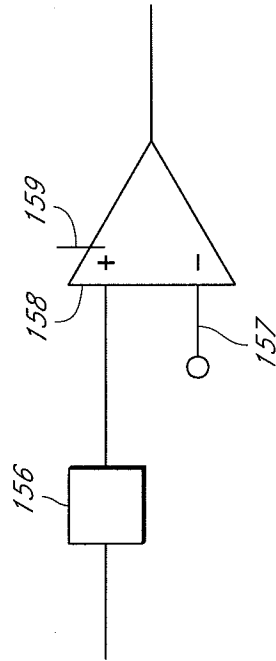


FIG. 7

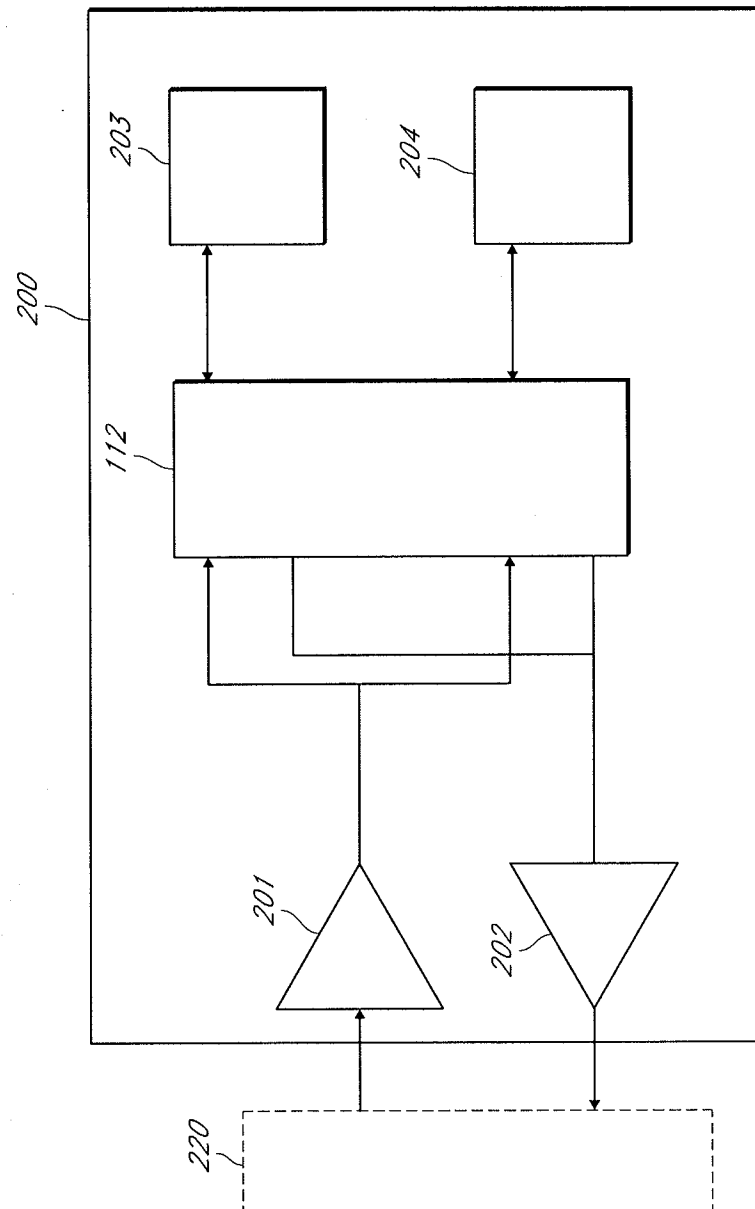


FIG. 8

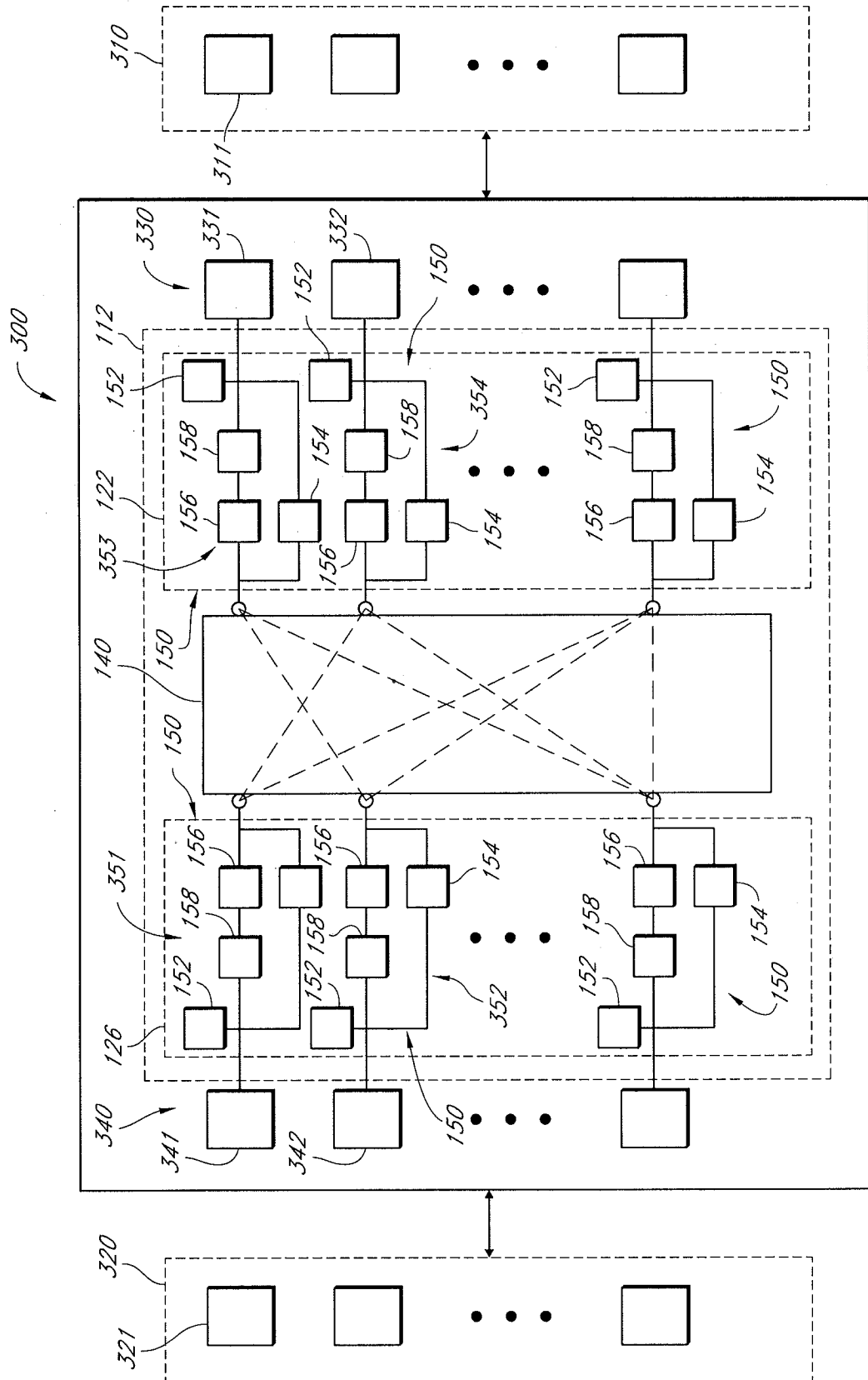


FIG. 9

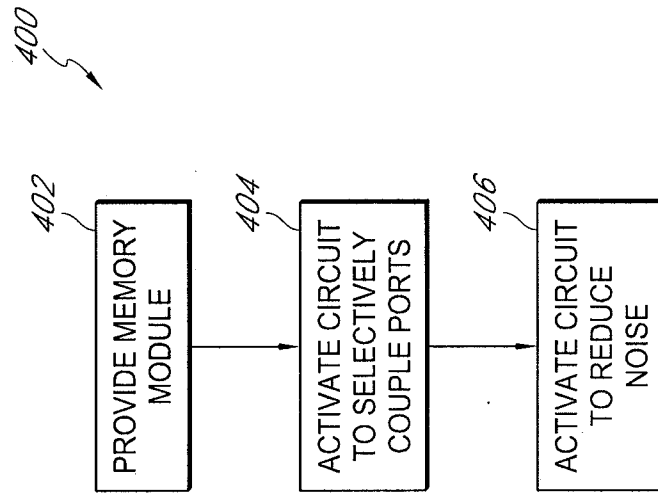


FIG. 10

Electronic Patent Application Fee Transmittal				
Application Number:				
Filing Date:				
Title of Invention:		MEMORY MODULE WITH LOAD ISOLATION AND NOISE REDUCTION		
First Named Inventor/Applicant Name:		HYUN LEE		
Filer:		Jamie Jie Zheng		
Attorney Docket Number:		NETL.045C3		
Filed as Small Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Utility filing Fee (Electronic filing)	4011	1	70	70
Utility Search Fee	2111	1	300	300
Utility Examination Fee	2311	1	360	360
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				730

Electronic Acknowledgement Receipt

EFS ID:	22381456
Application Number:	14715491
International Application Number:	
Confirmation Number:	8143
Title of Invention:	MEMORY MODULE WITH LOAD ISOLATION AND NOISE REDUCTION
First Named Inventor/Applicant Name:	HYUN LEE
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	NETL.045C3
Receipt Date:	18-MAY-2015
Filing Date:	
Time Stamp:	21:15:15
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Application Data Sheet	2015-05-18_ApplicationDataSheet_NETL045C3.pdf	1170656 c21d30da61b322217d9acdb89926d686e5b97aee	no	8

Warnings:

Information:

2		2015-05-18_Specification_NETL045C3.pdf	257686 ccb47fd89e5989c3892125286e983b8daa7988ea	yes	28
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Specification		1	23	
	Claims		24	27	
	Abstract		28	28	
Warnings:					
Information:					
3	Drawings-only black and white line drawings	2015-05-18_Drawings_NETL045C3.pdf	83908 ba4c96ef7f0139521178c281676b8c4e65c49a98	no	9
Warnings:					
Information:					
4	Fee Worksheet (SB06)	fee-info.pdf	34758 348ab1b2cd7bf96f7d8efb3c505520be1ee996d7	no	2
Warnings:					
Information:					
Total Files Size (in bytes):			1547008		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					



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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
14/715,491	05/18/2015	Hyun Lee	NETL.045C3

CONFIRMATION NO. 8143

FORMALITIES LETTER



OC000000075406430

79141

The Law Office of Jamie Zheng, Ph.D Esq.
P.O. Box 60573
Palo Alto, CA 94306

Date Mailed: 05/29/2015

NOTICE TO FILE MISSING PARTS OF NONPROVISIONAL APPLICATION

FILED UNDER 37 CFR 1.53(b)

Filing Date Granted

Items Required To Avoid Abandonment:

An application number and filing date have been accorded to this application. The item(s) indicated below, however, are missing. Applicant is given **TWO MONTHS** from the date of this Notice within which to file all required items below to avoid abandonment. Extensions of time may be obtained by filing a petition accompanied by the extension fee under the provisions of 37 CFR 1.136(a).

- The statutory basic filing fee is missing.
- The application search fee must be submitted.
- The application examination fee must be submitted.
- Surcharge as set forth in 37 CFR 1.16(f) must be submitted.

The surcharge is due for any one of:

- late submission of the basic filing fee, search fee, or examination fee,
- late submission of inventor's oath or declaration,
- filing an application that does not contain at least one claim on filing, or
- submission of an application filed by reference to a previously filed application.

SUMMARY OF FEES DUE:

The fee(s) required within **TWO MONTHS** from the date of this Notice to avoid abandonment is/are itemized below. Small entity discount is in effect. If applicant is qualified for micro entity status, an acceptable Certification of Micro Entity Status must be submitted to establish micro entity status. (See 37 CFR 1.29 and forms PTO/SB/15A and 15B.)

- \$ **70** basic filing fee.
- \$ **70** surcharge.
- \$ **300** search fee.
- \$ **360** examination fee.
- \$(**0**) previous unapplied payment amount.
- \$ **800** TOTAL FEE BALANCE DUE.

Items Required To Avoid Processing Delays:

Applicant is notified that the above-identified application contains the deficiencies noted below. No period for reply is set forth in this notice for correction of these deficiencies. However, if a deficiency relates to the inventor's

oath or declaration, the applicant must file an oath or declaration in compliance with 37 CFR 1.63, or a substitute statement in compliance with 37 CFR 1.64, executed by or with respect to each actual inventor no later than the expiration of the time period set in the "Notice of Allowability" to avoid abandonment. See 37 CFR 1.53(f).

- A properly executed inventor's oath or declaration has not been received for the following inventor(s):

Hyun Lee
Jayesh R. Bhakta
Jeffrey C. Solomon
Mario Jesus Martinez
Chi-She Chen

Replies must be received in the USPTO within the set time period or must include a proper Certificate of Mailing or Transmission under 37 CFR 1.8 with a mailing or transmission date within the set time period. For more information and a suggested format, see Form PTO/SB/92 and MPEP 512.

Replies should be mailed to:

Mail Stop Missing Parts
Commissioner for Patents
P.O. Box 1450
Alexandria VA 22313-1450

Registered users of EFS-Web may alternatively submit their reply to this notice via EFS-Web, including a copy of this Notice and selecting the document description "Applicant response to Pre-Exam Formalities Notice".
<https://portal.uspto.gov/authenticate/AuthenticateUserLocalEPF.html>

For more information about EFS-Web please call the USPTO Electronic Business Center at 1-866-217-9197 or visit our website at <http://www.uspto.gov/ebc>.

If you are not using EFS-Web to submit your reply, you must include a copy of this notice.

Questions about the contents of this notice and the requirements it sets forth should be directed to the Office of Data Management, Application Assistance Unit, at **(571) 272-4000** or **(571) 272-4200** or **1-888-786-0101**.

/nhassani/

PATENT APPLICATION FEE DETERMINATION RECORD

Substitute for Form PTO-875

Application or Docket Number
14/715,491

APPLICATION AS FILED - PART I

(Column 1)

(Column 2)

SMALL ENTITY

OR

OTHER THAN SMALL ENTITY

FOR	NUMBER FILED	NUMBER EXTRA
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A
SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A
TOTAL CLAIMS (37 CFR 1.16(i))	20 minus 20 =	*
INDEPENDENT CLAIMS (37 CFR 1.16(h))	2 minus 3 =	*
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).	
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))		

RATE(\$)	FEE(\$)
N/A	70
N/A	300
N/A	360
x 40 =	0.00
x 210 =	0.00
	0.00
	0.00
TOTAL	730

RATE(\$)	FEE(\$)
N/A	
N/A	
N/A	
TOTAL	

* If the difference in column 1 is less than zero, enter "0" in column 2.

APPLICATION AS AMENDED - PART II

(Column 1)

(Column 2)

(Column 3)

SMALL ENTITY

OR

OTHER THAN SMALL ENTITY

AMENDMENT A		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA
	Total (37 CFR 1.16(i))	*	Minus	**	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=
	Application Size Fee (37 CFR 1.16(s))				
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))				

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

(Column 1)

(Column 2)

(Column 3)

AMENDMENT B		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA
	Total (37 CFR 1.16(i))	*	Minus	**	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=
	Application Size Fee (37 CFR 1.16(s))				
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))				

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.



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APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
14/715,491	05/18/2015	2827	0.00	NETL.045C3	20	2

CONFIRMATION NO. 8143

79141

The Law Office of Jamie Zheng, Ph.D Esq.
P.O. Box 60573
Palo Alto, CA 94306

FILING RECEIPT



Date Mailed: 05/29/2015

Receipt is acknowledged of this non-provisional patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Inventor(s)

Hyun Lee, Ladera Ranch, CA;
Jayesh R. Bhakta, Cerritos, CA;
Jeffrey C. Solomon, Irvine, CA;
Mario Jesus Martinez, Laguna Niguel, CA;
Chi-She Chen, Walnut, CA;

Applicant(s)

Netlist, Inc., Irvine, CA;

Power of Attorney: None

Domestic Priority data as claimed by applicant

This application is a CON of 14/324,990 07/07/2014 PAT 9037809
which is a CON of 13/412,243 03/05/2012 PAT 8782350

Foreign Applications for which priority is claimed (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.) - None.

Foreign application information must be provided in an Application Data Sheet in order to constitute a claim to foreign priority. See 37 CFR 1.55 and 1.76.

If Required, Foreign Filing License Granted: 05/27/2015

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 14/715,491**

Projected Publication Date: Request for Non-Publication Acknowledged

Non-Publication Request: Yes

Early Publication Request: No

**** SMALL ENTITY ****

Title

Circuit Providing Load Isolation and Noise Reduction

Preliminary Class

365

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

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For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4258).

LICENSE FOR FOREIGN FILING UNDER
Title 35, United States Code, Section 184
Title 37, Code of Federal Regulations, 5.11 & 5.15

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This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).

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UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
14/715,491	05/18/2015	Hyun Lee	NETL.045C3

CONFIRMATION NO. 8143

IMPROPER CFR REQUEST



OC000000075406426

79141

The Law Office of Jamie Zheng, Ph.D Esq.
P.O. Box 60573
Palo Alto, CA 94306

Date Mailed: 05/29/2015

RESPONSE TO REQUEST FOR CORRECTED FILING RECEIPT

Continuity, Priority Claims, Petitions, and Non-Publication Requests

In response to your request for a corrected Filing Receipt, the Office is unable to comply with your request because:

- One or more of the benefit claims under 35 U.S.C. § 120 cannot be included on the Filing Receipt since applicant did not specify whether the application is a continuation, divisional or continuation-in-part of the prior application. Applicant must submit a new application data sheet (ADS) that sets forth the relationship, and the ADS must be accompanied by a petition under 37 CFR 1.78 if filed after the time period set forth in 37 CFR 1.78.

Questions about the contents of this notice and the requirements it sets forth should be directed to the Office of Data Management, Application Assistance Unit, at (571) 272-4000 or (571) 272-4200 or 1-888-786-0101.

/tmekuria/

Electronic Patent Application Fee Transmittal

Application Number:	14715491			
Filing Date:	18-May-2015			
Title of Invention:	Circuit Providing Load Isolation and Noise Reduction			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Jamie Jie Zheng			
Attorney Docket Number:	NETL.045C3			
Filed as Small Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Utility filing Fee (Electronic filing)	4011	1	70	70
Utility Search Fee	2111	1	300	300
Utility Examination Fee	2311	1	360	360
Pages:				
Claims:				
Miscellaneous-Filing:				
Late Filing Fee for Oath or Declaration	2051	1	70	70
Petition:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				
Extension - 2 months with \$0 paid	2252	1	300	300
Miscellaneous:				
Total in USD (\$)				1100

Electronic Acknowledgement Receipt

EFS ID:	23574324
Application Number:	14715491
International Application Number:	
Confirmation Number:	8143
Title of Invention:	Circuit Providing Load Isolation and Noise Reduction
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	NETL.045C3
Receipt Date:	22-SEP-2015
Filing Date:	18-MAY-2015
Time Stamp:	23:59:16
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	Credit Card
Payment was successfully received in RAM	\$ 1100
RAM confirmation Number	6922
Deposit Account	
Authorized User	

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Fee Worksheet (SB06)	fee-info.pdf	38742	no	2
			2f365f734a512051b3c0e14c910110f9ab8cb59b		

Warnings:

Information:

Total Files Size (in bytes):	38742
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This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

Customer No. 79141
Attorney Docket No.: NETL045C2

DECLARATION FOR UTILITY OR DESIGN APPLICATION	
Title of Invention	CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION
DECLARATION	
As a below named inventor, I hereby declare that:	
This declaration is directed to:	
☐ The attached application, or	
☒ United States application or PCT international application number <u>14/324,990</u> filed on <u>July 7, 2014</u>	
The above-identified application was made or authorized to be made by me.	
I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.	
I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.	
Note to Inventor: 37 C.F.R. § 1.63(c) states: "A person may not execute an oath or declaration for an application unless that person has reviewed and understands the contents of the application, including the claims, and is aware of the duty to disclose to the Office all information known to the person to be material to patentability as defined in § 1.56."	
First Inventor (Legal Name): <u>Hyun Lee</u>	
Signature: <u>[Signature]</u>	Date: <u>3-24-2015</u>
Second Inventor (Legal Name): <u>Jayesh R. Bhakta</u>	
Signature: <u>[Signature]</u>	Date: <u>3-24-2015</u>
Third Inventor (Legal Name): <u>Jeffery C. Solomon</u>	
Signature: _____	Date: _____
Fourth Inventor (Legal Name): <u>Mario Jesus Martinez</u>	
Signature: <u>[Signature]</u>	Date: <u>3-24-2015</u>
Fifth Inventor (Legal Name): <u>Chi She Chen</u>	
Signature: <u>[Signature]</u>	Date: <u>3-24-2015</u>

**DECLARATION FOR
UTILITY OR DESIGN APPLICATION**

**Title of
Invention**

**CIRCUIT PROVIDING LOAD ISOLATION AND NOISE
REDUCTION**

DECLARATION

As a below named inventor, I hereby declare that:

This declaration
is directed to:

☐

The attached application, or

☒

United States application or PCT international application
number 14/324,990 filed on July 7, 2014.

The above-identified application was made or authorized to be made by me.

I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001
by fine or imprisonment of not more than five (5) years, or both.

Note to Inventor: 37 C.F.R. § 1.63(c) states: "A person may not execute an oath or declaration for an application
unless that person has reviewed and understands the contents of the application, including the claims, and is aware
of the duty to disclose to the Office all information known to the person to be material to patentability as defined in
§ 1.56."

First Inventor (Legal Name): Hyun Lee

Signature: _____

Date: _____

Second Inventor (Legal Name): Jayesh R. Bhakta

Signature: Jayesh Bhakta

Date: 3-24-2015

Third Inventor (Legal Name): Jeffery C. Solomon

Signature: Jeffery C. Solomon

Date: 04-06-2015

Fourth Inventor (Legal Name): Mario Jesus Martínez

Signature: Mario Jesus Martínez

Date: 3-24-2015

Fifth Inventor (Legal Name): Chi She Chen

Signature: Chi She Chen

Date: 3-24-2015

Electronic Acknowledgement Receipt

EFS ID:	23574338
Application Number:	14715491
International Application Number:	
Confirmation Number:	8143
Title of Invention:	Circuit Providing Load Isolation and Noise Reduction
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	NETL.045C3
Receipt Date:	23-SEP-2015
Filing Date:	18-MAY-2015
Time Stamp:	00:10:54
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Oath or Declaration filed	2015-03-24_Declaration_NETL045C2.pdf	567329 3bc3fbfa9565252cb049792c19bf747753650b4f	no	2

Warnings:

The page size in the PDF is too large. The pages should be 8.5 x 11 or A4. If this PDF is submitted, the pages will be resized upon entry into the Image File Wrapper and may affect subsequent processing

Information:

Total Files Size (in bytes):

567329

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New International Application Filed with the USPTO as a Receiving Office

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UNITED STATES PATENT AND TRADEMARK OFFICE

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United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
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www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
14/715,491	05/18/2015	2827	800	NETL.045C3	20	2

CONFIRMATION NO. 8143

UPDATED FILING RECEIPT

79141

The Law Office of Jamie Zheng, Ph.D Esq.
P.O. Box 60573
Palo Alto, CA 94306



Date Mailed: 09/30/2015

Receipt is acknowledged of this non-provisional patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Inventor(s)

Hyun Lee, Ladera Ranch, CA;
Jayesh R. Bhakta, Cerritos, CA;
Jeffrey C. Solomon, Irvine, CA;
Mario Jesus Martinez, Laguna Niguel, CA;
Chi-She Chen, Walnut, CA;

Applicant(s)

Netlist, Inc., Irvine, CA;

Power of Attorney: None

Domestic Priority data as claimed by applicant

This application is a CON of 14/324,990 07/07/2014 PAT 9037809
which is a CON of 13/412,243 03/05/2012 PAT 8782350

Foreign Applications for which priority is claimed (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.) - None.

Foreign application information must be provided in an Application Data Sheet in order to constitute a claim to foreign priority. See 37 CFR 1.55 and 1.76.

If Required, Foreign Filing License Granted: 05/27/2015

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 14/715,491**

Projected Publication Date: Request for Non-Publication Acknowledged

Non-Publication Request: Yes

Early Publication Request: No

**** SMALL ENTITY ****

Title

Circuit Providing Load Isolation and Noise Reduction

Preliminary Class

365

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

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Title 35, United States Code, Section 184
Title 37, Code of Federal Regulations, 5.11 & 5.15

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PATENT APPLICATION FEE DETERMINATION RECORD

Substitute for Form PTO-875

Application or Docket Number
14/715,491

APPLICATION AS FILED - PART I

(Column 1)

(Column 2)

SMALL ENTITY

OR

OTHER THAN SMALL ENTITY

FOR	NUMBER FILED	NUMBER EXTRA
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A
SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A
TOTAL CLAIMS (37 CFR 1.16(i))	20 minus 20 =	*
INDEPENDENT CLAIMS (37 CFR 1.16(h))	2 minus 3 =	*
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).	
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))		

RATE(\$)	FEE(\$)
N/A	70
N/A	300
N/A	360
x 40 =	0.00
x 210 =	0.00
	0.00
	0.00
TOTAL	730

RATE(\$)	FEE(\$)
N/A	
N/A	
N/A	
TOTAL	

* If the difference in column 1 is less than zero, enter "0" in column 2.

APPLICATION AS AMENDED - PART II

(Column 1)

(Column 2)

(Column 3)

SMALL ENTITY

OR

OTHER THAN SMALL ENTITY

AMENDMENT A		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA
	Total (37 CFR 1.16(i))	*	Minus	**	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=
	Application Size Fee (37 CFR 1.16(s))				
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))				

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

(Column 1)

(Column 2)

(Column 3)

AMENDMENT B		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA
	Total (37 CFR 1.16(i))	*	Minus	**	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=
	Application Size Fee (37 CFR 1.16(s))				
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))				

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Applicant	:	Netlist, Inc.
App. No.	:	14/715,491
Filed	:	May 18, 2015
For	:	CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION
Examiner	:	UNKNOWN
Art Unit	:	2827
Conf. No.	:	8143
Docket No.	:	NETL.045C3

PRELIMINARY AMENDMENT**Mail Stop Amendment**

Commissioner for Patents

P.O. Box 1450

Alexandria, VA 22313-1450

Dear Sir:

Applicant submits the following amendments and remarks in the above-identified patent application.

The Commissioner is hereby authorized to charge any required fee(s) to Deposit Account No. 50-5963.

Amendments to the Specification: Begin on page 2 of this paper.

Amendments to the Claims: Begin on page 3 of this paper.

Remarks: Begin on page 8 of this paper.

In the Specification:

Please replace the title with the following:

MEMORY MODULE AND CIRCUIT PROVIDING LOAD ISOLATION AND NOISE
REDUCTION

In the Claims:

1. (Currently Amended) A memory module operable to communicate data with a system memory controller via a memory bus in response to address and control signals from the system memory controller, comprising:

a printed circuit board comprising at least one connector configured to be operatively coupled to the memory bus, the at least one connector including a plurality of edge connections distributed along one or more edges of the printed circuit board;

a plurality of memory devices on the printed circuit board, the plurality of memory devices being arranged in multiple ranks, including a first rank and a second rank, each rank comprising an independent set of memory devices that can be accessed by the system memory controller to access a full bit-width of the memory bus;

a circuit coupled between the at least one connector and the plurality of memory devices, the circuit comprising a first set of ports coupled to respective ones of the plurality of edge connections, the first set of ports including a set of bidirectional ports for conducting data signals, each port of the set of bidirectional ports in the first set of ports including a signal waveform reshaping circuit, the circuit further comprising a second set of ports coupled to the plurality of memory devices, the second set of ports including a first subset of ports coupled to the first rank and a second subset of ports coupled to the second rank, the circuit further comprising a switching sub-circuit configured to operatively couple either the first subset of ports or the second subset of ports to respective ones of the set of bidirectional ports ~~and to isolate the second subset of ports from the set of bidirectional ports in response to a set of address and control signals from the system memory controller, the set of address and control signals causing the memory module to select the first rank and not the second rank for memory access by the system memory controller;~~ and

control circuitry configured to receive address and control signals from the memory controller and to control the switching sub-circuit and the the signal waveform reshaping circuits in the the set of bidirectional ports ~~programmable impedance matching circuits~~ in response to one or more signals in the address and control signals, the signal waveform reshaping circuits being configured to reshape data signals between the

memory module and the plurality of memory devices, the address and control signals causing the memory module to select either the first rank or the second rank for memory access by the system memory controller.

2. (Original) The memory module of claim 1, wherein each port in the second set of ports includes an impedance matching circuit that is pre-set for matching impedance associated with a respective one of the plurality of memory devices.
3. (Original) The memory module of claim 2, wherein the first set of ports includes a set of uni-directional ports in addition to the set of bidirectional ports.
4. (Original) The memory module of claim 2, wherein each port of the set of bidirectional ports in the first set of ports further includes a programmable impedance matching circuit having at least one variable impedance that can be programmed to match a characteristic impedance of a corresponding signal path in the system.
5. (Original) The memory module of claim 1, wherein each of the programmable impedance matching circuits comprises at least one network of programmable resistors that is dynamically controllable.
6. (Original) The memory module of claim 1, wherein the circuit further comprises self-adjusting damper circuits corresponding to respective ones of the plurality of edge connections.
7. (Cancelled)
8. (Cancelled)
9. (Original) The memory module of claim 8, wherein the circuit comprises first dynamically programmable repeater/amplifier elements corresponding to respective ones of the plurality of edge connections.
10. (Currently Amended) The memory module of claim 9-8, wherein the circuit comprises second dynamically programmable repeater/amplifier elements corresponding to respective ones of the second set of ports.

11. (Original) The memory module of claim 10, wherein a number of ports in the second set of ports is larger than a number of edge connections in the edge connector.
12. (Currently Amended) The memory module of claim 1, wherein the circuit is further configured to provide a known static value to each port in the second set ~~subset~~ of ports ~~while the second subset of ports are being isolated from~~ that is not coupled to one of the set of bidirectional ports by the switching sub-circuit.
13. (Original) The memory module of claim 1, wherein the number of bi-directional ports in the first set of ports corresponds to a bit width of the memory bus.
14. (Original) The memory module of claim 1, wherein the circuit comprises multiple physically separate integrated circuit packages each having a data width corresponding to a data width of one of the plurality of memory devices.
15. (Original) The memory module of claim 1, wherein the circuit comprises multiple physically separate integrated circuit packages each having a data width corresponding to a data width of multiple memory devices of the plurality of memory devices.
16. (Currently Amended) A circuit mountable on a memory module, the memory module operable to communicate data with a system memory controller via a memory bus in response to address and control signals from the system memory controller, the memory module including a plurality of memory devices on a printed circuit board, the plurality of memory devices being arranged in multiple ranks, including a first rank and a second rank, each rank comprising an independent set of memory devices that can be accessed by the system memory controller to access a full bit-width of the memory bus, the printed circuit board comprising at least one connector configured to be operatively coupled to the memory bus, the at least one connector including a plurality of edge connections distributed along one or more edges of the printed circuit board, the circuit comprising:
- a first set of ports coupled to respective ones of the plurality of edge connections, the first set of ports including a set of bidirectional ports for conducting data signals, each port of the set of bidirectional ports in the first set of ports including a signal waveform

reshaping circuit;

a second set of ports coupled to the plurality of memory devices, the second set of ports including a first subset of ports coupled to the first rank and a second subset of ports coupled to the second rank; and

a switching sub-circuit configured to operatively couple either the first subset of ports or the second subset of ports to respective ones of the set of bidirectional ports ~~and to isolate the second subset of ports from the set of bidirectional ports~~ in response to a set of address and control signals from the system memory controller, the set of address and control signals causing the memory module to select either the first rank or ~~and not~~ the second rank for memory access by the system memory controller.

17. (Original) The circuit of claim 16, wherein each port in the second set of ports includes an impedance matching circuit that is pre-set for matching impedance associated with a respective one of the plurality of memory devices.

18. (Currently Amended) The circuit of claim 16, wherein the circuit is further configured to provide a known static value to each port in the second set ~~subset~~ of ports that is not coupled to the one of ~~while the second subset of ports are being isolated from~~ the set of bidirectional ports by the switching sub-circuit.

19. (Original) The circuit of claim 6, wherein the circuit comprises multiple physically separate integrated circuit packages each having a data width corresponding to a data width of one of the plurality of memory devices.

20. (Original) The circuit of claim 16, wherein the circuit comprises multiple physically separate integrated circuit packages each having a data width corresponding to a data width of multiple memory devices of the plurality of memory devices.

21. (New) The memory module of claim 1, wherein each signal reshaping circuit includes a repeater/amplifier element configured to receive an input signal and to generate an output signal having earlier transition trigger points and faster slew rates than the input signal.

22. (New) The circuit of claim 16, wherein each signal reshaping circuit includes a repeater/amplifier element configured to receive an input signal and to generate an output signal having earlier transition trigger points and faster slew rates than the input signal.

REMARKS

Claims 7-8 are cancelled. Claims 1, 10, 12, 16, and 18 are amended. New claims 21-22 are presented. No new matter is added. After entry of this amendment, the pending claims are 1-6, 9-22.

The Examiner is invited to call the undersigned attorney at (650) 521-4828, if a telephone call could help resolve any issues.

Respectfully submitted,

Date: July 3, 2016

/Jamie J. Zheng/

Jamie J. Zheng
Customer No. 79141

51167

(Reg. No.)

Electronic Acknowledgement Receipt

EFS ID:	26247754
Application Number:	14715491
International Application Number:	
Confirmation Number:	8143
Title of Invention:	Circuit Providing Load Isolation and Noise Reduction
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	NETL.045C3
Receipt Date:	04-JUL-2016
Filing Date:	18-MAY-2015
Time Stamp:	02:58:35
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Preliminary Amendment	2016-07-03_Prelim_NETL045C3.pdf	94456	no	8
			d1e5fdcaa6505a163698ba88089f9bc15d6031a6		

Warnings:

Information:	
Total Files Size (in bytes):	94456
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PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875	Application or Docket Number 14/715,491	Filing Date 05/18/2015	☐ To be Mailed
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ENTITY: ☐ LARGE ☒ SMALL ☐ MICRO

APPLICATION AS FILED – PART I

(Column 1)

(Column 2)

FOR	NUMBER FILED	NUMBER EXTRA	RATE (\$)	FEE (\$)
☒ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A	140
☐ SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A	N/A	
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A	
TOTAL CLAIMS (37 CFR 1.16(i))	minus 20 =	*	X \$ =	
INDEPENDENT CLAIMS (37 CFR 1.16(h))	minus 3 =	*	X \$ =	
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).			
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))				
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL	140

APPLICATION AS AMENDED – PART II

(Column 1)

(Column 2)

(Column 3)

AMENDMENT	07/04/2016	CLAIMS REMAINING AFTER AMENDMENT	Minus	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)
	Total (37 CFR 1.16(i))	* 20	Minus	** 20	= 0	X \$40 =	0
	Independent (37 CFR 1.16(h))	* 2	Minus	***3	= 0	X \$210 =	0
	☐ Application Size Fee (37 CFR 1.16(s))						
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						
						TOTAL ADD'L FEE	0

(Column 1)

(Column 2)

(Column 3)

AMENDMENT	CLAIMS REMAINING AFTER AMENDMENT	Minus	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)
	Total (37 CFR 1.16(i))	*	Minus	**	=	X \$ =
	Independent (37 CFR 1.16(h))	*	Minus	***	=	X \$ =
	☐ Application Size Fee (37 CFR 1.16(s))					
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))					
						TOTAL ADD'L FEE

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.

LIE
GLORIA J. TRAMMELL

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Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	14/715,491
				Filing Date	May 18, 2015
				First Named Inventor	Hyun Lee
				Art Unit	2827
				Examiner Name	Unknown
Sheet	1	of	3	Attorney Docket Number	NETL.045C3

US PATENT DOCUMENTS					
Examiner Initials *	Cite No. ¹	Document Number	Publication Date (MM-DD-YYYY)	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number-Kind Code ² (if known)			
	1.	2002/0038405 A1	03-28-2002	Leddige, M. et al.	
	2.	2009/0296503 A1	12-03-2009	Chu, M. et al.	
	3.	2011/0085408 A1	04-14-2011	Solomon, J. et al.	
	4.	2008/0162790 A1	07-03-2008	IM	

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²
	5.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Related Matters, filed April 28, 2015	
	6.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Patent Owner Response to Petition, filed February 22, 2016	
	7.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Petitioner's Objections to Evidence, filed February 29, 2016	
	8.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Petitioner's Reply, filed May 19, 2016	
	9.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Petitioner Request for Oral Argument, filed June 2, 2016	
	10.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, PATENT OWNER MOTION FOR OBSERVATIONS, filed June 2, 2016	
	11.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, PATENT OWNER REQUEST FOR ORAL ARGUMENT, filed June 2, 2016	
	12.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, ORDER Trial Hearing, filed June 8, 2016	
	13.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Patent Owner's SUBMISSION ON PROPRIETY OF PETITIONER REPLY, filed June 9, 2016	
	14.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Petitioner's Response to Motion for Observations, filed June 10, 2016	
	15.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Petitioner's Response to Netlist's Submission, filed June 13, 2016	
	16.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Patent Owner's Objections to Petitioner's Demonstrative Exhibits, filed June 23, 2016	
	17.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Patent Owner's Demonstrative Exhibits, filed June 24, 2016	

EXAMINER

DATE CONSIDERED

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				Application Number	14/715,491
				Filing Date	May 18, 2015
				First Named Inventor	Hyun Lee
				Art Unit	2827
				Examiner Name	Unknown
Sheet	2	of	3	Attorney Docket Number	NETL.045C3

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²
	18.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Petitioner's Updated Exhibit List, filed June 24, 2016	
	19.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Supplemental Declaration of Dr. Jagannathan, filed May 19, 2016	
	20.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, May 10, 2016 Deposition of Carl Sechen, filed May 19, 2016	
	21.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Petitioner's Demonstrative Exhibits, filed June 24, 2016	
	22.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Exhibit 2001, "Declaration of Professor Carl Sechen," filed February 22, 2016	
	23.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Exhibit 2002, "Videotaped Deposition of Srinivasan Jagannathan, Ph.D on February 12, 2016," filed February 22, 2016	
	24.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Exhibit 2003, "Videotaped Deposition of Srinivasan Jagannathan, Ph.D on April 20, 2015," filed February 22, 2016	
	25.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Exhibit 2007, "Deposition of Dr. Srinivasan Jagannathan on May 25, 2016," filed June 2, 2016	
	26.	Inter Partes Review of U.S. Patent No. 8081536, Case No. IPR2015-01021, Patent Owner Response to Petition, filed February 22, 2016	
	27.	Inter Partes Review of U.S. Patent No. 8081536, Case No. IPR2015-01021, Petitioner's Objections to Evidence, filed February 29, 2016	
	28.	Inter Partes Review of U.S. Patent No. 8081536, Case No. IPR2015-01021, Petitioner's Reply, filed May 19, 2016	
	29.	Inter Partes Review of U.S. Patent No. 8081536, Case No. IPR2015-01021, Petitioner Request for Oral Argument, filed June 2, 2016	
	30.	Inter Partes Review of U.S. Patent No. 8081536, Case No. IPR2015-01021, PATENT OWNER MOTION FOR OBSERVATIONS, filed June 2, 2016	
	31.	Inter Partes Review of U.S. Patent No. 8081536, Case No. IPR2015-01021, PATENT OWNER REQUEST FOR ORAL ARGUMENT, filed June 2, 2016	
	32.	Inter Partes Review of U.S. Patent No. 8081536, Case No. IPR2015-01021, Patent Owner's SUBMISSION ON PROPRIETY OF PETITIONER REPLY, filed June 9, 2016	
	33.	Inter Partes Review of U.S. Patent No. 8081536, Case No. IPR2015-01021, Petitioner's Response to Motion for Observations, filed June 10, 2016	
	34.	Inter Partes Review of U.S. Patent No. 8081536, Case No. IPR2015-01021, Petitioner's Response to Patent Owner's Submission, filed June 13, 2016	

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	14/715,491
				Filing Date	May 18, 2015
				First Named Inventor	Hyun Lee
				Art Unit	2827
				Examiner Name	Unknown
Sheet	3	of	3	Attorney Docket Number	NETL.045C3

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²
	35.	Inter Partes Review of U.S. Patent No. 8081536, Case No. IPR2015-01021, Patent Owner's Objections to Petitioner's Demonstrative Exhibits, filed June 23, 2016	
	36.	Inter Partes Review of U.S. Patent No. 8081536, Case No. IPR2015-01021, Patent Owner's Demonstrative Exhibits, filed June 24, 2016	
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	38.	Inter Partes Review of U.S. Patent No. 8081536, Case No. IPR2015-01021, Supplemental Declaration of Dr. Jagannathan, filed May 19, 2016	
	39.	Inter Partes Review of U.S. Patent No. 8081536, Case No. IPR2015-01021, Exhibit 2001, "Declaration of Professor Carl Sechen," filed February 22, 2016	
	40.	International Search Report and Written Opinion, International Application No. PCT/US14/48517, filed July 28, 2014, dated October 27, 2014	
	41.	Inter Partes Review Case No. IPR2014-01029, Petitioner's Request for Rehearing pursuant to 37 C.F.R. § 42.71, filed on January 15, 2015.	
	42.	Inter Partes Review Case No. IPR2014-01029, Decision Denying Request for Rehearing, Issued on March 3, 2015.	
	43.	Inter Partes Review Case No. IPR2014-01369, Decision Denying Institution of Inter Partes Review 37 C.F.R. § 42.108, issued March 9, 2014.	
	44.	Examination Report, European Patent Application No. 10730021.2, dated April 14, 2014.	
	45.	Response to Examination Report, European Patent Application No. 10730021.2, dated June 4, 2014.	
	46.	Examination Report, European Patent Application No. 10730021.2, dated April 29, 2015.	
	47.	Response to Examination Report, European Patent Application No. 10730021.2, dated April 29, 2015, filed November 4, 2015.	
	48.	English Translation of the Notice of Grounds for Rejection, Korean Patent Application No. 2012-7004038, dated May 11, 2016	
	49.	Notice of Allowance, U.S. Pat. Appl. No. 13/970,606, filed August 20, 2013, dated June 27, 2016	

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EFS ID:	26289819
Application Number:	14715491
International Application Number:	
Confirmation Number:	8143
Title of Invention:	MEMORY MODULE AND CIRCUIT PROVIDING LOAD ISOLTION AND NOISE REDUCTION
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	NETL.045C3
Receipt Date:	07-JUL-2016
Filing Date:	18-MAY-2015
Time Stamp:	20:38:12
Application Type:	Utility under 35 USC 111(a)

Payment information:

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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Transmittal Letter	2016-07-07_IDS_Cover_NETL045C3.pdf	50194 40361ab1c449a4a3a0176341a6d5e11e8d8c6a0c	no	2

Warnings:

Information:					
2	Information Disclosure Statement (IDS) Form (SB08)	2016-07-07_IDS_NETL045C3.pdf	133140 92abcb038b0649a18812cf0f5fffe20bffa779d	no	3
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Information:					
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3	Non Patent Literature	11-04-2015_Rsps_ER_NT001EP.pdf	1201884 6ce107ca9f466a51d1aef5c2cff9cab7d18f5c7c	no	26
Warnings:					
Information:					
4	Non Patent Literature	2016-05-11_Eng_Trans_NGR_NT001KR.pdf	183171 1bd82675735359e2c4e2487f593949917116f0f0	no	5
Warnings:					
Information:					
5	Non Patent Literature	2016-06-27_NOA_NETL048CPC1.pdf	4054649 16c330249c95548259ee6c37db691c0381d933df	no	68
Warnings:					
Information:					
Total Files Size (in bytes):			5623038		
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INFORMATION DISCLOSURE STATEMENT

Applicant	: NETLIST, INC.
App. No.	: 14/715,491
Filed	: May 18, 2015
For	: CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION
Examiner	: UNKNOWN
Art Unit	: 2827
Conf No.	: 8143

Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Dear Sir:

Enclosed for filing in the above-identified application is a Form PTO/SB/08a/b Equivalent listing 49 references. Copies of some of the foreign documents and/or their translation and non-patent literature are submitted herewith. Copies of the rest of the foreign documents and/or their translation and non-patent literature have been submitted in commonly-owned patent applications. Specifically, Items 5-39 have been submitted in U.S. Pat. Appl. No. 14/715,486 (filed May 18, 2015) on June 29, 2016; Items 40-43 are submitted in U.S. Pat. Appl. No. 13/970,606 (filed August 20, 2013,) on March 18, 2015; Items 44-46 are submitted in U.S. Pat. Appl. No. 13/970,606 on October 27, 2015; all of which are accessible via PAIR.

The applicants would appreciate the Examiner initialing and returning the Form PTO/SB/08a/b Equivalent indicating that the information has been considered and made of record herein.

The information contained in this Information Disclosure Statement is not to be construed as a representation that: (i) a complete search has been made; (ii) additional

Application No.: 14/715,491
Filing Date: May 18, 2015

information material to this application does not exist; (iii) the information, protocols, results and the like reported by third parties are accurate or enabling; or (iv) the above information constitutes prior art to the subject invention.

This Information Disclosure Statement is being filed before the receipt of a first Office Action on the merits, and presumably no fee is required. If a first Office Action on the merits was mailed before the mailing date of this Statement, the Commissioner is authorized to charge the fee set forth in 37 C.F.R. § 1.17(p) to Deposit Account No. 50-5963.

Respectfully submitted,

Dated: July 7, 2016

/Jamie J. Zheng/

Jamie J. Zheng
Customer No. 79141
(650) 521-4828

51167

(Reg. No.)

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Filing Date	May 18, 2015																
First Named Inventor	Hyun Lee																
Art Unit	2827																
Examiner Name	Unknown																
Attorney Docket Number	NETL.045C3																
Sheet	1	of	3														

US PATENT DOCUMENTS					
Examiner Initials *	Cite No. ¹	Document Number	Publication Date (MM-DD-YYYY)	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number-Kind Code ² (if known)			
	1.	6381140	04-30-2002	Liao, Sheng-Chi	
	2.	7254036	08-07-2007	Pauley et al.	
	3.	20090248969	10-01-2009	Wu; Larry	
	4.	20100125681	05-20-2010	Patel; Shwetel A.	

FOREIGN PATENT DOCUMENTS						
Examiner Initials *	Cite No. ¹	Foreign Patent Document	Publication Date (MM-DD-YYYY)	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear	T ⁶
	5.	WO99/30240	06-17-1999	Intel Corporation		

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²
	6.	Non-Final Office Action, U.S. Patent Application No. 13/970,606, filed August 20, 2013, dated November 23, 2015	
	7.	Response to Non-Final Office Action, U.S. Patent Application No. 13/970,606, filed August 20, 2013, dated March 23, 2016	
	8.	Inter Partes Review Case No. 2014-00882, Patent Owner Response, filed May 8, 2015	
	9.	Inter Partes Review Case No. 2014-00882, Petitioner Diablo Technologies, Inc.'s Reply to Patent Owner Netlist, Inc.'s Response, filed June 19, 2015	
	10.	Inter Partes Review Case No. 2014-00882, Petitioner's Request for Oral Argument, filed July 1, 2015	
	11.	Inter Partes Review Case No. 2014-00882, Patent Owner's Request for Oral Argument, filed July 3, 2015	
	12.	Inter Partes Review Case No. 2014-00882, Order - Request for Oral Argument - 37 C.F.R. 42.70, filed July 10, 2015	
	13.	Inter Partes Review Case No. 2014-00882, Record of Oral Hearing, filed November 16, 2015	
	14.	Inter Partes Review Case No. 2014-00882, Final Written Decision 35 USC 318 and 37 CFR 42.73, filed December 14, 2015	
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Sheet	2	of	3														

NON PATENT LITERATURE DOCUMENTS			
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	16.	Inter Partes Review Case No. 2014-00882, Exhibit 1022, 'Professor Carl Sechen Deposition Transcript,' filed June,19,2015	
	17.	Inter Partes Review Case No. 2014-00882, Exhibit 1023, 'Dr. Srinivasan Jagannathan Supplemental Declaration,' filed June 19,2015	
	18.	Inter Partes Review Case No. 2014-00882, Exhibit 2002, 'Declaration of Professor Carl Sechen,' filed May 8,2015	
	19.	Inter Partes Review Case No. 2014-00882, Exhibit 2003, 'Transcript of Videotaped Deposition of Srinivasan Jagannathan, Ph. D,' filed May 8,2015	
	20.	Microsoft Computer Dictionary, Fifth Edition, Microsoft Press, p. 334, 2002	
	21.	Jacob, Bruce, et al. "Memory Systems - Cache, DRAM, Disk," Elsevier Inc. 2008, Chapters 7 and 10	
	22.	U.S. District Court Northern District of California, Case No. 4.13-cn-05889-YGR, Netlist Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Joint Claim Construction and Prehearing Statement Pursuant to Patent L. T. 4-3, filed August 27, 2014, in 28 pages	
	23.	Excerpts from the Authoritative Dictionary of IEEE Standard Terms, Seventh Edition, Standards Information Network, IEEE Press (2000), pp. 349-411	
	24.	Excerpts from the Authoritative Dictionary of IEEE Standard Terms, Seventh Edition, Standards Information Network, IEEE Press (2000), pp. 133-265	
	25.	Selected Entries in Oxford English Dictionary, Oxford University Press 2014, http://www.oed.com/view/Entry/175039?redirectedFrom=selectively#eid	
	26.	Web search results from the Oxford English Dictionary, Oxford University Press 2014, http://www.oed.com/search?searchType=dictionary&q=select&_searchBin=Search	
	27.	Inter Partes Review Case No. 2014-00883, Patent Owner Response, filed May 8, 2015	
	28.	Inter Partes Review Case No. 2014-00883, Petitioner Diablo Technologies, Inc.'s Reply to Patent Owner Netlist, Inc.'s Response, filed June 19, 2015	
	29.	Inter Partes Review Case No. 2014-00883, Petitioner's Request for Oral Argument, filed July 1, 2015	
	30.	Inter Partes Review Case No. 2014-00883, Patent Owner's Request for Oral Hearing, filed July 3, 2015	
	31.	Inter Partes Review Case No. 2014-00883, Order - Request for Oral Argument - 37 C.F.R. 42.70, filed July 10, 2015	
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				Application Number	14/715,491
				Filing Date	May 18, 2015
				First Named Inventor	Hyun Lee
				Art Unit	2827
				Examiner Name	Unknown
Sheet	3	of	3	Attorney Docket Number	NETL.045C3

NON PATENT LITERATURE DOCUMENTS			
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	35.	Inter Partes Review Case No. 2014-00883, Exhibit 1025, 'Professor Carl Sechen Deposition Transcript,' filed June 19,2015	
	36.	Inter Partes Review Case No. 2014-00883, Exhibit 1026, 'Dr. Srinivasan Jagannathan Supplemental Declaration,' filed June 19,2015	
	37.	Inter Partes Review Case No. 2014-01011, Patent Owner Response, filed May 8, 2015	
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	45.	Inter Partes Review Case No. 2014-01011, Exhibit 1027, 'Professor Carl Sechen Deposition Transcript,' filed June 19,2015	
	46.	Inter Partes Review Case No. 2014-01011, Exhibit 1028, 'Dr. Srinivasan Jagannathan Supplemental Declaration,' filed June 19,2015	

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Electronic Acknowledgement Receipt

EFS ID:	26290367
Application Number:	14715491
International Application Number:	
Confirmation Number:	8143
Title of Invention:	MEMORY MODULE AND CIRCUIT PROVIDING LOAD ISOLTION AND NOISE REDUCTION
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	NETL.045C3
Receipt Date:	08-JUL-2016
Filing Date:	18-MAY-2015
Time Stamp:	00:31:22
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Transmittal Letter	2016-07-07_IDS_2_Cover_NETL 045C3.pdf	49459	no	2
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Warnings:

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2	Information Disclosure Statement (IDS) Form (SB08)	2016-07-07_IDS_2_NETL045C3.pdf	94879 00b4425f4bf401b03ca1a8c55b64b60e42d4b02b	no	3
Warnings:					
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3	Non Patent Literature	2015-11-23_NFOA_NETL048CP C1.pdf	4155462 35b8c4f3b724d8b3e3f65fa2a1c1257016da bb25	no	79
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Information:					
4	Non Patent Literature	2016-03-23_Rsps_NFOA_NETL0 48CPC1.pdf	118678 0f3c0dba6a043802851f4422a0b58d52f88e 9d5a	no	9
Warnings:					
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Total Files Size (in bytes):			4418478		
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App. No.	: 14/715,491
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Examiner	: UNKNOWN
Art Unit	: 2827
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Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Dear Sir:

Enclosed for filing in the above-identified application is a Form PTO/SB/08a/b Equivalent listing 46 references. Copies of some of the foreign documents and/or their translation and non-patent literature are submitted herewith. Copies of the rest of the foreign documents and/or their translation and non-patent literature have been submitted in commonly-owned U.S. Pat. Appl. No. 14/715,486 (filed May 18, 2015) on February 21, 2016, which is accessible via PAIR.

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Application No.: 14/715,491
Filing Date: May 18, 2015

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Respectfully submitted,

Dated: July 7, 2016

/Jamie J. Zheng/

Jamie J. Zheng
Customer No. 79141
(650) 521-4828

51167
(Reg. No.)

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				Application Number	14/715,491
				Filing Date	May 18, 2015
				First Named Inventor	Hyun Lee
				Art Unit	2827
				Examiner Name	Unknown
Sheet	1	of	3	Attorney Docket Number	NETL.045C3

US PATENT DOCUMENTS					
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		Number-Kind Code ² (if known)			
	1.	4,739,473	1988-04-19	Ng	
	2.	5,313,624	1994-05-17	Harriman	
	3.	5,630,096	1997-05-13	Zuravleff	
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	5.	6,061,754	2000-05-09	Cepulis	
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	7.	6,223,650	2001-05-15	Johnson	
	8.	6,262,938	2001-07-17	Lee	
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	13.	7,191,302	2007-03-13	Usami	
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	16.	7,990,746	2011-08-02	Rajan	
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	18.	8,335,894	2012-12-18	Rajan	
	19.	8,856,464	2014-10-07	Karamcheti	
	20.	20010008006	2001-07-12	Klein	
	21.	20030070052	2003-04-10	Lai	
	22.	20080025137	2008-01-31	Rajan et al.	
	23.	20090248969	10-01-2009	Wu; Larry	
	24.	20100125681	05-20-2010	Patel; Shwetal A.	

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Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²
	25.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Record of Oral Hearing, dated July 26, 2016	

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				Art Unit	2827
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Sheet	2	of	3	Attorney Docket Number	NETL.045C3

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	26.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Final Written Decision, issued September 28, 2016	
	27.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Exhibit 3001 to Final Decision, issued September 28, 2016	
	28.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Exhibit 3001-2 to Final Decision, issued September 28, 2016	
	29.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Exhibit 3002 to Final Decision, issued September 28, 2016	
	30.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Exhibit 3003 to Final Decision, issued September 28, 2016	
	31.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Exhibit 3004 to Final Decision, issued September 28, 2016	
	32.	Inter Partes Review of U.S. Patent No. 8081536, Case No. IPR2015-01021, Final Decision, issued September 28, 2016	
	33.	Inter Partes Review of U.S. Patent No. 8756364, Case No. IPR2017-00549, Petition for Inter Partes Review, filed December 30, 2016	
	34.	Inter Partes Review of U.S. Patent No. 8756364, Case No. IPR2017-00549, Exhibit 1003, "Declaration of Harold S. Stone," filed December 30, 2016	
	35.	Inter Partes Review of U.S. Patent No. 8516185, Case No. IPR2017-00577, Petition for Inter Partes Review, filed January 5, 2017	
	36.	Inter Partes Review of U.S. Patent No. 8516185, Case No. IPR2017-00577, Exhibit 1003, "Declaration of Harold S. Stone," filed January 5, 2017	
	37.	Inter Partes Review of U.S. Patent No. 8756364, Case No. IPR2017-00549, Exhibit 1008, Declaration of John J. Kelly Regarding Records of Joint Electron Device Engineering Council (JEDEC), filed December 30, 2016	
	38.	JEDEC Standard Double Data Rate (DDR) SDRAM Specification, JESD79 (June 2000)	
	39.	Texas Instruments SN74LS245 octal bus transceivers with 3-state outputs datasheet, 2002	
	40.	Stone, H.S. Microcomputer Interfacing, Reading, MA: Addison Wesley, 1982	
	41.	128Mbit SDRAM, Samsung datasheet K4S281632C, Rev. 00. March 2000	

EXAMINER

DATE CONSIDERED

* EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.¹ Applicant's unique citation designation number (optional).² See Kinds Codes of USPTO Patents Documents at www.uspto.gov or MPEP 901.04. ³ Enter Office that issued the document, by the two-letter code (WIPO Standard ST. 3). ⁴ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁵ Kind of document by the appropriate symbols as indicated on the document and WIPO Standard ST. 16 if possible. ⁶ Applicant is to place a check mark here if English language Translation is attached.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	14/715,491
				Filing Date	May 18, 2015
				First Named Inventor	Hyun Lee
				Art Unit	2827
				Examiner Name	Unknown
Sheet	3	of	3	Attorney Docket Number	NETL.045C3

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²
	42.	G. Moore, "Cramming more components onto integrated circuits," Electronics, volume 38, no. 8, April 19, 1965	

EXAMINER**DATE CONSIDERED**

* EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.¹ Applicant's unique citation designation number (optional).² See Kinds Codes of USPTO Patents Documents at www.uspto.gov or MPEP 901.04. ³ Enter Office that issued the document, by the two-letter code (WIPO Standard ST. 3). ⁴ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁵ Kind of document by the appropriate symbols as indicated on the document and WIPO Standard ST. 16 if possible. ⁶ Applicant is to place a check mark here if English language Translation is attached.

Electronic Acknowledgement Receipt

EFS ID:	28144438
Application Number:	14715491
International Application Number:	
Confirmation Number:	8143
Title of Invention:	MEMORY MODULE AND CIRCUIT PROVIDING LOAD ISOLTION AND NOISE REDUCTION
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	NETL.045C3
Receipt Date:	24-JAN-2017
Filing Date:	18-MAY-2015
Time Stamp:	02:16:13
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
------------------------	----

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Transmittal Letter	2017-01-23_IDS_Cover_NETL045C3.pdf	48823 a5e139ea9edf29e4ab559cc21efd7138b2dcdf99	no	2

Warnings:

Information:					
2	Information Disclosure Statement (IDS) Form (SB08)	2017-01-23_IDS_NETL045C3. pdf	99161	no	3
			caeb110b40b7a4bfbcc4db9207d177f5cd7264387		
Warnings:					
Information:					
This is not an USPTO supplied IDS fillable form					
Total Files Size (in bytes):			147984		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

INFORMATION DISCLOSURE STATEMENT

Applicant	: NETLIST, INC.
App. No.	: 14/715,491
Filed	: May 18, 2015
For	: CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION
Examiner	: UNKNOWN
Art Unit	: 2827
Conf No.	: 8143

Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Dear Sir:

Enclosed for filing in the above-identified application is a Form PTO/SB/08a/b Equivalent listing 42 references. Copies of the non-patent literature have been submitted on even date herewith in commonly-owned U.S. Pat. Appl. No. 14/715,486 (filed May 18, 2015), which is accessible via PAIR.

The applicants would appreciate the Examiner initialing and returning the Form PTO/SB/08a/b Equivalent indicating that the information has been considered and made of record herein.

The information contained in this Information Disclosure Statement is not to be construed as a representation that: (i) a complete search has been made; (ii) additional information material to this application does not exist; (iii) the information, protocols, results and the like reported by third parties are accurate or enabling; or (iv) the above information constitutes prior art to the subject invention.

This Information Disclosure Statement is being filed before the receipt of a first Office Action on the merits, and presumably no fee is required. If a first Office Action on the merits

Application No.: 14/715,491
Filing Date: May 18, 2015

was mailed before the mailing date of this Statement, the Commissioner is authorized to charge the fee set forth in 37 C.F.R. § 1.17(p) to Deposit Account No. 50-5963.

Respectfully submitted,

Dated: January 23, 2017

/Jamie J. Zheng/

Jamie J. Zheng
Customer No. 79141
(650) 521-4828

51167

(Reg. No.)



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

NOTICE OF ALLOWANCE AND FEE(S) DUE

79141 7590 06/15/2017
Jamie J. Zheng, Ph.D Esq.
MASCHOFF BRENNAN
1389 Center Drive
Suite 300
Park City, UT 84098

EXAMINER

BANSAL, GURTEJ

ART UNIT

PAPER NUMBER

2139

DATE MAILED: 06/15/2017

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
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14/715,491

05/18/2015

Hyun Lee

N3044.10007US08

8143

TITLE OF INVENTION: MEMORY MODULE AND CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	SMALL	\$480	\$0	\$0	\$480	09/15/2017

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.

If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 1/2 the amount of undiscounted fees, and micro entity fees are 1/2 the amount of small entity fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed and an extra copy of the form should be submitted. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Utility patents issuing on applications filed on or after Dec. 12, 1980 may require payment of maintenance fees. It is patentee's responsibility to ensure timely payment of maintenance fees when due.

PART B - FEE(S) TRANSMITTAL

**Complete and send this form, together with applicable fee(s), to: Mail Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450
or Fax (571)-273-2885**

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

79141 7590 06/15/2017
Jamie J. Zheng, Ph.D Esq.
MASCHOFF BRENNAN
 1389 Center Drive
 Suite 300
 Park City, UT 84098

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being facsimile transmitted to the USPTO (571) 273-2885, on the date indicated below.

(Depositor's name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/715,491	05/18/2015	Hyun Lee	N3044.10007US08	8143

TITLE OF INVENTION: MEMORY MODULE AND CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	SMALL	\$480	\$0	\$0	\$480	09/15/2017

EXAMINER	ART UNIT	CLASS-SUBCLASS
BANSAL, GURTEJ	2139	711-149000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363). ☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached. ☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-02 or more recent) attached. Use of a Customer Number is required.	2. For printing on the patent front page, list (1) The names of up to 3 registered patent attorneys or agents OR, alternatively, 1 _____ (2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed. 2 _____ 3 _____
---	---

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE (B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. The following fee(s) are submitted:

- ☐ Issue Fee
☐ Publication Fee (No small entity discount permitted)
☐ Advance Order - # of Copies _____

4b. Payment of Fee(s): (Please first reapply any previously paid issue fee shown above)

- ☐ A check is enclosed.
☐ Payment by credit card. Form PTO-2038 is attached.
☐ The director is hereby authorized to charge the required fee(s), any deficiency, or credits any overpayment, to Deposit Account Number _____ (enclose an extra copy of this form).

5. Change in Entity Status (from status indicated above)

- ☐ Applicant certifying micro entity status. See 37 CFR 1.29
☐ Applicant asserting small entity status. See 37 CFR 1.27
☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____ Date _____
 Typed or printed name _____ Registration No. _____



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/715,491	05/18/2015	Hyun Lee	N3044.10007US08	8143

79141 7590 06/15/2017
Jamie J. Zheng, Ph.D Esq.
MASCHOFF BRENNAN
1389 Center Drive
Suite 300
Park City, UT 84098

EXAMINER

BANSAL, GURTEJ

ART UNIT PAPER NUMBER

2139

DATE MAILED: 06/15/2017

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450. Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

<i>Examiner-Initiated Interview Summary</i>	Application No. 14/715,491	Applicant(s) LEE ET AL.	
	Examiner GURTEJ BANSAL	Art Unit 2139	

All participants (applicant, applicant's representative, PTO personnel):

(1) GURTEJ BANSAL. (3) ____.

(2) JAMIE ZHENG. (4) ____.

Date of Interview: 05 June 2017.

Type: ☒ Telephonic ☐ Video Conference
 ☐ Personal [copy given to: ☐ applicant ☐ applicant's representative]

Exhibit shown or demonstration conducted: ☐ Yes ☒ No.
 If Yes, brief description: ____.

Issues Discussed ☐101 ☐112 ☐102 ☐103 ☒Others
 (For each of the checked box(es) above, please describe below the issue and detailed description of the discussion)

Claim(s) discussed: 1.

Identification of prior art discussed: N/A.

Substance of Interview
 (For each issue discussed, provide a detailed description and indicate if agreement was reached. Some topics may include: identification or clarification of a reference or a portion thereof, claim interpretation, proposed amendments, arguments of any applied references etc...)

Examiner discussed the claimed invention in view of Applicant submitted prior art. Examiner suggested amending the claims to distinguish the claims from the cited prior art. Applicant agreed and authorized the Examiner to make the amendments via an Examiner's amendment.

Applicant recordation instructions: It is not necessary for applicant to provide a separate record of the substance of interview.

Examiner recordation instructions: Examiners must summarize the substance of any interview of record. A complete and proper recordation of the substance of an interview should include the items listed in MPEP 713.04 for complete and proper recordation including the identification of the general thrust of each argument or issue discussed, a general indication of any other pertinent matters discussed regarding patentability and the general results or outcome of the interview, to include an indication as to whether or not agreement was reached on the issues raised.

☐ Attachment

/GURTEJ BANSAL/ Primary Examiner, Art Unit 2139	
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Notice of Allowability	Application No. 14/715,491	Applicant(s) LEE ET AL.	
	Examiner GURTEJ BANSAL	Art Unit 2139	AIA (First Inventor to File) Status No

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to Examiner initiated interview dated June 5, 2017.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.
2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.
3. ☒ The allowed claim(s) is/are 1,4-6,9-15 and 23-29. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.
4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some *c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. _____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.

THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.
Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).
6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

- | | |
|--|--|
| 1. ☐ Notice of References Cited (PTO-892) | 5. ☒ Examiner's Amendment/Comment |
| 2. ☒ Information Disclosure Statements (PTO/SB/08),
Paper No./Mail Date <u>See Continuation Sheet</u> | 6. ☒ Examiner's Statement of Reasons for Allowance |
| 3. ☐ Examiner's Comment Regarding Requirement for Deposit of Biological Material | 7. ☐ Other _____. |
| 4. ☒ Interview Summary (PTO-413),
Paper No./Mail Date <u>20170606</u> . | |

/GURTEJ BANSAL/
Primary Examiner, Art Unit 2139

Continuation of Attachment(s) 2. Information Disclosure Statements (PTO/SB/08), Paper No./Mail Date: 07/07/2016; 07/08/2016; 01/24/2017.

EXAMINER'S AMENDMENT

Claims 1, 4-6, 9-15 and 23-29 are allowed.

An examiner's amendment to the record appears below. Should the changes and/or additions be unacceptable to applicant, an amendment may be filed as provided by 37 CFR 1.312. To ensure consideration of such an amendment, it MUST be submitted no later than the payment of the issue fee.

Authorization for this examiner's amendment was given in an interview with Jamie Zheng on June 5, 2017.

The application has been amended as follows:

IN THE CLAIMS

1. (Currently Amended) A memory module operable to communicate data with a system memory controller via a memory bus in response to address and control signals from the system memory controller, comprising:

a printed circuit board comprising at least one connector configured to be operatively coupled to the memory bus, the at least one connector including a plurality of edge connections distributed along one or more edges of the printed circuit board;

a plurality of memory devices on the printed circuit board, the plurality of memory devices being arranged in multiple ranks, ~~including a first rank and a second rank~~, each rank comprising an independent set of memory devices that can be accessed by the system memory controller to access a full bit-width of the memory bus;

at least one circuit coupled between the at least one connector and the plurality of memory devices, the at least one circuit comprising a first set of port connections ~~ports~~ coupled to respective ones of the plurality of edge connections and a second set of port connections coupled to the plurality of memory devices, each circuit of the at least one circuit further comprising a set of correction circuits, each correction circuit of the set of correction circuits being configured to make corrections in one or more signals transmitted between a respective port connection of the first set of port connections and a corresponding port connection in the second set of port connections, the each correction circuit of the set of correction circuits including at least one programmable impedance matching circuit, ~~the first set of ports including a set of bidirectional ports for conducting data signals, each port of the set of bidirectional ports in the first set of ports including a signal waveform reshaping circuit, the circuit further comprising a second set of ports coupled to the plurality of memory devices, the second set of ports including a first subset of ports coupled to the first rank and a second subset of ports coupled to the second rank, the circuit further comprising a switching sub-circuit configured to operatively couple either the first subset of ports or the second subset of ports to respective ones of the set of bidirectional ports; and~~

control circuitry configured to receive the address and control signals from the system memory controller and to control ~~the switching sub-circuit and the the signal waveform reshaping circuits in the the set of bidirectional ports~~ the plurality of memory devices in response to the address and control signals, ~~the signal waveform reshaping circuits being configured to reshape data signals between the memory module and the plurality of memory devices,~~ the address and control signals wherein the control circuitry is further configured to dynamically control the at least one programmable impedance matching circuit in the each correction circuit of the set of correction circuits in the each circuit of the at least one circuit based on which of the multiple ranks is selected to communicate data with the system memory controller in response to the address and control signals ~~causing the memory module to select either the first rank or the second rank for memory access by the system memory controller.~~

2. (Cancelled).

3. (Cancelled).

4. (Currently Amended) The memory module of claim 1-2, wherein the at least one impedance matching circuit in the each correction circuit of the set of correction circuits in the each circuit of the at least one circuit has dynamically programmable impedances whose values may differ depending on which of the multiple ranks is selected to communicate data with the system memory controller in response to the address and control signals ~~each port of the set of bidirectional ports in the first set of ports further includes a programmable impedance matching circuit having at least one~~

~~variable impedance that can be programmed to match a characteristic impedance of a corresponding signal path in the system.~~

5. (Currently Amended) The memory module of claim 1, wherein the at least one impedance matching circuit in the each correction circuit of the set of correction circuits in the each circuit of the at least one circuit ~~each of the programmable impedance matching circuits~~ comprises at least one network of programmable resistors ~~resistors~~ that is dynamically controllable.

6. (Currently Amended) The memory module of claim 1, wherein the each correction circuit of the set of correction circuits in the each circuit of the at least one circuit further comprises one or more self-adjusting damper circuits ~~corresponding to respective ones of the plurality of edge connections.~~

7. (Cancelled)

8. (Cancelled)

9. (Currently Amended) The memory module of claim 1-8, wherein the each correction circuit of the set of correction circuits in the each circuit of the at least one circuit further comprises first one or more dynamically programmable repeater/amplifier elements ~~corresponding to respective ones of the plurality of edge connections.~~

10. (Currently Amended) The memory module of claim 1, wherein the each correction circuit of the set of correction circuits in the each circuit of the at least one circuit further includes a delay circuit that is configured to generate delayed versions of the signals transmitted between the respective port connection of the first set of port connections and the corresponding port connection in the second set of port

~~connections⁹, wherein the circuit comprises second dynamically programmable
repeater/amplifier elements corresponding to respective ones of the second set of ports.~~

11. (Original) The memory module of claim ~~1-10~~, wherein the at least one circuit includes signal paths, wherein the control circuitry is configured to activate a subset of the signal paths in response to the address and control signals~~a number of ports in the second set of ports is larger than a number of edge connections in the edge connector.~~

12. (Currently Amended) The memory module of claim ~~11-4~~, wherein the at least one circuit is further configured to provide a known static value to each port connection in the second set of ~~port connections~~ ports that is not in an active signal path~~coupled to one of the set of bidirectional ports by the switching sub-circuit.~~

13. (Currently Amended) The memory module of claim 1, wherein the number of port connections ~~bi-directional ports~~ in the first set of port connections ~~ports~~ corresponds to a bit width of the memory bus.

14. (Currently Amended) The memory module of claim 1, wherein the at least one circuit comprises multiple physically separate integrated circuit packages each having a data width corresponding to a data width of one of the plurality of memory devices.

15. (Currently Amended) The memory module of claim 1, wherein the at least one circuit comprises multiple physically separate integrated circuit packages each having a data width corresponding to a data width of multiple memory devices of the plurality of memory devices

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16. (Cancelled)

17. (Cancelled)

18. (Cancelled)

19. (Cancelled)

20. (Cancelled)

21. (Cancelled)

22. (Cancelled)

23. (New) The memory module of claim 1, wherein the control circuitry is configured to activate signal paths in the each circuit of the at least one circuit in response to the address and control signals.

24. (New) The memory module of claim 1, wherein the control circuitry is configured to send at least one control signal to the at least one circuit in response to the address and control signals.

25. (New) The memory module of claim 1, wherein the each correction circuit of the set of correction circuits in the each circuit of the at least one circuit includes one or more circuit elements, and wherein the control circuitry is configured to activate the one or more circuit elements in response to the address and control signals.

26. (New) The memory module of claim 1, wherein each circuit in the at least one circuit has a data width of 4, 8 or 16 bits.

27. (New) The memory module of claim 1, wherein the at least one circuit is configured to improve signal integrity in signals transmitted between the at least one connector and the plurality of memory devices.

28. (New) The memory module of claim 1, wherein the at least one circuit is configured to improve timing of signals transmitted between the at least one connector and the plurality of memory devices.

29. (New) The memory module of claim 1, the at least one circuit is configured to improve power consumption by signals transmitted between the at least one connector and the plurality of memory devices.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to GURTEJ BANSAL whose telephone number is (571)270-5588. The examiner can normally be reached on Monday - Friday, 7:30 a.m. - 5:00 p.m., EST.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Reginald Bragdon can be reached on (571)272-4204. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Art Unit: 2139

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

GURTEJ BANSAL
Primary Examiner
Art Unit 2139

/GURTEJ BANSAL/
Primary Examiner, Art Unit 2139

<i>Examiner-Initiated Interview Summary</i>	Application No. 14/715,491	Applicant(s) LEE ET AL.	
	Examiner GURTEJ BANSAL	Art Unit 2139	

All participants (applicant, applicant's representative, PTO personnel):

(1) GURTEJ BANSAL. (3) ____.

(2) JAMIE ZHENG. (4) ____.

Date of Interview: 05 June 2017.

Type: ☒ Telephonic ☐ Video Conference
 ☐ Personal [copy given to: ☐ applicant ☐ applicant's representative]

Exhibit shown or demonstration conducted: ☐ Yes ☒ No.
 If Yes, brief description: ____.

Issues Discussed ☐101 ☐112 ☐102 ☐103 ☒Others
 (For each of the checked box(es) above, please describe below the issue and detailed description of the discussion)

Claim(s) discussed: 1.

Identification of prior art discussed: N/A.

Substance of Interview
 (For each issue discussed, provide a detailed description and indicate if agreement was reached. Some topics may include: identification or clarification of a reference or a portion thereof, claim interpretation, proposed amendments, arguments of any applied references etc...)

Examiner discussed the claimed invention in view of Applicant submitted prior art. Examiner suggested amending the claims to distinguish the claims from the cited prior art. Applicant agreed and authorized the Examiner to make the amendments via an Examiner's amendment.

Applicant recordation instructions: It is not necessary for applicant to provide a separate record of the substance of interview.

Examiner recordation instructions: Examiners must summarize the substance of any interview of record. A complete and proper recordation of the substance of an interview should include the items listed in MPEP 713.04 for complete and proper recordation including the identification of the general thrust of each argument or issue discussed, a general indication of any other pertinent matters discussed regarding patentability and the general results or outcome of the interview, to include an indication as to whether or not agreement was reached on the issues raised.

☐ Attachment

/GURTEJ BANSAL/ Primary Examiner, Art Unit 2139	
--	--

On-chip error detection and **correction** system for an embedded non-volatile **memory** array and method of operation

P McGinn - US Patent 6,216,251, 2001 - Google Patents

... **correction** and detection methodology used on secure, embedded non-volatile smart card **memories**. ... number of programming pulses used to program rows of non-volatile **memory** cells. ... 2010, Oct 23, 2012, Lsi Corporation, Systems and methods for low latency **noise cancellation**. ... Cited by 174 Related articles All 2 versions Cite Save

An overview of peak-to-average power ratio **reduction** techniques for multicarrier transmission

SH Han, JH Lee - IEEE wireless communications, 2005 - ieeexplore.ieee.org

... In [45] iterative estimation and **cancellation** of clipping **noise** is proposed. This technique exploits the fact that clipping **noise** is generated by a known process that can be recreated at the receiver ... Both the transmitter and receiver store the permutation indices $\{\pi(n)\}$ in **memory**. ... Cited by 2159 Related articles All 7 versions Cite Save

Fixed-pattern **noise correction** circuitry for solid-state imager

GJ Michon - US Patent 5,317,407, 1994 - Google Patents

... The digital **memories** tend to require considerably larger semiconductor die areas than analog **memories**, and errors ... Linear capacitors and multiplexing switches are used to implement this analog **memory** in the invention. ... Column amplifier for high fixed pattern **noise reduction**. ... Cited by 80 Related articles All 2 versions Cite Save

CMOS imaging device with integrated flash **memory** image **correction** circuitry

R Hirt, M Rollender - US Patent 5,863,830, 1999 - Google Patents

... US6707493, Jun 29, 1999, Mar 16, 2004, Hyundai Electronics Industries Co., Ltd. Method and apparatus for error detection and **correction** in image sensor. ... CMOS image sensor with **noise cancellation**. ... CMOS imager with integrated non-volatile **memory**. ... Cited by 124 Related articles All 2 versions Cite Save

High performance, fault tolerant orthogonal shuffle **memory** and method

TP Haraszti - US Patent 5,612,964, 1997 - Google Patents

... 4, *, Novel Fault Tolerant Circuits For Configurable Mass **Memories**, TP Haraszti et al, Proceedings of ... 14, 2011, Sep 10, 2013, Lsi Corporation, Systems and methods for **memory** efficient data ... 19, 2011, Sep 17, 2013, Lsi Corporation, Systems and methods for **noise** injection driven ... Cited by 171 Related articles All 2 versions Cite Save

Pattern recognition training method and apparatus using inserted **noise** followed by **noise reduction**

L Deng, X Huang, MD Plumpe - US Patent 6,876,966, 2005 - Google Patents

... in step 802 can be the same for each type of **noisy** environment or may be tailored for the specific **noisy** environment upon ... **noise** in the test speech signal, **noise** comparator 1302 retrieves the set identifier for the matching signal from **memory** 1304 ... Event driven **noise cancellation**. ... Cited by 45 Related articles All 2 versions Cite Save

Architecture for soft decision decoding of linear block error **correcting** codes

LJ Weng - US Patent 6,381,726, 2002 - Google Patents

... 14, 2011, Sep 10, 2013, Lsi Corporation, Systems and methods for **memory** efficient data ... 2012, Jun 24, 2014, Lsi Corporation, Systems and methods for post processing gain **correction**. ... 2013, Jul 8, 2014, Lsi Corporation, Systems and methods for X-sample based **noise cancellation**. ... Cited by 162 Related articles All 2 versions Cite Save

Characterisation and **correction** of instrumental bias in inductively coupled plasma quadrupole mass spectrometry for accurate measurement of lead isotope ratios

IS Begley, BL Sharp - Journal of Analytical Atomic Spectrometry, 1997 - pubs.rsc.org

... analyte contamination relates to sample-to-sample **memory** ... that ratioing of simultaneously measured emission line intensities, obtained by ICP-AES, is effective in **cancellation** of flicker ... assure that **noise** is minimised in both Tl and Pb isotopes if the thallium **correction** method is ...

Cited by 134 Related articles All 5 versions Cite Save

Offset, gain and bad pixel **correction** in electronic scanning arrays

Y Lin, AF Calarco - US Patent 4,920,428, 1990 - Google Patents

... no attribute value would be necessary and the size of the pixel **correction memory** can be ... Oct 14, 2003, Apr 29, 2004, Kimble Dong, On-chip dead pixel **correction** in a ... US20070269133 *, May 4, 2007, Nov 22, 2007, Fuji Film Corporation, Image-data **noise reduction** apparatus and ...

Cited by 81 Related articles All 2 versions Cite Save

Reducing the effects of **noise** in non-volatile **memories** through multiple reads

CJ Gonzalez, DC Guterman - US Patent 7,177,195, 2007 - Google Patents

... **Memory** cell readout using successive approximation. US7848149 *, Feb 12, 2007, Dec 7, 2010, Sandisk Corporation, Reducing the effects of **noise** in non-volatile **memories** through multiple reads. ... **Reduction** of back pattern dependency effects in **memory** devices. ...

Cited by 76 Related articles All 5 versions Cite Save

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				Application Number	14/715,491
				Filing Date	May 18, 2015
				First Named Inventor	Hyun Lee
				Art Unit	2827
				Examiner Name	Unknown
Sheet	1	of	3	Attorney Docket Number	NETL.045C3

US PATENT DOCUMENTS					
Examiner Initials *	Cite No. ¹	Document Number	Publication Date (MM-DD-YYYY)	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number-Kind Code ² (if known)			
	1.	6381140	04-30-2002	Liao, Sheng-Chi	
	2.	7254036	08-07-2007	Pauley et al.	
	3.	20090248969	10-01-2009	Wu; Larry	
	4.	20100125681	05-20-2010	Patel; Shwetel A.	

FOREIGN PATENT DOCUMENTS						
Examiner Initials *	Cite No. ¹	Foreign Patent Document	Publication Date (MM-DD-YYYY)	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear	T ⁶
	5.	WO99/30240	06-17-1999	Intel Corporation		

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²
	6.	Non-Final Office Action, U.S. Patent Application No. 13/970,606, filed August 20, 2013, dated November 23, 2015	
	7.	Response to Non-Final Office Action, U.S. Patent Application No. 13/970,606, filed August 20, 2013, dated March 23, 2016	
	8.	Inter Partes Review Case No. 2014-00882, Patent Owner Response, filed May 8, 2015	
	9.	Inter Partes Review Case No. 2014-00882, Petitioner Diablo Technologies, Inc.'s Reply to Patent Owner Netlist, Inc.'s Response, filed June 19, 2015	
	10.	Inter Partes Review Case No. 2014-00882, Petitioner's Request for Oral Argument, filed July 1, 2015	
	11.	Inter Partes Review Case No. 2014-00882, Patent Owner's Request for Oral Argument, filed July 3, 2015	
	12.	Inter Partes Review Case No. 2014-00882, Order - Request for Oral Argument - 37 C.F.R. 42.70, filed July 10, 2015	
	13.	Inter Partes Review Case No. 2014-00882, Record of Oral Hearing, filed November 16, 2015	
	14.	Inter Partes Review Case No. 2014-00882, Final Written Decision 35 USC 318 and 37 CFR 42.73, filed December 14, 2015	
	15.	Inter Partes Review Case No. 2014-00882, Netlist Inc.'s Notice of Appeal, filed February 10, 2016	

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				Examiner Name	Unknown
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	17.	Inter Partes Review Case No. 2014-00882, Exhibit 1023, 'Dr. Srinivasan Jagannathan Supplemental Declaration,' filed June 19,2015	
	18.	Inter Partes Review Case No. 2014-00882, Exhibit 2002, 'Declaration of Professor Carl Sechen,' filed May 8,2015	
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	20.	Microsoft Computer Dictionary, Fifth Edition, Microsoft Press, p. 334, 2002	
	21.	Jacob, Bruce, et al. "Memory Systems - Cache, DRAM, Disk," Elsevier Inc. 2008, Chapters 7 and 10	
	22.	U.S. District Court Northern District of California, Case No. 4:13-cn-05889-YGR, Netlist Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Joint Claim Construction and Prehearing Statement Pursuant to Patent L. T. 4-3, filed August 27, 2014, in 28 pages	
	23.	Excerpts from the Authoritative Dictionary of IEEE Standard Terms, Seventh Edition, Standards Information Network, IEEE Press (2000), pp. 349-411	
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	25.	Selected Entries in Oxford English Dictionary, Oxford University Press 2014, http://www.oed.com/view/Entry/175039?redirectedFrom=selectively#eid	
	26.	Web search results from the Oxford English Dictionary, Oxford University Press 2014, http://www.oed.com/search?searchType=dictionary&q=select&_searchBin=Search	
	27.	Inter Partes Review Case No. 2014-00883, Patent Owner Response, filed May 8, 2015	
	28.	Inter Partes Review Case No. 2014-00883, Petitioner Diablo Technologies, Inc.'s Reply to Patent Owner Netlist, Inc.'s Response, filed June 19, 2015	
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	30.	Inter Partes Review Case No. 2014-00883, Patent Owner's Request for Oral Hearing, filed July 3, 2015	
	31.	Inter Partes Review Case No. 2014-00883, Order - Request for Oral Argument - 37 C.F.R. 42.70, filed July 10, 2015	
	32.	Inter Partes Review Case No. 2014-00883, Record of Oral Hearing, filed November 16, 2015	
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				First Named Inventor	Hyun Lee
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				Examiner Name	Unknown
Sheet	3	of	3	Attorney Docket Number	NETL.045C3

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	35.	Inter Partes Review Case No. 2014-00883, Exhibit 1025, 'Professor Carl Sechen Deposition Transcript,' filed June 19, 2015	
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	37.	Inter Partes Review Case No. 2014-01011, Patent Owner Response, filed May 8, 2015	
	38.	Inter Partes Review Case No. 2014-01011, Petitioner Diablo Technologies, Inc.'s Reply to Patent Owner Netlist, Inc.'s Response, filed June 19, 2015	
	39.	Inter Partes Review Case No. 2014-01011, Petitioner's Request for Oral Argument, filed July 1, 2015	
	40.	Inter Partes Review Case No. 2014-01011, Patent Owner's Request for Oral Hearing, filed July 3, 2015	
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	43.	Inter Partes Review Case No. 2014-01011, Final Written Decision - 35 USC 318 and 37 CFR 42.73, filed December 14, 2015	
	44.	Inter Partes Review Case No. 2014-01011, Netlist Inc.'s Notice of Appeal, filed February 10, 2016	
	45.	Inter Partes Review Case No. 2014-01011, Exhibit 1027, 'Professor Carl Sechen Deposition Transcript,' filed June 19, 2015	
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		Number-Kind Code ² (if known)			
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	13.	7,191,302	2007-03-13	Usami	
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NON PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²
	25.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Record of Oral Hearing, dated July 26, 2016	

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DATE CONSIDERED 06/09/2017

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	14/715,491
				Filing Date	May 18, 2015
				First Named Inventor	Hyun Lee
				Art Unit	2827
				Examiner Name	Unknown
Sheet	2	of	3	Attorney Docket Number	NETL.045C3

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	26.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Final Written Decision, issued September 28, 2016	
	27.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Exhibit 3001 to Final Decision, issued September 28, 2016	
	28.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Exhibit 3001-2 to Final Decision, issued September 28, 2016	
	29.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Exhibit 3002 to Final Decision, issued September 28, 2016	
	30.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Exhibit 3003 to Final Decision, issued September 28, 2016	
	31.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Exhibit 3004 to Final Decision, issued September 28, 2016	
	32.	Inter Partes Review of U.S. Patent No. 8081536, Case No. IPR2015-01021, Final Decision, issued September 28, 2016	
	33.	Inter Partes Review of U.S. Patent No. 8756364, Case No. IPR2017-00549, Petition for Inter Partes Review, filed December 30, 2016	
	34.	Inter Partes Review of U.S. Patent No. 8756364, Case No. IPR2017-00549, Exhibit 1003, "Declaration of Harold S. Stone," filed December 30, 2016	
	35.	Inter Partes Review of U.S. Patent No. 8516185, Case No. IPR2017-00577, Petition for Inter Partes Review, filed January 5, 2017	
	36.	Inter Partes Review of U.S. Patent No. 8516185, Case No. IPR2017-00577, Exhibit 1003, "Declaration of Harold S. Stone," filed January 5, 2017	
	37.	Inter Partes Review of U.S. Patent No. 8756364, Case No. IPR2017-00549, Exhibit 1008, Declaration of John J. Kelly Regarding Records of Joint Electron Device Engineering Council (JEDEC), filed December 30, 2016	
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				Application Number	14/715,491
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				Art Unit	2827
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	42.	G. Moore, "Cramming more components onto integrated circuits," Electronics, volume 38, no. 8, April 19, 1965	

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EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L1	0	select\$3 with rank and impedance with match\$3	US-PGPUB; USPAT	OR	OFF	2017/06/09 10:54
L2	0	select\$3 with rank and impedance with match\$3	US-PGPUB; USPAT	OR	OFF	2017/06/09 10:54
L3	597	select\$3 with rank and signal with correction	US-PGPUB; USPAT	OR	OFF	2017/06/09 11:00
L4	45	select\$3 with rank same signal with correction	US-PGPUB; USPAT	OR	OFF	2017/06/09 11:00
L5	8280	G06F12/1684.cpc. G11C5/02.cpc. G11C5/06.cpc. G11C8/18.cpc. G06F13/4027.cpc.	US-PGPUB; USPAT	OR	OFF	2017/06/09 11:08
L6	0	select\$3 with rank same load same noise same (cancel\$5 reduct\$3)	US-PGPUB; USPAT	OR	OFF	2017/06/09 11:30
L7	6165	load same noise same (cancel\$5 reduct\$3)	US-PGPUB; USPAT	OR	OFF	2017/06/09 11:35
L8	0	load same noise same (cancel\$5 reduct\$3) same rank with memory	US-PGPUB; USPAT	OR	OFF	2017/06/09 11:35
L15	28	(US-20010008006-\$ US-20030070052-\$ US-20080025137-\$ US-20090248969-\$ US-20100125681-\$ US-4739473-\$ US-5313624-\$ US-5630096-\$ US-5905401-\$ US-6061754-\$ US-6141245-\$ US-6223650-\$ US-6262938-\$ US-6546476-\$ US-6636446-\$ US-6704910-\$ US-7024518-\$ US-7191302-\$ US-7225303-\$ US-7730254-\$ US-7990746-\$ US-7881150-\$ US-8244971-\$ US-8335894-\$ US-8856464-\$ US-8081536-\$ US-8516185-\$ US-8756364-\$).DID.	US-PGPUB; USPAT	OR	OFF	2017/06/09 11:47
L16	4	"6381140".pn. "7254036".pn. "20090248969".pn. "20100125681".pn.	US-PGPUB; USPAT	OR	OFF	2017/06/09 11:47
S1	7	"5559970".pn. "6700823".pn. "6812869".pn. "6981089".pn. "7136958".pn. "7184916".pn. "20070013429".pn.	US-PGPUB; USPAT	OR	OFF	2013/12/28 15:58
S2	496	bidirectional with port with (crossbar switch)	US-PGPUB; USPAT	OR	OFF	2013/12/28 16:02
S3	89	bidirectional with port with (crossbar switch) same memory	US-PGPUB; USPAT	OR	OFF	2013/12/28 16:02

S4	366	bidirectional with ports with (crossbar switch)	US-PGPUB; USPAT	OR	OFF	2013/12/28 16:05
S5	48	bidirectional with ports with (switch\$3 with circuit)	US-PGPUB; USPAT	OR	OFF	2013/12/28 16:15
S6	2546	bi\$1directional near2 ports	US-PGPUB; USPAT	OR	OFF	2013/12/28 16:20
S7	666	bi\$1directional near2 ports same (crossbar switch\$3)	US-PGPUB; USPAT	OR	OFF	2013/12/28 16:40
S8	1	"7130308".pn.	US-PGPUB; USPAT	OR	OFF	2013/12/28 17:14
S9	786	711/149.cds.	US-PGPUB; USPAT	OR	OFF	2013/12/28 22:03
S10	78	port with noise with reduc\$4 and PCB	US-PGPUB; USPAT	OR	OFF	2013/12/28 22:16
S11	20	signal with deteriorat\$3 with lossy	US-PGPUB; USPAT	OR	OFF	2014/05/04 17:59
S12	1	signal with deteriorat\$3 with switch same lossy	US-PGPUB; USPAT	OR	OFF	2014/05/04 18:01
S13	16	signal with loss with switch same lossy	US-PGPUB; USPAT	OR	OFF	2014/05/04 18:01
S14	19	signal with (loss deteriorat\$3 distort\$3) with switch same lossy	US-PGPUB; USPAT	OR	OFF	2014/05/04 18:16
S15	1227	signal with (loss deteriorat\$3 distort\$3) same lossy	US-PGPUB; USPAT	OR	OFF	2014/05/04 18:16
S16	14	signal with (loss deteriorat\$3 distort\$3) with lossy same (correct\$3 ameliorat\$3)	US-PGPUB; USPAT	OR	OFF	2014/05/04 18:17
S17	2	signal with (loss deteriorat\$3 distort\$3) with lossy same (switch\$3 with circuit)	US-PGPUB; USPAT	OR	OFF	2014/05/04 18:18
S18	529	signal with (loss deteriorat\$3 distort\$3) with lossy	US-PGPUB; USPAT	OR	OFF	2014/05/04 18:20
S19	11	signal with (loss deteriorat\$3 distort\$3) with lossy same port	US-PGPUB; USPAT	OR	OFF	2014/05/04 18:20
S20	19	signal with (loss deteriorat\$3 distort\$3) with lossy same port	US-PGPUB; USPAT	OR	ON	2014/05/04 18:20
S21	116	signal with (loss deteriorat\$3 distort\$3) with lossy same circuit	US-PGPUB; USPAT	OR	ON	2014/05/04 18:22
S22	8	signal with (loss deteriorat\$3 distort\$3) with lossy same discontin\$6	US-PGPUB; USPAT	OR	ON	2014/05/04 18:23

S23	66	signal with (loss deteriorat\$3 distort\$3) with lossy same (repeater amplifier)	US-PGPUB; USPAT	OR	ON	2014/05/04 18:36
S24	47	transmission with (loss deteriorat\$3 distort\$3) with lossy same (repeater amplifier)	US-PGPUB; USPAT	OR	ON	2014/05/04 18:39
S25	13	signal with (loss deteriorat\$3 distort\$3) with lossy same port	US-PGPUB; USPAT	OR	OFF	2014/05/23 18:39
S26	20	signal with deteriorat\$3 with lossy	US-PGPUB; USPAT	OR	OFF	2014/05/23 18:41
S27	34	signal with discontinuit\$3 with lossy	US-PGPUB; USPAT	OR	OFF	2014/05/23 18:43
S28	3171	365/191.ccls. 365/196.ccls. 711/149.ccls.	US-PGPUB; USPAT	OR	OFF	2014/05/23 18:48
S32	1	"20030070052".pn.	US-PGPUB; USPAT	OR	OFF	2015/09/30 14:47
S33	22	(isolat\$3 mux select\$3) with rand with memory	US-PGPUB; USPAT	OR	OFF	2015/09/30 14:58
S34	1589	(isolat\$3 mux select\$3) with rank with memory	US-PGPUB; USPAT	OR	OFF	2015/09/30 14:58
S35	551	signal with waveform with reshap\$3	US-PGPUB; USPAT	OR	OFF	2015/09/30 15:11
S36	6	signal with waveform with reshap\$3 with loss	US-PGPUB; USPAT	OR	OFF	2015/09/30 15:12
S37	0	signal with waveform with reshap\$3 with rank	US-PGPUB; USPAT	OR	OFF	2015/09/30 15:14
S38	0	signal with waveform with reshap\$3 same rank	US-PGPUB; USPAT	OR	OFF	2015/09/30 15:14
S39	43	signal with waveform with reshap\$3 same memory	US-PGPUB; USPAT	OR	OFF	2015/09/30 15:15
S40	1	"6959369".pn.	US-PGPUB; USPAT	OR	OFF	2015/10/02 12:27
S41	1	"6732244".pn.	US-PGPUB; USPAT	OR	OFF	2015/10/02 12:29
S42	0	impedence near3 match\$3 same rank with memory	US-PGPUB; USPAT	OR	OFF	2017/06/09 00:49
S43	668	impedence near3 match\$3	US-PGPUB; USPAT	OR	OFF	2017/06/09 00:56
S44	0	impedence near3 match\$3 and rank with memory	US-PGPUB; USPAT	OR	OFF	2017/06/09 00:56

S45	9	impedence near3 match\$3 same memory	US-PGPUB; USPAT	OR	OFF	2017/06/09 00:56
S46	0	impedence near3 match\$3 same port\$1 same3 memory	US-PGPUB; USPAT	OR	OFF	2017/06/09 01:10
S47	0	impedence with match\$3 same port\$1 same3 memory	US-PGPUB; USPAT	OR	OFF	2017/06/09 01:10
S48	19	impedence with match\$3 and memory with port\$1	US-PGPUB; USPAT	OR	OFF	2017/06/09 01:52

EAST Search History (Interference)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L9	4615	G06F12/1684.cpc. G11C5/02.cpc. G11C5/06.cpc. G11C8/18.cpc. G06F13/4027.cpc.	USPAT	OR	OFF	2017/06/09 11:37
L10	4903	correction.clm. adj circuit.clm.	USPAT	OR	OFF	2017/06/09 11:38
L11	15	L9 and L10	USPAT	OR	OFF	2017/06/09 11:38
L12	0	L9 and L10 and impedence.clm.	USPAT	OR	OFF	2017/06/09 11:38
L13	2	"6381140".pn. "7254036".pn. "20090248969".pn. "20100125681".pn.	USPAT	OR	OFF	2017/06/09 11:47
L14	23	"Term Removed"	USPAT	OR	OFF	2017/06/09 11:47
S29	2421	365/191.ccls. 365/196.ccls. 711/149.ccls.	USPAT	OR	OFF	2014/05/23 18:51
S30	1498	lossy.clm.	USPAT	OR	OFF	2014/05/23 18:52
S31	0	S29 and S30	USPAT	OR	OFF	2014/05/23 18:52

6/ 9/ 2017 11:47:41 AM
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INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	14/715,491
				Filing Date	May 18, 2015
				First Named Inventor	Hyun Lee
				Art Unit	2827
				Examiner Name	Unknown
Sheet	1	of	3	Attorney Docket Number	NETL.045C3

US PATENT DOCUMENTS					
Examiner Initials *	Cite No. ¹	Document Number	Publication Date (MM-DD-YYYY)	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number-Kind Code ² (if known)			
	1.	2002/0038405 A1	03-28-2002	Leddige, M. et al.	
	2.	2009/0296503 A1	12-03-2009	Chu, M. et al.	
	3.	2011/0085408 A1	04-14-2011	Solomon, J. et al.	
	4.	2008/0162790 A1	07-03-2008	IM	

NON PATENT LITERATURE DOCUMENTS			
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	5.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Related Matters, filed April 28, 2015	
	6.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Patent Owner Response to Petition, filed February 22, 2016	
	7.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Petitioner's Objections to Evidence, filed February 29, 2016	
	8.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Petitioner's Reply, filed May 19, 2016	
	9.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Petitioner Request for Oral Argument, filed June 2, 2016	
	10.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, PATENT OWNER MOTION FOR OBSERVATIONS, filed June 2, 2016	
	11.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, PATENT OWNER REQUEST FOR ORAL ARGUMENT, filed June 2, 2016	
	12.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, ORDER Trial Hearing, filed June 8, 2016	
	13.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Patent Owner's SUBMISSION ON PROPRIETY OF PETITIONER REPLY, filed June 9, 2016	
	14.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Petitioner's Response to Motion for Observations, filed June 10, 2016	
	15.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Petitioner's Response to Netlist's Submission, filed June 13, 2016	
	16.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Patent Owner's Objections to Petitioner's Demonstrative Exhibits, filed June 23, 2016	
	17.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Patent Owner's Demonstrative Exhibits, filed June 24, 2016	

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	18.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Petitioner's Updated Exhibit List, filed June 24, 2016	
	19.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Supplemental Declaration of Dr. Jagannathan, filed May 19, 2016	
	20.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, May 10, 2016 Deposition of Carl Sechen, filed May 19, 2016	
	21.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Petitioner's Demonstrative Exhibits, filed June 24, 2016	
	22.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Exhibit 2001, "Declaration of Professor Carl Sechen," filed February 22, 2016	
	23.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Exhibit 2002, "Videotaped Deposition of Srinivasan Jagannathan, Ph.D on February 12, 2016," filed February 22, 2016	
	24.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Exhibit 2003, "Videotaped Deposition of Srinivasan Jagannathan, Ph.D on April 20, 2015," filed February 22, 2016	
	25.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Exhibit 2007, "Deposition of Dr. Srinivasan Jagannathan on May 25, 2016," filed June 2, 2016	
	26.	Inter Partes Review of U.S. Patent No. 8081536, Case No. IPR2015-01021, Patent Owner Response to Petition, filed February 22, 2016	
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	28.	Inter Partes Review of U.S. Patent No. 8081536, Case No. IPR2015-01021, Petitioner's Reply, filed May 19, 2016	
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	34.	Inter Partes Review of U.S. Patent No. 8081536, Case No. IPR2015-01021, Petitioner's Response to Patent Owner's Submission, filed June 13, 2016	

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	14/715,491
				Filing Date	May 18, 2015
				First Named Inventor	Hyun Lee
				Art Unit	2827
				Examiner Name	Unknown
Sheet	3	of	3	Attorney Docket Number	NETL.045C3

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²
	35.	Inter Partes Review of U.S. Patent No. 8081536, Case No. IPR2015-01021, Patent Owner's Objections to Petitioner's Demonstrative Exhibits, filed June 23, 2016	
	36.	Inter Partes Review of U.S. Patent No. 8081536, Case No. IPR2015-01021, Patent Owner's Demonstrative Exhibits, filed June 24, 2016	
	37.	Inter Partes Review of U.S. Patent No. 8081536, Case No. IPR2015-01021, Petitioner's Updated Exhibit List, filed June 24, 2016	
	38.	Inter Partes Review of U.S. Patent No. 8081536, Case No. IPR2015-01021, Supplemental Declaration of Dr. Jagannathan, filed May 19, 2016	
	39.	Inter Partes Review of U.S. Patent No. 8081536, Case No. IPR2015-01021, Exhibit 2001, "Declaration of Professor Carl Sechen," filed February 22, 2016	
	40.	International Search Report and Written Opinion, International Application No. PCT/US14/48517, filed July 28, 2014, dated October 27, 2014	
	41.	Inter Partes Review Case No. IPR2014-01029, Petitioner's Request for Rehearing pursuant to 37 C.F.R. § 42.71, filed on January 15, 2015.	
	42.	Inter Partes Review Case No. IPR2014-01029, Decision Denying Request for Rehearing, Issued on March 3, 2015.	
	43.	Inter Partes Review Case No. IPR2014-01369, Decision Denying Institution of Inter Partes Review 37 C.F.R. § 42.108, issued March 9, 2014.	
	44.	Examination Report, European Patent Application No. 10730021.2, dated April 14, 2014.	
	45.	Response to Examination Report, European Patent Application No. 10730021.2, dated June 4, 2014.	
	46.	Examination Report, European Patent Application No. 10730021.2, dated April 29, 2015.	
	47.	Response to Examination Report, European Patent Application No. 10730021.2, dated April 29, 2015, filed November 4, 2015.	
	48.	English Translation of the Notice of Grounds for Rejection, Korean Patent Application No. 2012-7004038, dated May 11, 2016	
	49.	Notice of Allowance, U.S. Pat. Appl. No. 13/970,606, filed August 20, 2013, dated June 27, 2016	

EXAMINER /GURTEJ BANSAL/

DATE CONSIDERED 06/09/2017

* EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.¹ Applicant's unique citation designation number (optional).² See Kinds Codes of USPTO Patents Documents at www.uspto.gov or MPEP 901.04. ³ Enter Office that issued the document, by the two-letter code (WIPO Standard ST. 3). ⁴ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁵ Kind of document by the appropriate symbols as indicated on the document and WIPO Standard ST. 16 if possible. ⁶ Applicant is to place a check mark here if English language Translation is attached.

ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /G.B./

EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L1	0	select\$3 with rank and impedance with match\$3	US-PGPUB; USPAT	OR	OFF	2017/06/09 10:54
L2	0	select\$3 with rank and impedance with match\$3	US-PGPUB; USPAT	OR	OFF	2017/06/09 10:54
L3	597	select\$3 with rank and signal with correction	US-PGPUB; USPAT	OR	OFF	2017/06/09 11:00
L4	45	select\$3 with rank same signal with correction	US-PGPUB; USPAT	OR	OFF	2017/06/09 11:00
L5	8280	G06F12/1684.cpc. G11C5/02.cpc. G11C5/06.cpc. G11C8/18.cpc. G06F13/4027.cpc.	US-PGPUB; USPAT	OR	OFF	2017/06/09 11:08
L6	0	select\$3 with rank same load same noise same (cancel\$5 reduct\$3)	US-PGPUB; USPAT	OR	OFF	2017/06/09 11:30
L7	6165	load same noise same (cancel\$5 reduct\$3)	US-PGPUB; USPAT	OR	OFF	2017/06/09 11:35
L8	0	load same noise same (cancel\$5 reduct\$3) same rank with memory	US-PGPUB; USPAT	OR	OFF	2017/06/09 11:35
S1	7	"5559970".pn. "6700823".pn. "6812869".pn. "6981089".pn. "7136958".pn. "7184916".pn. "20070013429".pn.	US-PGPUB; USPAT	OR	OFF	2013/12/28 15:58
S2	496	bidirectional with port with (crossbar switch)	US-PGPUB; USPAT	OR	OFF	2013/12/28 16:02
S3	89	bidirectional with port with (crossbar switch) same memory	US-PGPUB; USPAT	OR	OFF	2013/12/28 16:02
S4	366	bidirectional with ports with (crossbar switch)	US-PGPUB; USPAT	OR	OFF	2013/12/28 16:05
S5	48	bidirectional with ports with (switch\$3 with circuit)	US-PGPUB; USPAT	OR	OFF	2013/12/28 16:15
S6	2546	bi\$1directional near2 ports	US-PGPUB; USPAT	OR	OFF	2013/12/28 16:20
S7	666	bi\$1directional near2 ports same (crossbar switch\$3)	US-PGPUB; USPAT	OR	OFF	2013/12/28 16:40
S8	1	"7130308".pn.	US-PGPUB;	OR	OFF	2013/12/28 17:14

			USPAT			
S9	786	711/149.cds.	US-PGPUB; USPAT	OR	OFF	2013/12/28 22:03
S10	78	port with noise with reduc\$4 and PCB	US-PGPUB; USPAT	OR	OFF	2013/12/28 22:16
S11	20	signal with deteriorat\$3 with lossy	US-PGPUB; USPAT	OR	OFF	2014/05/04 17:59
S12	1	signal with deteriorat\$3 with switch same lossy	US-PGPUB; USPAT	OR	OFF	2014/05/04 18:01
S13	16	signal with loss with switch same lossy	US-PGPUB; USPAT	OR	OFF	2014/05/04 18:01
S14	19	signal with (loss deteriorat\$3 distort\$3) with switch same lossy	US-PGPUB; USPAT	OR	OFF	2014/05/04 18:16
S15	1227	signal with (loss deteriorat\$3 distort\$3) same lossy	US-PGPUB; USPAT	OR	OFF	2014/05/04 18:16
S16	14	signal with (loss deteriorat\$3 distort\$3) with lossy same (correct\$3 ameliorat\$3)	US-PGPUB; USPAT	OR	OFF	2014/05/04 18:17
S17	2	signal with (loss deteriorat\$3 distort\$3) with lossy same (switch\$3 with circuit)	US-PGPUB; USPAT	OR	OFF	2014/05/04 18:18
S18	529	signal with (loss deteriorat\$3 distort\$3) with lossy	US-PGPUB; USPAT	OR	OFF	2014/05/04 18:20
S19	11	signal with (loss deteriorat\$3 distort\$3) with lossy same port	US-PGPUB; USPAT	OR	OFF	2014/05/04 18:20
S20	19	signal with (loss deteriorat\$3 distort\$3) with lossy same port	US-PGPUB; USPAT	OR	ON	2014/05/04 18:20
S21	116	signal with (loss deteriorat\$3 distort\$3) with lossy same circuit	US-PGPUB; USPAT	OR	ON	2014/05/04 18:22
S22	8	signal with (loss deteriorat\$3 distort\$3) with lossy same discontin\$6	US-PGPUB; USPAT	OR	ON	2014/05/04 18:23
S23	66	signal with (loss deteriorat\$3 distort\$3) with lossy same (repeater amplifier)	US-PGPUB; USPAT	OR	ON	2014/05/04 18:36
S24	47	transmission with (loss deteriorat\$3 distort\$3) with lossy same (repeater amplifier)	US-PGPUB; USPAT	OR	ON	2014/05/04 18:39
S25	13	signal with (loss deteriorat\$3 distort\$3) with lossy same port	US-PGPUB; USPAT	OR	OFF	2014/05/23 18:39
S26	20	signal with deteriorat\$3 with lossy	US-PGPUB; USPAT	OR	OFF	2014/05/23 18:41
S27	34	signal with discontinuit\$3 with lossy	US-PGPUB;	OR	OFF	2014/05/23 18:43

			USPAT			
S28	3171	365/191.ccls. 365/196.ccls. 711/149.ccls.	US-PGPUB; USPAT	OR	OFF	2014/05/23 18:48
S32	1	"20030070052".pn.	US-PGPUB; USPAT	OR	OFF	2015/09/30 14:47
S33	22	(isolat\$3 mux select\$3) with rand with memory	US-PGPUB; USPAT	OR	OFF	2015/09/30 14:58
S34	1589	(isolat\$3 mux select\$3) with rank with memory	US-PGPUB; USPAT	OR	OFF	2015/09/30 14:58
S35	551	signal with waveform with reshap\$3	US-PGPUB; USPAT	OR	OFF	2015/09/30 15:11
S36	6	signal with waveform with reshap\$3 with loss	US-PGPUB; USPAT	OR	OFF	2015/09/30 15:12
S37	0	signal with waveform with reshap\$3 with rank	US-PGPUB; USPAT	OR	OFF	2015/09/30 15:14
S38	0	signal with waveform with reshap\$3 same rank	US-PGPUB; USPAT	OR	OFF	2015/09/30 15:14
S39	43	signal with waveform with reshap\$3 same memory	US-PGPUB; USPAT	OR	OFF	2015/09/30 15:15
S40	1	"6959369".pn.	US-PGPUB; USPAT	OR	OFF	2015/10/02 12:27
S41	1	"6732244".pn.	US-PGPUB; USPAT	OR	OFF	2015/10/02 12:29
S42	0	impedence near3 match\$3 same rank with memory	US-PGPUB; USPAT	OR	OFF	2017/06/09 00:49
S43	668	impedence near3 match\$3	US-PGPUB; USPAT	OR	OFF	2017/06/09 00:56
S44	0	impedence near3 match\$3 and rank with memory	US-PGPUB; USPAT	OR	OFF	2017/06/09 00:56
S45	9	impedence near3 match\$3 same memory	US-PGPUB; USPAT	OR	OFF	2017/06/09 00:56
S46	0	impedence near3 match\$3 same port\$1 same3 memory	US-PGPUB; USPAT	OR	OFF	2017/06/09 01:10
S47	0	impedence with match\$3 same port\$1 same3 memory	US-PGPUB; USPAT	OR	OFF	2017/06/09 01:10
S48	19	impedence with match\$3 and memory with port\$1	US-PGPUB; USPAT	OR	OFF	2017/06/09 01:52

EAST Search History (Interference)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L9	4615	G06F12/1684.cpc. G11C5/02.cpc. G11C5/06.cpc. G11C8/18.cpc. G06F13/4027.cpc.	USPAT	OR	OFF	2017/06/09 11:37
L10	4903	correction.clm. adj circuit.clm.	USPAT	OR	OFF	2017/06/09 11:38
L11	15	L9 and L10	USPAT	OR	OFF	2017/06/09 11:38
L12	0	L9 and L10 and impedance.clm.	USPAT	OR	OFF	2017/06/09 11:38
S29	2421	365/191.ccls. 365/196.ccls. 711/149.ccls.	USPAT	OR	OFF	2014/05/23 18:51
S30	1498	lossy.clm.	USPAT	OR	OFF	2014/05/23 18:52
S31	0	S29 and S30	USPAT	OR	OFF	2014/05/23 18:52

6/ 9/ 2017 11:38:53 AM
C:\ Users\ gbansal\ Documents\ EAST\ Workspaces\ 13412243.wsp

Issue Classification 	Application/Control No. 14715491	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner GURTEJ BANSAL	Art Unit 2139

☐ Claims renumbered in the same order as presented by applicant															
☐ CPA															
☐ T.D.															
☐ R.1.47															
Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original
1	1		17												
	2		18												
	3		19												
2	4		20												
3	5		21												
4	6		22												
	7	10	23												
	8	11	24												
5	9	12	25												
6	10	13	26												
17	11	14	27												
18	12	15	28												
7	13	16	29												
8	14														
9	15														
	16														

NONE	Total Claims Allowed: 18	
(Assistant Examiner)	(Date)	
/GURTEJ BANSAL/ Primary Examiner.Art Unit 2139	06/09/2017	O.G. Print Claim(s) 1
(Primary Examiner)	(Date)	O.G. Print Figure 10

Search Notes 	Application/Control No. 14715491	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner GURTEJ BANSAL	Art Unit 2139

CPC- SEARCHED		
Symbol	Date	Examiner
G06F12/1684.cpc. G11C5/02.cpc. G11C5/06.cpc. G11C8/18.cpc. G06F13/4027.cpc.	06/09/2017	GB

CPC COMBINATION SETS - SEARCHED		
Symbol	Date	Examiner

US CLASSIFICATION SEARCHED			
Class	Subclass	Date	Examiner

SEARCH NOTES		
Search Notes	Date	Examiner
Text Search in EAST (See Attached)	06/09/2017	GB
Inventor Search in PE2E	06/09/2017	GB
NPL Search in Google Scholar (See Attached)	06/09/2017	GB

INTERFERENCE SEARCH			
US Class/ CPC Symbol	US Subclass / CPC Group	Date	Examiner
	G06F12/1684.cpc. G11C5/02.cpc. G11C5/06.cpc. G11C8/18.cpc. G06F13/4027.cpc.	06/09/2017	GB

	/GURTEJ BANSAL/ Primary Examiner.Art Unit 2139
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BIB DATA SHEET

CONFIRMATION NO. 8143

SERIAL NUMBER	FILING or 371(c) DATE	CLASS	GROUP ART UNIT	ATTORNEY DOCKET NO.		
14/715,491	05/18/2015	711	2139	N3044.10007US08		
APPLICANTS Netlist, Inc., Irvine, CA; INVENTORS Hyun Lee, Ladera Ranch, CA; Jayesh R. Bhakta, Cerritos, CA; Jeffrey C. Solomon, Irvine, CA; Mario Jesus Martinez, Laguna Niguel, CA; Chi-She Chen, Walnut, CA; ** CONTINUING DATA ***** This application is a CON of 14/324,990 07/07/2014 PAT 9037809 which is a CON of 13/412,243 03/05/2012 PAT 8782350 ** FOREIGN APPLICATIONS ***** ** IF REQUIRED, FOREIGN FILING LICENSE GRANTED ** ** SMALL ENTITY ** 05/27/2015						
Foreign Priority claimed ☐ Yes ☒ No 35 USC 119(a-d) conditions met ☐ Yes ☒ No Verified and Acknowledged <u>/GURTEJ BANSAL/</u> Examiner's Signature		☐ Met after Allowance Initials	STATE OR COUNTRY CA	SHEETS DRAWINGS 9	TOTAL CLAIMS 20	INDEPENDENT CLAIMS 2
ADDRESS Jamie J. Zheng, Ph.D Esq. MASCHOFF BRENNAN 1389 Center Drive Suite 300 Park City, UT 84098 UNITED STATES						
TITLE MEMORY MODULE AND CIRCUIT PROVIDING LOAD ISOLTION AND NOISE REDUCTION						
FILING FEE RECEIVED 800	FEES: Authority has been given in Paper No. _____ to charge/credit DEPOSIT ACCOUNT No. _____ for following:			☐ All Fees ☐ 1.16 Fees (Filing) ☐ 1.17 Fees (Processing Ext. of time) ☐ 1.18 Fees (Issue) ☐ Other _____ ☐ Credit		

CORRECTED ADS FORM

Application Number	14715491
Title of Invention	MEMORY MODULE AND CIRCUIT PROVIDING LOAD ISOLTION AND NOISE REDUCTION

Inventor Information

****If no data is shown, no data has been corrected****

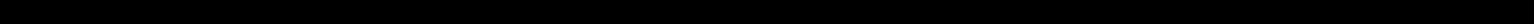
	Data of Record	Updated Data
Order Number		
Name		

Residence Information

Residency		
City		
State		
Country of Residence		

Mailing Address of Inventor

Address 1		
Address 2		
City,State/Province, Postal Code		
Country		



Application Information

	Data of Record	Updated Data
Title of Invention	MEMORY MODULE AND CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION	
Attorney Docket Number	N3044.10007US08	
Entity Type	Small	

Domestic Benefit/National Stage Information

****If no data is shown, no data has been corrected****

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, 365(c), or 386(c) or indicate National Stage entry from a PCT application. Providing this information in the application data sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78(a).

	Data of Record	Updated Data
Prior Application Status	patented	
Application Number	14715491	
Continuity Type	CON	
Prior Application Number	14324990	
Filing Date (YYYY-MM-DD)	2014-07-07	
Patent Number	9037809	
Issue Date (YYYY-MM-DD)	2015-05-19	

		Data of Record	Updated Data
	Prior Application Status	patented	
	Application Number	14324990	
	Continuity Type	CON	
	Prior Application Number	13412243	
	Filing Date (YYYY-MM-DD)	2012-03-05	
	Patent Number	8782350	
	Issue Date (YYYY-MM-DD)	2014-07-15	

		Data of Record	Updated Data
	Prior Application Status		<u>patented</u>
	Application Number		<u>13412243</u>
	Continuity Type		<u>CON</u>
	Prior Application Number		<u>12422853</u>
	Filing Date (YYYY-MM-DD)		<u>2009-04-13</u>
	Patent Number		<u>8154901</u>
	Issue Date (YYYY-MM-DD)		<u>2012-04-10</u>

		Data of Record	Updated Data
	Prior Application Status		<u>expired</u>
	Application Number		<u>12422853</u>
	Continuity Type		<u>PRO</u>
	Prior Application Number		<u>61044801</u>
	Filing Date (YYYY-MM-DD)		<u>2008-04-14</u>
	Patent Number		
	Issue Date (YYYY-MM-DD)		

		Data of Record	Updated Data
	Prior Application Status		<u>expired</u>
	Application Number		<u>12422853</u>
	Continuity Type		<u>PRO</u>
	Prior Application Number		<u>61044825</u>
	Filing Date (YYYY-MM-DD)		<u>2008-04-14</u>
	Patent Number		
	Issue Date (YYYY-MM-DD)		

		Data of Record	Updated Data
	Prior Application Status		<u>expired</u>
	Application Number		<u>12422853</u>
	Continuity Type		<u>PRO</u>
	Prior Application Number		<u>61044839</u>
	Filing Date (YYYY-MM-DD)		<u>2008-04-14</u>
	Patent Number		
	Issue Date (YYYY-MM-DD)		

Foreign Priority Information

****If no data is shown, no data has been corrected****

This section allows for the applicant to claim priority to a foreign application. Providing this information in the application data sheet constitutes the claim for priority as required by 35 U.S.C. 119(b) and 37 CFR 1.55. When priority is claimed to a foreign application that is eligible for retrieval under the priority document exchange program (PDX) the information will be used by the Office to automatically attempt retrieval pursuant to 37 CFR 1.55(i)(1) and (2). Under the PDX program, applicant bears the ultimate responsibility for ensuring that a copy of the foreign application is received by the Office from the participating foreign intellectual property office, or a certified copy of the foreign priority application is filed, within the time period specified in 37 CFR 1.55(g)(1).

		Data of Record	Updated Data
	Application Number		
	Country		
	Filing Date		
	Access Code		

Applicant Information

****If no data is shown, no data has been corrected****

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

	Data of Record	Updated Data
Applicant Type		
If applicant is the legal representative, indicate the authority to file the patent application, the inventor is		
Name of the Deceased or Legally Incapacitated Inventor		
Applicant is an Organization		
Name		
Organization Name		
Address 1		
Address 2		
City,State/Province,Postal Code		
Country		
Phone Number		
Fax Number		

Email Address		
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Assignee Information including Non-Applicant Assignee Information

****If no data is shown, no data has been corrected****
Providing this information in the application data sheet does not substitute for compliance with any requirement of part 3 of Title 37 of the CFR to have an assignment recorded in the Office

	Data of Record	Updated Data
Order		
Applicant is an Organization		
Name		
Organization Name		

Mailing Address

Address 1		
Address 2		
City,State/Province,Postal Code		
Country		
Phone Number		
Fax Number		
Email Address		

Signature

NOTE: This Application Data Sheet must be signed in accordance with 37 CFR 1.33(b).

This Application Data Sheet **must** be signed by a patent practitioner if one or more of the applicants is a **juristic entity** (e.g., corporation or association). If the applicant is two or more joint inventors, this form must be signed by a patent practitioner, **all** joint inventors who are the applicant, or one or more joint inventor-applicants who have been given power of attorney (e.g., see USPTO Form PTO/AIA/81) on behalf of **all** joint inventor-applicants.

See 37 CFR 1.4(d) for the manner of making signatures and certifications.

Corrected ADS 1.0

Signature	/Jamie J. Zheng/	Registration Number	51167
First Name	Jamie	Last Name	Zheng

Electronic Patent Application Fee Transmittal

Application Number:	14715491			
Filing Date:	18-May-2015			
Title of Invention:	MEMORY MODULE AND CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION			
First Named Inventor/Applicant Name:				
Filer:	Jamie Jie Zheng/Brian Gonzalez			
Attorney Docket Number:	N3044.10007US08			
Filed as Small Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
PROCESSING FEE, EXCEPT PROV. APPLS.	2830	1	70	70
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				70

Electronic Acknowledgement Receipt

EFS ID:	30384219
Application Number:	14715491
International Application Number:	
Confirmation Number:	8143
Title of Invention:	MEMORY MODULE AND CIRCUIT PROVIDING LOAD ISOLTION AND NOISE REDUCTION
First Named Inventor/Applicant Name:	
Customer Number:	79141
Filer:	Jamie Jie Zheng/Brian Gonzalez
Filer Authorized By:	Jamie Jie Zheng
Attorney Docket Number:	N3044.10007US08
Receipt Date:	15-SEP-2017
Filing Date:	18-MAY-2015
Time Stamp:	18:51:18
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	CARD
Payment was successfully received in RAM	\$ 70
RAM confirmation Number	091817INTEFSW18511700
Deposit Account	505394
Authorized User	Brian Gonzalez

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

37 CFR 1.21 (Miscellaneous fees and charges)

--

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Application Data Sheet to update/ correct info	CorrectedADS.pdf	94156	no	7
			73f39768666b3c3ae7149e60b73599523c620a98		

Warnings:

Information:

2	Fee Worksheet (SB06)	fee-info.pdf	30476	no	2
			88c6b914b9defa95786d5fc7e2d2f1425080d5a4		

Warnings:

Information:

Total Files Size (in bytes):			124632
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This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), to: **Mail** **Mail Stop ISSUE FEE**
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450
or Fax **(571)-273-2885**

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

79141 7590 06/15/2017
Jamie J. Zheng, Ph.D Esq.
MASCHOFF BRENNAN
1389 Center Drive
Suite 300
Park City, UT 84098

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being facsimile transmitted to the USPTO (571) 273-2885, on the date indicated below.

(Depositor's name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/715,491	05/18/2015	Hyun Lee	N3044.10007US08	8143

TITLE OF INVENTION: MEMORY MODULE AND CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	SMALL	\$480	\$0	\$0	\$480	09/15/2017

EXAMINER	ART UNIT	CLASS-SUBCLASS
BANSAL, GURTEJ	2139	711-149000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

- ☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.
- ☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

- (1) The names of up to 3 registered patent attorneys or agents OR, alternatively,
- (2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

1 **MASCHOFF BRENNAN**

2 _____

3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Netlist, Inc.

Irvine, California

Please check the appropriate assignee category or categories (will not be printed on the patent): ☐ Individual ☒ Corporation or other private group entity ☐ Government

4a. The following fee(s) are submitted:

- ☒ Issue Fee
- ☐ Publication Fee (No small entity discount permitted)
- ☐ Advance Order - # of Copies _____

4b. Payment of Fee(s): (Please first reapply any previously paid issue fee shown above)

- ☐ A check is enclosed.
- ☐ Payment by credit card. Form PTO-2038 is attached.
- ☒ The director is hereby authorized to charge the required fee(s), any deficiency, or credits any overpayment, to Deposit Account Number **50-5394** (enclose an extra copy of this form).

5. Change in Entity Status (from status indicated above)

- ☐ Applicant certifying micro entity status. See 37 CFR 1.29
- ☐ Applicant asserting small entity status. See 37 CFR 1.27
- ☒ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature /Jamie J. Zheng/

Date September 15, 2017

Typed or printed name Jamie J. Zheng

Registration No. 51167

Electronic Patent Application Fee Transmittal

Application Number:	14715491			
Filing Date:	18-May-2015			
Title of Invention:	MEMORY MODULE AND CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Jamie Jie Zheng/Brian Gonzalez			
Attorney Docket Number:	N3044.10007US08			
Filed as Large Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
UTILITY APPL ISSUE FEE	1501	1	960	960

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				960

Electronic Acknowledgement Receipt

EFS ID:	30384834
Application Number:	14715491
International Application Number:	
Confirmation Number:	8143
Title of Invention:	MEMORY MODULE AND CIRCUIT PROVIDING LOAD ISOLTION AND NOISE REDUCTION
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng/Brian Gonzalez
Filer Authorized By:	Jamie Jie Zheng
Attorney Docket Number:	N3044.10007US08
Receipt Date:	15-SEP-2017
Filing Date:	18-MAY-2015
Time Stamp:	18:57:48
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	CARD
Payment was successfully received in RAM	\$960
RAM confirmation Number	091817INTEFSW18581900
Deposit Account	505394
Authorized User	Brian Gonzalez

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:
37 CFR 1.20 (Post Issuance fees)

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Post Allowance Communication - Incoming	2017-09-15_Comments_N3044-10007US08.pdf	226133	no	1
			363babb6b87aefa0d0d5087c5e0cfeffe610d0de		
Warnings:					
Information:					
2	Issue Fee Payment (PTO-85B)	2017-09-15_Issue-Fee_N3044-10007US08.pdf	140894	no	1
			2e1dab3da14fb6fcbcb89c92bf37ffc8d752b269		
Warnings:					
Information:					
3	Fee Worksheet (SB06)	fee-info.pdf	30785	no	2
			3aadab1ecef06c65ff3abd30aa335da0b09002c		
Warnings:					
Information:					
Total Files Size (in bytes):			397812		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re application of :	Hyun Lee	Art Unit 2139
Application No.:	14/715,491	
Filed:	May 18, 2015	
Conf. No.:	8143	
For:	MEMORY MODULE AND CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION	
Examiner:	BANSAL, GURTEJ	
Customer No.:	79141	
Attorney Docket No.:	N3044.10007US08	

COMMENTS ON STATEMENT OF REASONS FOR ALLOWANCE

Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Dear Sir:

Applicant acknowledges Examiner's statements of Reasons for Allowance of the above-referenced patent application and agrees that the claimed subject matter is patentable. However, Applicant takes no position regarding the Reasons for Allowance presented by the Examiner other than the positions Applicant may have previously taken during prosecution. Therefore, the Examiner's Reasons for Allowance should not be attributed to Applicant as an indication of the basis for Applicant's belief that the claims are patentable. Furthermore, Applicant respectfully asserts that there may also be additional reasons for patentability of the claimed subject matter not explicitly stated in this record and Applicant does not waive its rights to such arguments by not further addressing such reasons herein.

Dated this 15th day of September, 2017.

Respectfully submitted,

/Jamie J. Zheng, Reg. no. 51167/

JAMIE J. ZHENG
Registration No. 51,167
Attorney for Applicant
Telephone No. (650) 521-4828

Document code: WFEE

United States Patent and Trademark Office
Sales Receipt for Accounting Date: 09/22/2017

ERIMANDO	ADJ #00000001	Mailroom Dt: 09/15/2017	
	Seq No: 5572	Sales Acctg Dt: 09/18/2017	14715491
	01 FC : 2830	-70.00 OP	

Document code: WFEE

United States Patent and Trademark Office
Sales Receipt for Accounting Date: 09/22/2017

ERIMANDO RF #30202516 Mailroom Dt: 09/22/2017 14715491

Credit Card Refund Total: \$70.00

American Express

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UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
14/715,491	05/18/2015	Hyun Lee	N3044.10007US08

CONFIRMATION NO. 8143
IMPROPER CFR REQUEST



OC000000094144182

79141
Jamie J. Zheng, Ph.D Esq.
MASCHOFF BRENNAN
1389 Center Drive
Suite 300
Park City, UT 84098

Date Mailed: 09/25/2017

RESPONSE TO REQUEST FOR CORRECTED FILING RECEIPT

Continuity, Priority Claims, Petitions, and Non-Publication Requests

In response to your request for a corrected Filing Receipt, the Office is unable to comply with your request because:

- The priority or continuity claim has not been entered because it was not filed during the required time period. Applicant may wish to consider filing a petition to accept an unintentionally delayed claim for priority. See 37 CFR 1.55 or 1.78.

Questions about the contents of this notice and the requirements it sets forth should be directed to the Office of Data Management, Application Assistance Unit, at (571) 272-4000 or (571) 272-4200 or 1-888-786-0101.

/ggasgedom/



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/715,491	05/18/2015	Hyun Lee	N3044.10007US08	8143

79141 7590 10/18/2017

Jamie J. Zheng, Ph.D Esq.
MASCHOFF BRENNAN
1389 Center Drive
Suite 300
Park City, UT 84098

EXAMINER

BANSAL, GURTEJ

ART UNIT	PAPER NUMBER
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2139

NOTIFICATION DATE	DELIVERY MODE
-------------------	---------------

10/18/2017

ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

d@mabr.com
bisraelsen@mabr.com

**UNITED STATES DEPARTMENT OF COMMERCE****U.S. Patent and Trademark Office**

Address : COMMISSIONER FOR PATENTS

P.O. Box 1450

Alexandria, Virginia 22313-1450

APPLICATION NO./ CONTROL NO.	FILING DATE	FIRST NAMED INVENTOR / PATENT IN REEXAMINATION	ATTORNEY DOCKET NO.
14/715,491	18 May, 2015	LEE ET AL.	N3044.10007US08

Jamie J. Zheng, Ph.D Esq. MASCHOFF BRENNAN 1389 Center Drive Suite 300 Park City, UT 84098		EXAMINER	
		GURTEJ BANSAL	
		ART UNIT	PAPER
		2139	20171014

DATE MAILED:

Please find below and/or attached an Office communication concerning this application or proceeding.

Commissioner for Patents

The priority or continuity claim has not been entered because it was not filed during the required time period. Applicant may wish to consider filing a petition to accept an unintentionally delayed claim for priority. See CFR 1.55 or 1.67.

/GURTEJ BANSAL/
Primary Examiner, Art Unit 2139

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re application of	Hyun Lee	Art Unit 2139
Application No.:	14/715,491	
Filed:	May 18, 2015	
Conf. No.:	8143	
For:	MEMORY MODULE AND CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION	
Examiner:	BANSAL, GURTEJ	
Customer No.:	79141	
Attorney Docket No.:	N3044.10007US08	

PETITION FOR UNINTENTIONALLY DELAYED DOMESTIC BENEFIT CLAIM
UNDER 37 CFR 1.78

Mail Stop Petitions
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Dear Commissioner,

The above-referenced patent application was filed on May 18, 2015, and claims priority to the following:

This application is a CON of 14/324,990 07/07/2014 PAT 9037809
which is a CON of 13/412,243 03/05/2012 PAT 8782350
which is a CON of 12/422,853 04/13/2009 PAT 8154901
which claims benefit of 61/044,825 04/14/2008
and claims benefit of 61/044,801 04/14/2008
and claims benefit of 61/044,839 04/14/2008

The above listing of priority data can be found in paragraph [001] of the originally filed specification, filed on May 18, 2015. The originally filed ADS also included the above listing of priority data except an error in the Continuity Type for Application No. 13/412,243.

This Petition is accompanied by a marked-up Supplemental ADS. The Supplemental ADS has been corrected to indicate Application no. 13/412,243 is a Continuation of 12/422,853.

Applicant hereby states that the entire delay between the date the benefit claim was due under 37 CFR 1.78 and the date of this Petition was unintentional. In part, the priority claim in the ADS and in the paragraph [001] of the specification shows that the priority claim was made, but included an unintentional error, which was discovered only recently. Thus, the delay was unintentional.

This petition is being accompanied with the petition fee under 37 CFR 1.17(m). The USPTO is authorized to charge any additional fees required for this petition to Deposit Account No. 50-5394.

Dated this 18th day of October, 2017.

Respectfully submitted,

/Jamie J. Zheng, Reg. no. 51167/

JAMIE J. ZHENG
Registration No. 51,167
Attorney for Applicant
Telephone No. (650) 521-4828

Electronic Patent Application Fee Transmittal

Application Number:	14715491			
Filing Date:	18-May-2015			
Title of Invention:	MEMORY MODULE AND CIRCUIT PROVIDING LOAD ISOLTION AND NOISE REDUCTION			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Jamie Jie Zheng/Brian Gonzalez			
Attorney Docket Number:	N3044.10007US08			
Filed as Large Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
PET. DELAY SUB OR RESTORE PRIORITY-CLAIM	1454	1	1700	1700
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				1700

Electronic Acknowledgement Receipt

EFS ID:	30695849
Application Number:	14715491
International Application Number:	
Confirmation Number:	8143
Title of Invention:	MEMORY MODULE AND CIRCUIT PROVIDING LOAD ISOLTION AND NOISE REDUCTION
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng/Brian Gonzalez
Filer Authorized By:	Jamie Jie Zheng
Attorney Docket Number:	N3044.10007US08
Receipt Date:	18-OCT-2017
Filing Date:	18-MAY-2015
Time Stamp:	17:29:28
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	CARD
Payment was successfully received in RAM	\$ 1700
RAM confirmation Number	101917INTEFSW17303700
Deposit Account	505394
Authorized User	Brian Gonzalez

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

37 CFR 1.21 (Miscellaneous fees and charges)

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Application Data Sheet	2017-10-18_ADS_N3044-10007US08.pdf	363371	no	8
			caf0ea9a57bc33fcc6e6ddf7089e4291a8e7efc5		
Warnings:					
Information:					
This is not an USPTO supplied ADS fillable form					
2	Petition for review by the Office of Petitions	2017-10-18_Petition-Unintentionally-Delayed-Priority-Claim_N3044-10007US08.pdf	229334	no	2
			a5ab92836c5f467bdb5b0a42478cfc69f85e792		
Warnings:					
Information:					
3	Fee Worksheet (SB06)	fee-info.pdf	30491	no	2
			fbba1a883b3867e466b1b88306edae41f3e1917a		
Warnings:					
Information:					
Total Files Size (in bytes):			623196		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.045C3
		Application Number	
Title of Invention	Circuit Providing Load Isolation and Noise Reduction		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76. This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.			

Secrecy Order 37 CFR 5.2

☐ Portions or all of the application associated with this Application Data Sheet may fall under a Secrecy Order pursuant to 37 CFR 5.2 (Paper filers only. Applications that fall under Secrecy Order may not be filed electronically.)
--

Inventor Information:

Inventor 1					Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Hyun		Lee		
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service					
City	Ladera Ranch	State/Province	CA	Country of Residence ⁱ	US
Mailing Address of Inventor:					
Address 1	21 Thalia				
Address 2					
City	Ladera Ranch	State/Province	CA		
Postal Code	92694	Country ⁱ	US		
Inventor 2					Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Jayesh	R.	Bhakta		
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service					
City	Cerritos	State/Province	CA	Country of Residence ⁱ	US
Mailing Address of Inventor:					
Address 1	12220 Rose Street				
Address 2					
City	Cerritos	State/Province	CA		
Postal Code	90703	Country ⁱ	US		
Inventor 3					Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Jeffrey	C.	Solomon		
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service					

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.045C3
		Application Number	
Title of Invention	Circuit Providing Load Isolation and Noise Reduction		

City	Irvine	State/Province	CA	Country of Residence i	US
-------------	--------	-----------------------	----	-------------------------------	----

Mailing Address of Inventor:

Address 1	16 Silver Fir				
Address 2					
City	Irvine	State/Province	CA		
Postal Code	92604	Country i	US		

Inventor 4

Remove

Legal Name

Prefix	Given Name	Middle Name	Family Name	Suffix
	Mario	Jesus	Martinez	

Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service

City	Laguna Niguel	State/Province	CA	Country of Residence i	US
-------------	---------------	-----------------------	----	-------------------------------	----

Mailing Address of Inventor:

Address 1	19 Lindall Street				
Address 2					
City	Laguna Niguel	State/Province	CA		
Postal Code	92677	Country i	US		

Inventor 5

Remove

Legal Name

Prefix	Given Name	Middle Name	Family Name	Suffix
	Chi-She		Chen	

Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service

City	Walnut	State/Province	CA	Country of Residence i	US
-------------	--------	-----------------------	----	-------------------------------	----

Mailing Address of Inventor:

Address 1	944 Crystal Water Lane				
Address 2					
City	Walnut	State/Province	CA		
Postal Code	91789	Country i	US		

All Inventors Must Be Listed - Additional Inventor Information blocks may be generated within this form by selecting the **Add** button.

Add

Correspondence Information:Enter either Customer Number or complete the Correspondence Information section below.
For further information see 37 CFR 1.33(a).☐ An Address is being provided for the correspondence Information of this application.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.045C3
		Application Number	
Title of Invention	Circuit Providing Load Isolation and Noise Reduction		
Customer Number	79141		
Email Address	jz@jzpatent.com	Add Email	Remove Email

Application Information:

Title of the Invention	Circuit Providing Load Isolation and Noise Reduction		
Attorney Docket Number	NETL.045C3	Small Entity Status Claimed	☒
Application Type	Nonprovisional		
Subject Matter	Utility		
Suggested Class (if any)		Sub Class (if any)	
Suggested Technology Center (if any)			
Total Number of Drawing Sheets (if any)	9	Suggested Figure for Publication (if any)	

Publication Information:

☐	Request Early Publication (Fee required at time of Request 37 CFR 1.219)
☒	Request Not to Publish. I hereby request that the attached application not be published under 35 U.S.C. 122(b) and certify that the invention disclosed in the attached application has not and will not be the subject of an application filed in another country, or under a multilateral international agreement, that requires publication at eighteen months after filing.

Representative Information:

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32). Either enter Customer Number or complete the Representative Name section below. If both sections are completed the customer Number will be used for the Representative Information during processing.			
Please Select One:	☒ Customer Number	☐ US Patent Practitioner	☐ Limited Recognition (37 CFR 11.9)
Customer Number	79141		

Domestic Benefit/National Stage Information:

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, or 365(c) or indicate National Stage entry from a PCT application. Providing this information in the application data sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78.			
Prior Application Status	Pending	Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)
	Continuation of	14324990	2014-07-07
Prior Application Status	Patented	Remove	

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.045C3
		Application Number	
Title of Invention	Circuit Providing Load Isolation and Noise Reduction		

Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)	Patent Number	Issue Date (YYYY-MM-DD)
14324990	Continuation of	13412243	2012-03-05	8782350	2014-07-15
Prior Application Status		Patented	Remove		
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)	Patent Number	Issue Date (YYYY-MM-DD)
13412243	non-provisional of Continuation of	12422853	2009-04-13	8154901	2012-04-10
Prior Application Status		Expired	Remove		
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)		
12422853	non provisional of	61044839	2008-04-14		
Prior Application Status		Expired	Remove		
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)		
12422853	non provisional of	61044825	2008-04-14		
Prior Application Status		Expired	Remove		
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)		
12422853	non provisional of	61044801	2008-04-14		
Additional Domestic Benefit/National Stage Data may be generated within this form by selecting the Add button. Add					

Foreign Priority Information:

This section allows for the applicant to claim benefit of foreign priority and to identify any prior foreign application for which priority is not claimed. Providing this information in the application data sheet constitutes the claim for priority as required by 35 U.S.C. 119(b) and 37 CFR 1.55(a).			
			Remove
Application Number	Country i	Filing Date (YYYY-MM-DD)	Priority Claimed
			☒ Yes ☐ No
Additional Foreign Priority Data may be generated within this form by selecting the Add button. Add			

Authorization to Permit Access:

☐ Authorization to Permit Access to the Instant Application by the Participating Offices

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.045C3
		Application Number	
Title of Invention	Circuit Providing Load Isolation and Noise Reduction		

If checked, the undersigned hereby grants the USPTO authority to provide the European Patent Office (EPO), the Japan Patent Office (JPO), the Korean Intellectual Property Office (KIPO), the World Intellectual Property Office (WIPO), and any other intellectual property offices in which a foreign application claiming priority to the instant patent application is filed access to the instant patent application. See 37 CFR 1.14(c) and (h). This box should not be checked if the applicant does not wish the EPO, JPO, KIPO, WIPO, or other intellectual property office in which a foreign application claiming priority to the instant patent application is filed to have access to the instant patent application.

In accordance with 37 CFR 1.14(h)(3), access will be provided to a copy of the instant patent application with respect to: 1) the instant patent application-as-filed; 2) any foreign application to which the instant patent application claims priority under 35 U.S.C. 119(a)-(d) if a copy of the foreign application that satisfies the certified copy requirement of 37 CFR 1.55 has been filed in the instant patent application; and 3) any U.S. application-as-filed from which benefit is sought in the instant patent application.

In accordance with 37 CFR 1.14(c), access may be provided to information concerning the date of filing this Authorization.

Applicant Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Applicant 1

[Remove](#)

If the applicant is the inventor (or the remaining joint inventor or inventors under 37 CFR 1.45), this section should not be completed. The information to be provided in this section is the name and address of the legal representative who is the applicant under 37 CFR 1.43; or the name and address of the assignee, person to whom the inventor is under an obligation to assign the invention, or person who otherwise shows sufficient proprietary interest in the matter who is the applicant under 37 CFR 1.46. If the applicant is an applicant under 37 CFR 1.46 (assignee, person to whom the inventor is obligated to assign, or person who otherwise shows sufficient proprietary interest) together with one or more joint inventors, then the joint inventor or inventors who are also the applicant should be identified in this section.

[Clear](#)

- ☒ Assignee
 ☐ Legal Representative under 35 U.S.C. 117
 ☐ Joint Inventor
- ☐ Person to whom the inventor is obligated to assign.
 ☐ Person who shows sufficient proprietary interest

If applicant is the legal representative, indicate the authority to file the patent application, the inventor is:

Name of the Deceased or Legally Incapacitated Inventor :

If the Applicant is an Organization check here. ☒

Organization Name Netlist, Inc.

Mailing Address Information:

Address 1	175 Technology, Suite 150		
Address 2			
City	Irvine	State/Province	CA
Country ⁱ	US	Postal Code	92618
Phone Number	949-435-0025	Fax Number	

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.045C3
		Application Number	
Title of Invention	Circuit Providing Load Isolation and Noise Reduction		
Email Address	nwhitley@netlist.com		
Additional Applicant Data may be generated within this form by selecting the Add button.			Add

Non-Applicant Assignee Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Assignee 1				
Complete this section only if non-applicant assignee information is desired to be included on the patent application publication in accordance with 37 CFR 1.215(b). Do not include in this section an applicant under 37 CFR 1.46 (assignee, person to whom the inventor is obligated to assign, or person who otherwise shows sufficient proprietary interest), as the patent application publication will include the name of the applicant(s).				
				Remove
If the Assignee is an Organization check here. ☐				
Prefix	Given Name	Middle Name	Family Name	Suffix
Mailing Address Information:				
Address 1				
Address 2				
City		State/Province		
Country i		Postal Code		
Phone Number		Fax Number		
Email Address				
Additional Assignee Data may be generated within this form by selecting the Add button.				Add

Signature:

NOTE: This form must be signed in accordance with 37 CFR 1.33. See 37 CFR 1.4 for signature requirements and certifications				
Signature	/Jamie J. Zheng/		Date (YYYY-MM-DD)	2017-10-18
First Name	Jamie	Last Name	Zheng	Registration Number
				51167
Additional Signature may be generated within this form by selecting the Add button.				Add

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.045C3
		Application Number	
Title of Invention	Circuit Providing Load Isolation and Noise Reduction		

This collection of information is required by 37 CFR 1.76. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 23 minutes to complete, including gathering, preparing, and submitting the completed application data sheet form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	14/715,491
				Filing Date	May 18, 2015
				First Named Inventor	Hyun Lee
				Art Unit	2827
				Examiner Name	Unknown
Sheet	1	of	3	Attorney Docket Number	NETL.045C3

US PATENT DOCUMENTS					
Examiner Initials *	Cite No. ¹	Document Number	Publication Date (MM-DD-YYYY)	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number-Kind Code ² (if known)			
	1.	2002/0038405 A1	03-28-2002	Leddige, M. et al.	
	2.	2009/0296503 A1	12-03-2009	Chu, M. et al.	
	3.	2011/0085408 A1	04-14-2011	Solomon, J. et al.	20110085406
	4.	2008/0162790 A1	07-03-2008	IM	

Change(s) applied
to document,
/M.H.E./
6/28/2017

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²
	5.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Related Matters, filed April 28, 2015	
	6.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Patent Owner Response to Petition, filed February 22, 2016	
	7.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Petitioner's Objections to Evidence, filed February 29, 2016	
	8.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Petitioner's Reply, filed May 19, 2016	
	9.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Petitioner Request for Oral Argument, filed June 2, 2016	
	10.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, PATENT OWNER MOTION FOR OBSERVATIONS, filed June 2, 2016	
	11.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, PATENT OWNER REQUEST FOR ORAL ARGUMENT, filed June 2, 2016	
	12.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, ORDER Trial Hearing, filed June 8, 2016	
	13.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Patent Owner's SUBMISSION ON PROPRIETY OF PETITIONER REPLY, filed June 9, 2016	
	14.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Petitioner's Response to Motion for Observations, filed June 10, 2016	
	15.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Petitioner's Response to Netlist's Submission, filed June 13, 2016	
	16.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Patent Owner's Objections to Petitioner's Demonstrative Exhibits, filed June 23, 2016	
	17.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Patent Owner's Demonstrative Exhibits, filed June 24, 2016	

EXAMINER /GURTEJ BANSAL/

DATE CONSIDERED 06/09/2017

* EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant. ¹ Applicant's unique citation designation number (optional). ² See Kinds Codes of USPTO Patents Documents at www.uspto.gov or MPEP 901.04. ³ Enter Office that issued the document, by the two-letter code (WIPO Standard ST. 3). ⁴ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁵ Kind of document by the appropriate symbols as indicated on the document and WIPO Standard ST. 16 if possible. ⁶ Applicant is to place a check mark here if English language Translation is attached.

ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /G.B./

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	14/715,491
				Filing Date	May 18, 2015
				First Named Inventor	Hyun Lee
				Art Unit	2827
Examiner Name	Unknown				
Sheet	1	of	3	Attorney Docket Number	NETL.045C3

US PATENT DOCUMENTS					
Examiner Initials *	Cite No. ¹	Document Number	Publication Date (MM-DD-YYYY)	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number-Kind Code ² (if known)			
	1.	4,739,473	1988-04-19	Ng	
	2.	5,313,624	1994-05-17	Harriman	
	3.	5,630,096	1997-05-13	Zuravleff	
	4.	5,905,401	1999-05-18	Sher	
	5.	6,061,754	2000-05-09	Cepulis	
	6.	6,141,245	2000-10-31	Bertin	
	7.	6,223,650	2001-05-15	Johnson	6,233,650
	8.	6,262,938	2001-07-17	Lee	
	9.	6,546,476	2003-04-08	Gillingham	
	10.	6,636,446	2003-10-21	Lee	
	11.	6,704,910	2004-03-09	Hong	
	12.	7,024,518	04-04-2006	Halbert et al	
	13.	7,191,302	2007-03-13	Usami	
	14.	7,225,303	2007-05-29	Choi	
	15.	7,730,254	2010-06-01	Risse	
	16.	7,990,746	2011-08-02	Rajan	
	17.	8,244,971	2012-08-24	Rajan	
	18.	8,335,894	2012-12-18	Rajan	
	19.	8,856,464	2014-10-07	Karamcheti	
	20.	20010008006	2001-07-12	Klein	
	21.	20030070052	2003-04-10	Lai	
	22.	20080025137	2008-01-31	Rajan et al.	
	23.	20090248969	10-01-2009	Wu, Larry	
	24.	20100125681	05-20-2010	Patel, Shwetal A.	

Change(s) applied
to document,
/M.H.E./
6/28/2017

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²
	25.	Inter Partes Review of U.S. Patent No. 7881150, Case No. IPR2015-01020, Record of Oral Hearing, dated July 26, 2016	

EXAMINER /GURTEJ BANSAL/

DATE CONSIDERED 06/09/2017

* EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant. ¹ Applicant's unique citation designation number (optional). ² See Kinds Codes of USPTO Patents Documents at www.uspto.gov or MPEP 901.04. ³ Enter Office that issued the document, by the two-letter code (WIPO Standard ST. 3). ⁴ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁵ Kind of document by the appropriate symbols as indicated on the document and WIPO Standard ST. 16 if possible. ⁶ Applicant is to place a check mark here if English language Translation is attached.

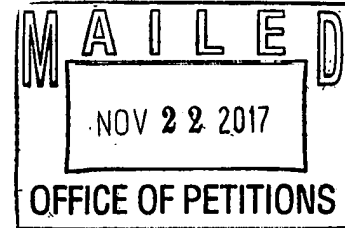
ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /G.B./



UNITED STATES PATENT AND TRADEMARK OFFICE

Commissioner for Patents
United States Patent and Trademark Office
P.O. Box 1450
Alexandria, VA 22313-1450
www.uspto.gov

Jamie J. Zheng, Ph.D Esq.
MASCHOFF BRENNAN
1389 Center Drive
Suite 300
Park City UT 84098



In re Application of :
Lee et al. :
Application No. 14/715,491 : ON PETITION
Filed: May 18, 2015 :
Atty. Docket No. N3044.10007US08 :

This is a decision on the constructive petition under 37 CFR 1.78(c) and (e), filed October 18, 2017, to accept an unintentionally delayed claim under 35 U.S.C. 119(e) and 120 for the benefit of priority to the prior-filed applications listed in the concurrently filed application data sheet (ADS).

The petition under 37 CFR 1.78(c) and (e) is **DISMISSED**.

A petition for acceptance of a claim for late priority under 37 CFR 1.78(c) and 1.78(e) is only applicable to those applications in which a proper benefit claim is filed after the expiration of the period specified in 37 CFR 1.78(a)(4) and 1.78(d)(3).

The petition under 37 CFR 1.78(c) and 1.78(e) must be accompanied by:

- (1) the reference required by 35 U.S.C. 120 and 119(e) and 37 CFR 1.78(d)(2) and 1.78(a)(3) of the prior-filed application, which must be filed in an Application Data Sheet¹;
- (2) the petition fee set forth in § 1.17(m); and
- (3) a statement that the entire delay between the date the claim was due under 37 CFR 1.78(d)(3) and 1.78(a)(4) and the date the claim was filed was unintentional. The Director may require additional where there is a question whether the delay was unintentional.

¹ 37 CFR 1.78(d)(2) explicitly requires the reference to be presented in an ADS.

The petition does not comply with (1) above. The ADS filed October 18, 2017, does not properly identify the information that is being changed.

37 CFR 1.76(b)(5) states:

(b) Bibliographic data. Bibliographic data as used in paragraph (a) of this section includes:

- (5) Domestic benefit information. This information includes the application number, the filing date, the status (including patent number if available), and relationship of each application for which a benefit is claimed under 35 U.S.C. 119(e), 120, 121, 365(c), or 386(c). Providing this information in the application data sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120 and §1.78.

37 CFR 1.76(c)(2) provides, in pertinent part:

An application data sheet providing corrected or updated information may include all of the sections listed in paragraph (b) of this section or only those sections containing changed or updated information. The application data sheet must include the section headings listed in paragraph (b) of this section for each section included in the application data sheet, and **must identify the information that is being changed, with underlining for insertions, and strike-through or brackets for text removed** ... (emphasis added).

MPEP 601.05(a) II.CORRECTING AND UPDATING AN ADS OR INFORMATION OTHERWISE OF RECORD states:

37 CFR 1.76(c) provides the procedure for correcting and updating not only an application data sheet (ADS), but also information otherwise of record (e.g., information provided on the most recent filing receipt). Any ADS filed after the filing date of the application is considered a corrected (or updated) ADS even if an ADS was not previously submitted. Such a corrected ADS must identify the information that is being changed with underlining for insertions and strike-through or brackets for text removed, except that identification of information being changed is not required for an ADS included with an initial submission under 35 U.S.C. 371. **In general, the identification of the information being changed should be made relative to the most recent filing receipt...** (emphasis added).

A review of the May 29, 2015 filing receipt reveals the current domestic applications for which benefit is claimed. Any changes to the domestic priority data of record must be made in compliance with 37 CFR 1.76(c). Petitioner has attempted to make changes relative to the contents of a prior filed ADS, rather than relative to the information of record as listed in the most recent filing receipt.

There should be no strike-through information pertaining to Prior Application No. 12/422,853, because there is no information of record to be removed.

The issue fee was paid on September 15, 2017. 37 CFR 1.312 no longer permits filing an amendment after the date the issue fee has been paid.

If reconsideration of this decision is filed after issuance, a certificate of correction with \$100 fee, a renewed petition under 37 CFR 1.78(c) and (e), and an Application Data Sheet (complying with the provisions 37 CFR 1.76(b)(5)) or a Corrected WebADS to correct the above matters are required.

By mail: Mail Stop PETITIONS
 Commissioner for Patents
 Post Office Box 1450
 Alexandria, VA 22313-1450

By fax: (571) 273-8300
ATTN: Office of Petitions

By internet: EFS-Web

Any questions concerning this matter may be directed to the undersigned at (571) 272-3230.

A handwritten signature in black ink, reading "Shirene Willis Brantley". The signature is written in a cursive, flowing style.

Shirene Willis Brantley
Attorney Advisor
Office of Petitions



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	ISSUE DATE	PATENT NO.	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/715,491	12/19/2017	9846659	N3044.10007US08	8143

79141 7590 11/29/2017
Jamie J. Zheng, Ph.D Esq.
MASCHOFF BRENNAN
1389 Center Drive
Suite 300
Park City, UT 84098

ISSUE NOTIFICATION

The projected patent number and issue date are specified above.

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment is 275 day(s). Any patent to issue from the above-identified application will include an indication of the adjustment on the front page.

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Application Information Retrieval (PAIR) WEB site (<http://pair.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Application Assistance Unit (AAU) of the Office of Data Management (ODM) at (571)-272-4200.

APPLICANT(s) (Please see PAIR WEB site <http://pair.uspto.gov> for additional applicants):

Hyun Lee, Ladera Ranch, CA;
Netlist, Inc., Irvine, CA;
Jayesh R. Bhakta, Cerritos, CA;
Jeffrey C. Solomon, Irvine, CA;
Mario Jesus Martinez, Laguna Niguel, CA;
Chi-She Chen, Walnut, CA;

The United States represents the largest, most dynamic marketplace in the world and is an unparalleled location for business investment, innovation, and commercialization of new technologies. The USA offers tremendous resources and advantages for those who invest and manufacture goods here. Through SelectUSA, our nation works to encourage and facilitate business investment. To learn more about why the USA is the best country in the world to develop technology, manufacture products, and grow your business, visit SelectUSA.gov.

REQUEST FOR CONTINUED EXAMINATION(RCE)TRANSMITTAL **(Submitted Only via EFS-Web)**

Application Number	14715491	Filing Date	2015-05-18	Docket Number (if applicable)	N3044.10007US08	Art Unit	2139
First Named Inventor	Hyun Lee			Examiner Name	BANSAL, GURTEJ		

This is a Request for Continued Examination (RCE) under 37 CFR 1.114 of the above-identified application.
 Request for Continued Examination (RCE) practice under 37 CFR 1.114 does not apply to any utility or plant application filed prior to June 8, 1995, to any international application that does not comply with the requirements of 35 U.S.C. 371, or to any design application. The Instruction Sheet for this form is located at WWW.USPTO.GOV.

SUBMISSION REQUIRED UNDER 37 CFR 1.114

Note: If the RCE is proper, any previously filed unentered amendments and amendments enclosed with the RCE will be entered in the order in which they were filed unless applicant instructs otherwise. If applicant does not wish to have any previously filed unentered amendment(s) entered, applicant must request non-entry of such amendment(s).

☐ Previously submitted. If a final Office action is outstanding, any amendments filed after the final Office action may be considered as a submission even if this box is not checked.

☐ Consider the arguments in the Appeal Brief or Reply Brief previously filed on _____

☐ Other _____

☒ Enclosed

☐ Amendment/Reply

☐ Information Disclosure Statement (IDS)

☐ Affidavit(s)/ Declaration(s)

☒ Other

Request To Correct Benefit Claim

MISCELLANEOUS

☐ Suspension of action on the above-identified application is requested under 37 CFR 1.103(c) for a period of months _____
 (Period of suspension shall not exceed 3 months; Fee under 37 CFR 1.17(i) required)

☐ Other _____

FEES

The RCE fee under 37 CFR 1.17(e) is required by 37 CFR 1.114 when the RCE is filed.

☒ The Director is hereby authorized to charge any underpayment of fees, or credit any overpayments, to
 Deposit Account No 505394

SIGNATURE OF APPLICANT, ATTORNEY, OR AGENT REQUIRED

☒ Patent Practitioner Signature

Applicant Signature

Signature of Registered U.S. Patent Practitioner			
Signature	JAMIE J. ZHENG/	Date (YYYY-MM-DD)	2017-12-15
Name	Jamie Zheng	Registration Number	51167

This collection of information is required by 37 CFR 1.114. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.045C3
		Application Number	
Title of Invention	Circuit Providing Load Isolation and Noise Reduction		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76. This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.			

Secrecy Order 37 CFR 5.2

☐ Portions or all of the application associated with this Application Data Sheet may fall under a Secrecy Order pursuant to 37 CFR 5.2 (Paper filers only. Applications that fall under Secrecy Order may not be filed electronically.)
--

Inventor Information:

Inventor 1					Remove	
Legal Name						
Prefix	Given Name	Middle Name	Family Name	Suffix		
	Hyun		Lee			
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service						
City	Ladera Ranch	State/Province	CA	Country of Residence ⁱ	US	
Mailing Address of Inventor:						
Address 1		21 Thalia				
Address 2						
City	Ladera Ranch	State/Province	CA			
Postal Code	92694	Country ⁱ	US			
Inventor 2					Remove	
Legal Name						
Prefix	Given Name	Middle Name	Family Name	Suffix		
	Jayesh	R.	Bhakta			
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service						
City	Cerritos	State/Province	CA	Country of Residence ⁱ	US	
Mailing Address of Inventor:						
Address 1		12220 Rose Street				
Address 2						
City	Cerritos	State/Province	CA			
Postal Code	90703	Country ⁱ	US			
Inventor 3					Remove	
Legal Name						
Prefix	Given Name	Middle Name	Family Name	Suffix		
	Jeffrey	C.	Solomon			
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service						

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.045C3
		Application Number	
Title of Invention	Circuit Providing Load Isolation and Noise Reduction		

City	Irvine	State/Province	CA	Country of Residence i	US
-------------	--------	-----------------------	----	-------------------------------	----

Mailing Address of Inventor:

Address 1	16 Silver Fir				
Address 2					
City	Irvine	State/Province	CA		
Postal Code	92604	Country i	US		

Inventor 4[Remove](#)**Legal Name**

Prefix	Given Name	Middle Name	Family Name	Suffix
	Mario	Jesus	Martinez	

Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service

City	Laguna Niguel	State/Province	CA	Country of Residence i	US
-------------	---------------	-----------------------	----	-------------------------------	----

Mailing Address of Inventor:

Address 1	19 Lindall Street				
Address 2					
City	Laguna Niguel	State/Province	CA		
Postal Code	92677	Country i	US		

Inventor 5[Remove](#)**Legal Name**

Prefix	Given Name	Middle Name	Family Name	Suffix
	Chi-She		Chen	

Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service

City	Walnut	State/Province	CA	Country of Residence i	US
-------------	--------	-----------------------	----	-------------------------------	----

Mailing Address of Inventor:

Address 1	944 Crystal Water Lane				
Address 2					
City	Walnut	State/Province	CA		
Postal Code	91789	Country i	US		

All Inventors Must Be Listed - Additional Inventor Information blocks may be generated within this form by selecting the **Add** button.[Add](#)**Correspondence Information:**Enter either Customer Number or complete the Correspondence Information section below.
For further information see 37 CFR 1.33(a).☐ An Address is being provided for the correspondence Information of this application.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.045C3
		Application Number	
Title of Invention	Circuit Providing Load Isolation and Noise Reduction		
Customer Number	79141		
Email Address	jz@jzpatent.com	Add Email	Remove Email

Application Information:

Title of the Invention	Circuit Providing Load Isolation and Noise Reduction		
Attorney Docket Number	NETL.045C3	Small Entity Status Claimed	☒
Application Type	Nonprovisional		
Subject Matter	Utility		
Suggested Class (if any)		Sub Class (if any)	
Suggested Technology Center (if any)			
Total Number of Drawing Sheets (if any)	9	Suggested Figure for Publication (if any)	

Publication Information:

☐	Request Early Publication (Fee required at time of Request 37 CFR 1.219)
☒	Request Not to Publish. I hereby request that the attached application not be published under 35 U.S.C. 122(b) and certify that the invention disclosed in the attached application has not and will not be the subject of an application filed in another country, or under a multilateral international agreement, that requires publication at eighteen months after filing.

Representative Information:

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32). Either enter Customer Number or complete the Representative Name section below. If both sections are completed the customer Number will be used for the Representative Information during processing.			
Please Select One:	☒ Customer Number	☐ US Patent Practitioner	☐ Limited Recognition (37 CFR 11.9)
Customer Number	79141		

Domestic Benefit/National Stage Information:

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, or 365(c) or indicate National Stage entry from a PCT application. Providing this information in the application data sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78.			
Prior Application Status	Pending	Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)
	Continuation of	14324990	2014-07-07
Prior Application Status	Patented	Remove	

Application Data Sheet 37 CFR 1.76		Attorney Docket Number		NETL.045C3	
		Application Number			
Title of Invention		Circuit Providing Load Isolation and Noise Reduction			

Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)	Patent Number	Issue Date (YYYY-MM-DD)
14324990	Continuation of	13412243	2012-03-05	8782350	2014-07-15
Prior Application Status		Patented		Remove	

Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)	Patent Number	Issue Date (YYYY-MM-DD)
13412243	Continuation of	12422853	2009-04-13	8154901	2012-04-10
Prior Application Status		Expired		Remove	

Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)
12422853	non provisional of	61044839	2008-04-14
Prior Application Status		Remove	

Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)
12422853	non provisional of	61044825	2008-04-14
Prior Application Status		Remove	

Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)
12422853	non provisional of	61044801	2008-04-14
Prior Application Status		Remove	

Additional Domestic Benefit/National Stage Data may be generated within this form by selecting the **Add** button.
 [Add](#)

Foreign Priority Information:

This section allows for the applicant to claim benefit of foreign priority and to identify any prior foreign application for which priority is not claimed. Providing this information in the application data sheet constitutes the claim for priority as required by 35 U.S.C. 119(b) and 37 CFR 1.55(a).

Remove			
Application Number	Country ⁱ	Filing Date (YYYY-MM-DD)	Priority Claimed
			☒ Yes ☐ No

Additional Foreign Priority Data may be generated within this form by selecting the **Add** button.
 [Add](#)

Authorization to Permit Access:

☐ Authorization to Permit Access to the Instant Application by the Participating Offices

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.045C3
		Application Number	
Title of Invention	Circuit Providing Load Isolation and Noise Reduction		

If checked, the undersigned hereby grants the USPTO authority to provide the European Patent Office (EPO), the Japan Patent Office (JPO), the Korean Intellectual Property Office (KIPO), the World Intellectual Property Office (WIPO), and any other intellectual property offices in which a foreign application claiming priority to the instant patent application is filed access to the instant patent application. See 37 CFR 1.14(c) and (h). This box should not be checked if the applicant does not wish the EPO, JPO, KIPO, WIPO, or other intellectual property office in which a foreign application claiming priority to the instant patent application is filed to have access to the instant patent application.

In accordance with 37 CFR 1.14(h)(3), access will be provided to a copy of the instant patent application with respect to: 1) the instant patent application-as-filed; 2) any foreign application to which the instant patent application claims priority under 35 U.S.C. 119(a)-(d) if a copy of the foreign application that satisfies the certified copy requirement of 37 CFR 1.55 has been filed in the instant patent application; and 3) any U.S. application-as-filed from which benefit is sought in the instant patent application.

In accordance with 37 CFR 1.14(c), access may be provided to information concerning the date of filing this Authorization.

Applicant Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.			
Applicant 1			Remove
If the applicant is the inventor (or the remaining joint inventor or inventors under 37 CFR 1.45), this section should not be completed. The information to be provided in this section is the name and address of the legal representative who is the applicant under 37 CFR 1.43; or the name and address of the assignee, person to whom the inventor is under an obligation to assign the invention, or person who otherwise shows sufficient proprietary interest in the matter who is the applicant under 37 CFR 1.46. If the applicant is an applicant under 37 CFR 1.46 (assignee, person to whom the inventor is obligated to assign, or person who otherwise shows sufficient proprietary interest) together with one or more joint inventors, then the joint inventor or inventors who are also the applicant should be identified in this section.			
Clear			
☒ Assignee	☐ Legal Representative under 35 U.S.C. 117	☐ Joint Inventor	
☐ Person to whom the inventor is obligated to assign.		☐ Person who shows sufficient proprietary interest	
If applicant is the legal representative, indicate the authority to file the patent application, the inventor is:			
Name of the Deceased or Legally Incapacitated Inventor :			
If the Applicant is an Organization check here. ☒			
Organization Name	Netlist, Inc.		
Mailing Address Information:			
Address 1	175 Technology, Suite 150		
Address 2			
City	Irvine	State/Province	CA
Country ⁱ	US	Postal Code	92618
Phone Number	949-435-0025	Fax Number	

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.045C3
		Application Number	
Title of Invention	Circuit Providing Load Isolation and Noise Reduction		
Email Address	nwhitley@netlist.com		
Additional Applicant Data may be generated within this form by selecting the Add button.			Add

Non-Applicant Assignee Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Assignee 1				
Complete this section only if non-applicant assignee information is desired to be included on the patent application publication in accordance with 37 CFR 1.215(b). Do not include in this section an applicant under 37 CFR 1.46 (assignee, person to whom the inventor is obligated to assign, or person who otherwise shows sufficient proprietary interest), as the patent application publication will include the name of the applicant(s).				
				Remove
If the Assignee is an Organization check here. ☐				
Prefix	Given Name	Middle Name	Family Name	Suffix
Mailing Address Information:				
Address 1				
Address 2				
City		State/Province		
Country i		Postal Code		
Phone Number		Fax Number		
Email Address				
Additional Assignee Data may be generated within this form by selecting the Add button.				Add

Signature:

NOTE: This form must be signed in accordance with 37 CFR 1.33. See 37 CFR 1.4 for signature requirements and certifications				
Signature	/Jamie J. Zheng/		Date (YYYY-MM-DD)	12-15-2017
First Name	Jamie	Last Name	Zheng	Registration Number 51167
Additional Signature may be generated within this form by selecting the Add button.				Add

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.045C3
		Application Number	
Title of Invention	Circuit Providing Load Isolation and Noise Reduction		

This collection of information is required by 37 CFR 1.76. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 23 minutes to complete, including gathering, preparing, and submitting the completed application data sheet form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

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1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Doc Code: PET.AUTO Document Description: Petition automatically granted by EFS-Web		PTO/SB/140 U.S. Patent and Trademark Office Department of Commerce	
Electronic Petition Request	PETITION TO WITHDRAW AN APPLICATION FROM ISSUE AFTER PAYMENT OF THE ISSUE FEE UNDER 37 CFR 1.313(c)		
Application Number	14715491		
Filing Date	18-May-2015		
First Named Inventor	Hyun Lee		
Art Unit	2139		
Examiner Name	GURTEJ BANSAL		
Attorney Docket Number	N3044.10007US08		
Title	MEMORY MODULE AND CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION		
An application may be withdrawn from issue for further action upon petition by the applicant. To request that the Office withdraw an application from issue, applicant must file a petition under this section including the fee set forth in § 1.17(h) and a showing of good and sufficient reasons why withdrawal of the application from issue is necessary. APPLICANT HEREBY PETITIONS TO WITHDRAW THIS APPLICATION FROM ISSUE UNDER 37 CFR 1.313(c). A grantable petition requires the following items: (1) Petition fee; and (2) One of the following reasons: (a) Unpatentability of one or more claims, which must be accompanied by an unequivocal statement that one or more claims are unpatentable, an amendment to such claim or claims, and an explanation as to how the amendment causes such claim or claims to be patentable; (b) Consideration of a request for continued examination in compliance with § 1.114 (for a utility or plant application only); or (c) Express abandonment of the application. Such express abandonment may be in favor of a continuing application, but not a CPA under 37 CFR 1.53(d).			
Petition Fee			
☐ Small Entity			
☐ Micro Entity			
☒ Regular Undiscounted			
Reason for withdrawal from issue			

- ☐ One or more claims are unpatentable
- ☒ Consideration of a request for continued examination (RCE) (List of Required Documents and Fees)
- ☐ Applicant hereby expressly abandons the instant application (any attorney/agent signing for this reason must have power of attorney pursuant to 37 CFR 1.32(b)).

RCE request, submission, and fee.

- ☐ I certify, in accordance with 37 CFR 1.4(d)(4) that :
- ☐ The RCE request ,submission, and fee have already been filed in the above-identified application on
- ☒ Are attached.

THIS PORTION MUST BE COMPLETED BY THE SIGNATORY OR SIGNATORIES

I certify, in accordance with 37 CFR 1.4(d)(4) that I am:

- ☐ An attorney or agent registered to practice before the Patent and Trademark Office who has been given power of attorney in this application.
- ☒ An attorney or agent registered to practice before the Patent and Trademark Office, acting in a representative capacity.
- ☐ A sole inventor
- ☐ A joint inventor; I certify that I am authorized to sign this submission on behalf of all of the inventors as evidenced by the power of attorney in the application
- ☐ A joint inventor; all of whom are signing this e-petition

Signature	/Jamie J. Zheng/
Name	Jamie J. Zheng
Registration Number	51167

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re application of	Hyun Lee	Art Unit 2139
Application No.:	14/715,491	
Filed:	May 18, 2015	
Conf. No.:	8143	
For:	MEMORY MODULE AND CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION	
Examiner:	BANSAL, GURTEJ	
Customer No.:	79141	
Attorney Docket No.:	N3044.10007US08	

**RENEWED PETITION FOR ACCEPTANCE OF UNINTENTIONALLY DELAYED
DOMESTIC BENEFIT CLAIM UNDER 37 CFR 1.78**

Mail Stop Petitions
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Dear Commissioner,

In response to the Decision dated November 22, 2017, on the Petition for Unintentionally Delayed Domestic Benefit Claim under 37 CFR 1.78, filed on October 18, 2017 (“Original Petition”), Applicants respectfully request reconsideration of the Original Petition with the submission of this Renewed Petition for Acceptance of Unintentionally Delayed Domestic Benefit Claim under 37 CFR 1.78(c) and 1.78(e) (“Renewed Petition”). Applicants hereby state that:

The entire delay between the date the benefit claim was due under 37 CFR 1.78(d)(3) and 1.78(a)(4) and the date the benefit claim was filed was unintentional.

The above-referenced patent application was filed on May 18, 2015, and claims priority to the following:

This application is a CON of 14/324,990 07/07/2014 PAT 9037809
which is a CON of 13/412,243 03/05/2012 PAT 8782350
which is a CON of 12/422,853 04/13/2009 PAT 8154901
which claims benefit of 61/044,825 04/14/2008
and claims benefit of 61/044,801 04/14/2008
and claims benefit of 61/044,839 04/14/2008

The above listing of priority data can be found in paragraph [001] of the originally filed specification, filed on May 18, 2015. The originally filed ADS also included the above listing of priority data except a clerical error in the Continuity Type for Application No. 13/412,243. This unintentional error in the originally filed ADS resulted in the incorrect priority claim in the above-referenced patent application, which was not discovered until recently. Thus, the entire delay between the date the benefit claim was due under 37 CFR 1.78(d)(3) and 1.78(a)(4) and the date the benefit claim was filed was unintentional.

This Renewed Petition is accompanied by a marked-up Supplemental ADS. The Supplemental ADS has been corrected to indicate that Application no. 13/412,243 is a Continuation of 12/422,853, filed 4/13/2009 and patented 4/10/2014 as US 8154901, which is a non-provisional of 61/044,839 filed 04/14/2008, 61/044,825 filed 04/14/2008, and 61/044,801 filed 04/14/2008.

Since this Renewed Petition is filed after payment of issue fee, it is also accompanied by a petition to withdraw the above-referenced patent application from issue.

The petition fee under 37 CFR 1.17(m) was submitted with the Original Petition. The USPTO is authorized to charge any additional fees required for this petition to Deposit Account No. 50-5394.

Dated this 15th day of December, 2017.

Respectfully submitted,

/Jamie J. Zheng, Reg. no. 51167/

JAMIE J. ZHENG
Registration No. 51,167
Attorney for Applicant
Telephone No. (650) 521-4828

Electronic Patent Application Fee Transmittal

Application Number:	14715491			
Filing Date:	18-May-2015			
Title of Invention:	MEMORY MODULE AND CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Jamie Jie Zheng/Brian Gonzalez			
Attorney Docket Number:	N3044.10007US08			
Filed as Large Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
PETITION FEE- 37 CFR 1.17(H) (GROUP III)	1464	1	140	140
RCE- 1st Request	1801	1	1200	1200
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				1340



UNITED STATES PATENT AND TRADEMARK OFFICE

Commissioner for Patents
United States Patent and Trademark Office
P.O. Box 1450
Alexandria, VA 22313-1450
www.uspto.gov

Decision Date : December 15, 2017

In re Application of :

Hyun Lee

DECISION ON PETITION

UNDER CFR 1.313(c)(2)

Application No : 14715491

Filed : 18-May-2015

Attorney Docket No : N3044.10007US08

This is an electronic decision on the petition under 37 CFR 1.313(c)(2), filed December 15, 2017 to withdraw the above-identified application from issue after payment of the issue fee.

The petition is **GRANTED**.

The above-identified application is withdrawn from issue for consideration of a submission under 37 CFR 1.114 (request for continued examination). See 37 CFR 1.313(c)(2).

Petitioner is advised that the issue fee paid in this application cannot be refunded. If, however, this application is again allowed, petitioner may request that it be applied towards the issue fee required by the new Notice of Allowance.

Telephone inquiries concerning this decision should be directed to the Patent Electronic Business Center (EBC) at 866-217-9197.

This application file is being referred to Technology Center AU 2139 for processing of the request for continuing examination under 37 CFR 1.114.

Office of Petitions

Electronic Acknowledgement Receipt

EFS ID:	31246525
Application Number:	14715491
International Application Number:	
Confirmation Number:	8143
Title of Invention:	MEMORY MODULE AND CIRCUIT PROVIDING LOAD ISOLTION AND NOISE REDUCTION
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng/Brian Gonzalez
Filer Authorized By:	Jamie Jie Zheng
Attorney Docket Number:	N3044.10007US08
Receipt Date:	15-DEC-2017
Filing Date:	18-MAY-2015
Time Stamp:	18:46:15
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	CARD
Payment was successfully received in RAM	\$ 1340
RAM confirmation Number	121817INTEFSW18461000
Deposit Account	505394
Authorized User	Brian Gonzalez

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:
37 CFR 1.21 (Miscellaneous fees and charges)

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Request for Continued Examination (RCE)	2017-12-15_RCE_N3044-10007 US08.pdf	1349928	no	3
			7e4220f5a90459e21153780d7e68c144d0cb3b10		
Warnings:					
Information:					
2	Transmittal Letter	2017-12-15_ADS_N3044-10007 US08.pdf	375876	no	8
			bf14f9663a1b56b777f69434052c90ee5fafa6a		
Warnings:					
Information:					
3	Petition automatically granted by EFS	petition-request.pdf	31628	no	2
			60506641d8b7905ea40eeb9fb7c4276c30ed435b		
Warnings:					
Information:					
4	Transmittal Letter	2017-12-15_Request-for-Reconsideration_N3044-10007 US08.pdf	239414	no	2
			04b948b7772e2d968c4e1bf4a06a4b3815d49e79		
Warnings:					
Information:					
5	Fee Worksheet (SB06)	fee-info.pdf	32300	no	2
			eea078369d47c6220f9b402a8c382f303c4237cb		
Warnings:					
Information:					
Total Files Size (in bytes):			2029146		

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



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APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
14/715,491	05/18/2015	2139	800	N3044.10007US08	20	2

CONFIRMATION NO. 8143

CORRECTED FILING RECEIPT



79141

Jamie J. Zheng, Ph.D Esq.
MASCHOFF BRENNAN
1389 Center Drive
Suite 300
Park City, UT 84098

Date Mailed: 01/19/2018

Receipt is acknowledged of this non-provisional patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Inventor(s)

Hyun Lee, Ladera Ranch, CA;
Jayesh R. Bhakta, Cerritos, CA;
Jeffrey C. Solomon, Irvine, CA;
Mario Jesus Martinez, Laguna Niguel, CA;
Chi-She Chen, Walnut, CA;

Applicant(s)

Netlist, Inc., Irvine, CA;

Power of Attorney: None

Domestic Priority data as claimed by applicant

This application is a CON of 14/324,990 07/07/2014 PAT 9037809
which is a CON of 13/412,243 03/05/2012 PAT 8782350
which is a CON of 12/422,853 04/13/2009 PAT 8154901
which claims benefit of 61/044,839 04/14/2008
and claims benefit of 61/044,825 04/14/2008
and claims benefit of 61/044,801 04/14/2008

Foreign Applications for which priority is claimed (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.) - None.

Foreign application information must be provided in an Application Data Sheet in order to constitute a claim to foreign priority. See 37 CFR 1.55 and 1.76.

Permission to Access Application via Priority Document Exchange: No

Permission to Access Search Results: No

Applicant may provide or rescind an authorization for access using Form PTO/SB/39 or Form PTO/SB/69 as appropriate.

If Required, Foreign Filing License Granted: 05/27/2015

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 14/715,491**

Projected Publication Date: Request for Non-Publication Acknowledged

Non-Publication Request: Yes

Early Publication Request: No

Title

MEMORY MODULE AND CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION

Preliminary Class

711

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

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For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific

countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4258).

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Title 35, United States Code, Section 184
Title 37, Code of Federal Regulations, 5.11 & 5.15

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This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).

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technology, manufacture products, deliver services, and grow your business, visit <http://www.SelectUSA.gov> or call +1-202-482-6800.



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Bib Data Sheet

CONFIRMATION NO. 8143

SERIAL NUMBER 14/715,491	FILING OR 371(c) DATE 05/18/2015 RULE	CLASS 711	GROUP ART UNIT 2139	ATTORNEY DOCKET NO. N3044.10007US08
AIA (First Inventor to File): YES				
INVENTORS Hyun Lee, Ladera Ranch, CA; Jayesh R. Bhakta, Cerritos, CA; Jeffrey C. Solomon, Irvine, CA; Mario Jesus Martinez, Laguna Niguel, CA; Chi-She Chen, Walnut, CA;				
APPLICANTS Netlist, Inc., Irvine, CA;				
** CONTINUING DATA ***** This application is a CON of 14/324,990 07/07/2014 PAT 9037809 which is a CON of 13/412,243 03/05/2012 PAT 8782350 which is a CON of 12/422,853 04/13/2009 PAT 8154901 which claims benefit of 61/044,839 04/14/2008 and claims benefit of 61/044,825 04/14/2008 and claims benefit of 61/044,801 04/14/2008				
** FOREIGN APPLICATIONS *****				
IF REQUIRED, FOREIGN FILING LICENSE GRANTED ** 05/27/2015				
Foreign Priority claimed ☐ yes ☐ no 35 USC 119 (a-d) conditions ☐ yes ☐ no ☐ Met after met Allowance Verified and Acknowledged		STATE OR COUNTRY CA	SHEETS DRAWING 9	TOTAL CLAIMS 20
Examiner's Signature _____ Initials _____				INDEPENDENT CLAIMS 2
ADDRESS 79141				
TITLE MEMORY MODULE AND CIRCUIT PROVIDING LOAD ISOLTION AND NOISE REDUCTION				

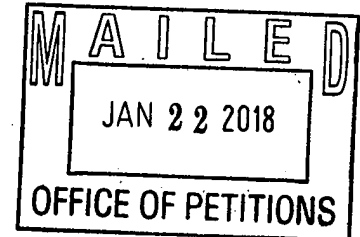
FILING FEE RECEIVED 800	FEES: Authority has been given in Paper No. _____ to charge/credit DEPOSIT ACCOUNT No. _____ for following:	<table border="1"><tr><td>☐ All Fees</td></tr><tr><td>☐ 1.16 Fees (Filing)</td></tr><tr><td>☐ 1.17 Fees (Processing Ext. of time)</td></tr><tr><td>☐ 1.18 Fees (Issue)</td></tr><tr><td>☐ Other _____</td></tr><tr><td>☐ Credit</td></tr></table>	☐ All Fees	☐ 1.16 Fees (Filing)	☐ 1.17 Fees (Processing Ext. of time)	☐ 1.18 Fees (Issue)	☐ Other _____	☐ Credit
☐ All Fees								
☐ 1.16 Fees (Filing)								
☐ 1.17 Fees (Processing Ext. of time)								
☐ 1.18 Fees (Issue)								
☐ Other _____								
☐ Credit								



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Jamie J. Zheng, Ph.D Esq.
MASCHOFF BRENNAN
1389 Center Drive
Suite 300
Park City UT 84098



In re Application of :
Lee et al. :
Application No. 14/715,491 : ON PETITION
Filed: May 18, 2015 :
Atty. Docket No. N3044.10007US08 :

This is a decision on the renewed petition under 37 CFR 1.78(c) and (e), filed December 15, 2017, to accept an unintentionally delayed claim under 35 U.S.C. 119(e) and 120 for the benefit of priority to the prior-filed applications listed in the concurrently filed application data sheet (ADS).

The petition under 37 CFR 1.78(c) and (e) is **GRANTED**.

A petition for acceptance of a claim for late priority under 37 CFR 1.78(c) and 1.78(e) is only applicable to those applications in which a proper benefit claim is filed after the expiration of the period specified in 37 CFR 1.78(a)(4) and 1.78(d)(3).

The petition under 37 CFR 1.78(c) and 1.78(e) must be accompanied by:

- (1) the reference required by 35 U.S.C. 120 and 119(e) and 37 CFR 1.78(d)(2) and 1.78(a)(3) of the prior-filed application, which must be filed in an Application Data Sheet¹;
- (2) the petition fee set forth in § 1.17(m); and
- (3) a statement that the entire delay between the date the claim was due under 37 CFR 1.78(d)(3) and 1.78(a)(4) and the date the claim was filed was unintentional. The Director may require additional where there is a question whether the delay was unintentional.

¹ 37 CFR 1.78(d)(2) explicitly requires the reference to be presented in an ADS.

A nonprovisional application in the priority chain claiming the benefit of each prior-filed provisional application must have been filed within twelve months of the filing date of the prior-filed provisional application.

All the above requirements having been satisfied, the late claim for benefit of priority under 35 U.S.C. 120 and 119(e) is accepted as being unintentionally delayed.

Petitioner is advised that this decision grants the petition to accept the unintentionally delayed domestic benefit claim to the prior filed applications because the petition requirements of 37 CFR 1.78(c) and (e) and the formal requirements for claiming domestic benefit (see MPEP 211.01 et. seq.) have been met. This acceptance should not be construed as meaning that this application is entitled to the benefit of the prior-filed applications. Whether a claimed invention in a nonprovisional application is entitled to the benefit of the filing date of a prior-filed application on the basis of the disclosure thereof is determined during examination if it becomes necessary to do so (e.g., intervening reference and interference proceeding). See MPEP 211.05.

A corrected Filing Receipt, which includes the priority claim to the prior-filed applications, accompanies this decision on petition.

Any questions concerning this decision may be directed to the undersigned at (571) 272-3230. All other inquiries concerning either the examination procedures or status of the application should be directed to the Technology Center.

This application is being forwarded to Technology Center Art Unit 2139 for consideration by the examiner of the claim under 35 U.S.C. 120 and 119(e) of the prior-filed nonprovisional and provisional applications.



Shirene Willis Brantley
Attorney Advisor
Office of Petitions

Enclosure: Corrected Filing Receipt



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United States Patent and Trademark Office
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APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
14/715,491	05/18/2015	2139	800	N3044.10007US08	20	2

CONFIRMATION NO. 8143

CORRECTED FILING RECEIPT



0000000098795807

79141

Jamie J. Zheng, Ph.D Esq.
MASCHOFF BRENNAN
1389 Center Drive
Suite 300
Park City, UT 84098

Date Mailed: 01/19/2018

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Inventor(s)

Hyun Lee, Ladera Ranch, CA;
Jayesh R. Bhakta, Cerritos, CA;
Jeffrey C. Solomon, Irvine, CA;
Mario Jesus Martinez, Laguna Niguel, CA;
Chi-She Chen, Walnut, CA;

Applicant(s)

Netlist, Inc., Irvine, CA;

Power of Attorney: None

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which is a CON of 13/412,243 03/05/2012 PAT 8782350
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Foreign application information must be provided in an Application Data Sheet in order to constitute a claim to foreign priority. See 37 CFR 1.55 and 1.76.

Permission to Access Application via Priority Document Exchange: No

Permission to Access Search Results: No

Applicant may provide or rescind an authorization for access using Form PTO/SB/39 or Form PTO/SB/69 as appropriate.

If Required, Foreign Filing License Granted: 05/27/2015

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 14/715,491**

Projected Publication Date: Request for Non-Publication Acknowledged

Non-Publication Request: Yes

Early Publication Request: No

Title

MEMORY MODULE AND CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION

Preliminary Class

711

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

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Title 37, Code of Federal Regulations, 5.11 & 5.15

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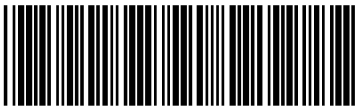
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technology, manufacture products, deliver services, and grow your business, visit <http://www.SelectUSA.gov> or call +1-202-482-6800.

Search Notes 	Application/Control No. 14/715,491	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner GURTEJ BANSAL	Art Unit 2139

CPC - Searched*		
Symbol	Date	Examiner
G06F12/1684.cpc. G11C5/02.cpc. G11C5/06.cpc. G11C8/18.cpc. G06F13/4027.cpc.	06/09/2017	GB

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

Search Notes		
Search Notes	Date	Examiner
Text Search in EAST (See Attached)	06/09/2017	GB
Inventor Search in PE2E	06/09/2017	GB
NPL Search in Google Scholar (See Attached)	06/09/2017	GB
Updated Text Search in EAST (See Attached)	02/02/2018	GB
Updated NPL Search in Google Scholar (See Attached)	02/02/2018	GB

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner
	G06F12/1684.cpc. G11C5/02.cpc. G11C5/06.cpc. G11C8/18.cpc. G06F13/4027.cpc.	06/09/2017	GB
	G06F12/1684.cpc. G11C5/02.cpc. G11C5/06.cpc. G11C8/18.cpc. G06F13/4027.cpc.	02/02/2018	GB

/GURTEJ BANSAL/ Primary Examiner.Art Unit 2139	
--	--

Issue Classification 	Application/Control No. 14/715,491	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner GURTEJ BANSAL	Art Unit 2139

CPC						
Symbol					Type	Version
G06F	/	13	/	1684	F	2013-01-01
G11C	/	5	/	02	I	2013-01-01
G11C	/	5	/	06	I	2013-01-01
G11C	/	8	/	18	I	2013-01-01
G06F	/	13	/	4027	I	2013-01-01

CPC Combination Sets						
Symbol				Type	Set	Ranking
	/		/			

NONE (Assistant Examiner) _____ (Date) _____		Total Claims Allowed: 18	
/GURTEJ BANSAL/ Primary Examiner, Art Unit 2139 (Primary Examiner) _____ (Date) 02 February 2018		O.G. Print Claim(s) 1	O.G. Print Figure 10

Issue Classification 	Application/Control No. 14/715,491	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner GURTEJ BANSAL	Art Unit 2139

INTERNATIONAL CLASSIFICATION					
CLAIMED					
G06F	/	12	/	00	
NON-CLAIMED					
	/		/		

US ORIGINAL CLASSIFICATION	
CLASS	SUBCLASS

CROSS REFERENCES(S)						
CLASS	SUBCLASS (ONE SUBCLASS PER BLOCK)					

NONE		Total Claims Allowed:	
(Assistant Examiner)	(Date)	18	
/GURTEJ BANSAL/ Primary Examiner, Art Unit 2139	02 February 2018	O.G. Print Claim(s)	O.G. Print Figure
(Primary Examiner)	(Date)	1	10

Issue Classification 	Application/Control No. 14/715,491	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner GURTEJ BANSAL	Art Unit 2139

☐ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47															
CLAIMS															
Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original
1	1	6	10		19	15	28								
	2	17	11		20	16	29								
	3	18	12		21										
2	4	7	13		22										
3	5	8	14	10	23										
4	6	9	15	11	24										
	7		16	12	25										
	8		17	13	26										
5	9		18	14	27										

NONE (Assistant Examiner) _____ (Date) _____		Total Claims Allowed: 18	
/GURTEJ BANSAL/ Primary Examiner, Art Unit 2139 (Primary Examiner) _____ (Date) _____		02 February 2018 O.G. Print Claim(s) 1	O.G. Print Figure 10



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
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Alexandria, Virginia 22313-1450
www.uspto.gov

NOTICE OF ALLOWANCE AND FEE(S) DUE

79141 7590 03/05/2018
Jamie J. Zheng, Ph.D Esq.
MASCHOFF BRENNAN
1389 Center Drive
Suite 300
Park City, UTAH 84098

EXAMINER

BANSAL, GURTEJ

ART UNIT

PAPER NUMBER

2139

DATE MAILED: 03/05/2018

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/715,491	05/18/2015	Hyun Lee	N3044.10007US08	8143

TITLE OF INVENTION: MEMORY MODULE AND CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
REGULAR	UNDISCOUNTED	\$1000	\$0.00	\$960.00	\$1000	06/05/2018

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.

If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 1/2 the amount of undiscounted fees, and micro entity fees are 1/2 the amount of small entity fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed and an extra copy of the form should be submitted. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Utility patents issuing on applications filed on or after Dec. 12, 1980 may require payment of maintenance fees. It is patentee's responsibility to ensure timely payment of maintenance fees when due.

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), to: **Mail** **Mail Stop ISSUE FEE**
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450
or Fax (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

79141 7590 03/05/2018
Jamie J. Zheng, Ph.D Esq.
MASCHOFF BRENNAN
1389 Center Drive
Suite 300
Park City, UTAH 84098

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being facsimile transmitted to the USPTO (571) 273-2885, on the date indicated below.

(Depositor's name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/715,491	05/18/2015	Hyun Lee	N3044.10007US08	8143

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REGULAR	UNDISCOUNTED	\$1000	\$0.00	\$960.00	\$1000	06/05/2018

EXAMINER	ART UNIT	CLASS-SUBCLASS
BANSAL, GURTEJ	2139	711-149000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.

☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

(1) The names of up to 3 registered patent attorneys or agents OR, alternatively,

(2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

1 _____
2 _____
3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. The following fee(s) are submitted:

4b. Payment of Fee(s): (**Please first reapply any previously paid issue fee shown above**)

☐ Issue Fee

☐ A check is enclosed.

☐ Publication Fee (No small entity discount permitted)

☐ Payment by credit card. Form PTO-2038 is attached.

☐ Advance Order - # of Copies _____

☐ The director is hereby authorized to charge the required fee(s), any deficiency, or credits any overpayment, to Deposit Account Number _____ (enclose an extra copy of this form).

5. **Change in Entity Status** (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29

☐ Applicant asserting small entity status. See 37 CFR 1.27

☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____

Date _____

Typed or printed name _____

Registration No. _____



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/715,491	05/18/2015	Hyun Lee	N3044.10007US08	8143

79141 7590 03/05/2018

Jamie J. Zheng, Ph.D Esq.
MASCHOFF BRENNAN
1389 Center Drive
Suite 300
Park City, UTAH 84098

EXAMINER

BANSAL, GURTEJ

ART UNIT	PAPER NUMBER
----------	--------------

2139

DATE MAILED: 03/05/2018

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450. Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b) (2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Notice of Allowability	Application No. 14/715,491	Applicant(s) Lee et al.	
	Examiner GURTEJ BANSAL	Art Unit 2139	AIA Status No

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to Examiner's amendment dated 6/15/17 and RCE dated 12/15/2017 .
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____ .

2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

3. ☒ The allowed claim(s) is/are 1,4-6,9-15 and 23-29 . As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information , please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a) ☐ All b) ☐ Some *c) ☐ None of the:

1. ☐ Certified copies of the priority documents have been received.

2. ☐ Certified copies of the priority documents have been received in Application No. _____ .

3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____ .

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file areply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.
THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____ .

Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).

6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

1. ☐ Notice of References Cited (PTO-892)	5. ☐ Examiner's Amendment/Comment
2. ☐ Information Disclosure Statements (PTO/SB/08), Paper No./Mail Date _____	6. ☐ Examiner's Statement of Reasons for Allowance
3. ☐ Examiner's Comment Regarding Requirement for Deposit of Biological Material _____	7. ☐ Other _____ .
4. ☐ Interview Summary (PTO-413), Paper No./Mail Date. _____ .	

/GURTEJ BANSAL/ Primary Examiner, Art Unit 2139	
--	--

Did you mean:

High power C+ L-band Erbium ASE source using optical circulator with double-pass and **bi-directional** pumping configuration

H Lin, CH Chang - Optics Express, 2004 - osapublishing.org

... pumping-efficiency L-band erbium-doped fiber ASE source using double-pass **bidirectional**-pumping configuration ... Fig. 1. Proposed double-pass and **bi-directional** configuration for high-power C+L- band ASE ... one-way mechanism of ASE source and used as an output **port** O1 ...

☆ 77 Cited by 30 Related articles All 7 versions

Slew rate control circuit for small computer system interface (SCSI) differential driver

B Jiang, SK Chan - US Patent 7,298,173, 2007 - Google Patents

... data bit propagates through the cascaded delay cells 46 and causes the driver output **ports** 62 to ... 9, families of **waveforms** are shown for the example output **port** circuit 40 having N=8 ... Each rising edge **waveform** is generated at the output **port** 42 with the cascaded delay cells 46 ...

☆ 77 Cited by 12 Related articles All 2 versions

Method for **bi-directional** data communication in a digital game **port**

RL Church, DA Nickless, ST Kaminsky... - US Patent ..., 2001 - Google Patents

... It is, therefore, an object of the invention to provide a **bi-directional** game/simulation system that ... RAM, as well as other conventional microprocessor elements such as I/O **ports** whose operation ... than 126 μ s assuming a worst case capacitor value on the PC game **port** of 10000 pF ...

☆ 77 Cited by 18 Related articles All 2 versions

Slew rate control circuit for small computer system interface (SCSI) differential driver

B Jiang, SK Chan - US Patent 7,719,314, 2010 - Google Patents

... bit propagates through the cascaded delay cells 46 and causes the driver output **ports** 62 to ... outputs of the plurality of predrivers to provide a predetermined slew rate at the output **port** ... en), 1987-02-17, 1989-12-19, Nec Corporation, Echo canceller for **bidirectional** transmission on ...

☆ 77 Cited by 9 Related articles All 2 versions

Optical **waveform** shaping

A Galvanauskas, Y Gianchandani... - US Patent App. 11 ..., 2007 - Google Patents

... means having data bus means, ie plural waveguides interconnected and providing an inherently **bidirectional** system by ... wave **shaping** system 20 may be used in an iterative fashion to generate **waveforms** and with different applied voltages, the output **waveform** may be ...

☆ 77 Cited by 24 Related articles All 2 versions

Low voltage differential signaling [LVDS] driver with pre-emphasis

AG Radelinow - US Patent 6,977,534, 2005 - Google Patents

... This **waveform** **shaping** is realized by a digital filter ... The **waveform** **shaping** is not well reproducible due to the spread of process parameters, supply voltage and temperature variations. Another disadvantage in prior art US Pat. No ...

☆ 77 Cited by 57 Related articles All 4 versions

Bi-directional all-optical regenerator

JA Fee, AN Robinson - US Patent 6,317,232, 2001 - Google Patents

... H04B10/297—**Bidirectional** amplification; H04B10/2971—A single amplifier for both directions. H—ELECTRICITY; ... data signal causes the clock pulse to emerge from the output **port** of the ... US09047637 Active US6317232B1 (en), 1998-03-25, 1998-03-25, **Bi-directional** all-optical ...

☆ 77 Cited by 17 Related articles All 2 versions

Optical modulator, optical short pulse generating device, optical **waveform shaping** device, and optical demultiplexer device

M Suzuki, N Edagawa, I Morita, S Yamamoto... - US Patent ..., 1999 - Google Patents

... absorption-type optical modulator 4 and the readmission of the signal to the **port** (e), n is ...

Furthermore, the optical **waveform shaping** device of the present invention eliminates the timing jitter of transmission optical signals and makes it possible to shape soliton **waveforms** ...

☆ 77 Cited by 12 Related articles All 2 versions >>

Optical arbitrary **waveform** generation and processing using spectral line-by-line pulse **shaping**

Z Jiang, D Leaird, A Weiner - US Patent App. 11/804,477, 2008 - Google Patents

... data bus means, ie plural waveguides interconnected and providing an inherently **bidirectional**

system by ... Thus, **waveform** manipulation at both macro and micro scale is possible simultaneously ...

output of the wavelength dependent optical modulator to an output **port**, wherein the ...

☆ 77 Cited by 17 Related articles All 2 versions >>

A zero-voltage switching three-**port** isolated full-bridge converter

H Al-Atrash, M Pepper, I Batarseh - ... Energy Conference, 2006 ..., 2006 - ieeexplore.ieee.org

... The two **ports** of the topology on the primary side of the transformer are inherently

bidirectional ... The third **port**, the loading **port**, is strictly a power sink if diode rectifiers are

used, and becomes **bidirectional** if synchronous rectification is utilized ...

☆ 77 Cited by 31 Related articles All 2 versions

Did you mean to search for:

EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L1	0	select\$3 with rank same load same noise same (cancel\$5 reduct\$3)	US-PGPUB; USPAT	OR	OFF	2018/02/03 11:31
L2	1	signal with deteriorat\$3 with switch same lossy	US-PGPUB; USPAT	OR	OFF	2018/02/03 11:33
L3	12	signal with (loss deteriorat\$3 distort\$3) with lossy same discontin\$6	US-PGPUB; USPAT	OR	ON	2018/02/03 11:41
L4	93	bidirectional with ports with (switch\$3 with circuit)	US-PGPUB; USPAT	OR	OFF	2018/02/03 12:03
S1	7	"5559970".pn. "6700823".pn. "6812869".pn. "6981089".pn. "7136958".pn. "7184916".pn. "20070013429".pn.	US-PGPUB; USPAT	OR	OFF	2013/12/28 15:58
S2	496	bidirectional with port with (crossbar switch)	US-PGPUB; USPAT	OR	OFF	2013/12/28 16:02
S3	89	bidirectional with port with (crossbar switch) same memory	US-PGPUB; USPAT	OR	OFF	2013/12/28 16:02
S4	366	bidirectional with ports with (crossbar switch)	US-PGPUB; USPAT	OR	OFF	2013/12/28 16:05
S5	48	bidirectional with ports with (switch\$3 with circuit)	US-PGPUB; USPAT	OR	OFF	2013/12/28 16:15
S6	2546	bi\$1directional near2 ports	US-PGPUB; USPAT	OR	OFF	2013/12/28 16:20
S7	666	bi\$1directional near2 ports same (crossbar switch\$3)	US-PGPUB; USPAT	OR	OFF	2013/12/28 16:40
S8	1	"7130308".pn.	US-PGPUB; USPAT	OR	OFF	2013/12/28 17:14
S9	786	711/149.ccls.	US-PGPUB; USPAT	OR	OFF	2013/12/28 22:03
S10	78	port with noise with reduc\$4 and PCB	US-PGPUB; USPAT	OR	OFF	2013/12/28 22:16
S11	20	signal with deteriorat\$3 with lossy	US-PGPUB; USPAT	OR	OFF	2014/05/04 17:59
S12	1	signal with deteriorat\$3 with switch same lossy	US-PGPUB;	OR	OFF	2014/05/04 18:01

			USPAT			
S13	16	signal with loss with switch same lossy	US-PGPUB; USPAT	OR	OFF	2014/05/04 18:01
S14	19	signal with (loss deteriorat\$3 distort\$3) with switch same lossy	US-PGPUB; USPAT	OR	OFF	2014/05/04 18:16
S15	1227	signal with (loss deteriorat\$3 distort\$3) same lossy	US-PGPUB; USPAT	OR	OFF	2014/05/04 18:16
S16	14	signal with (loss deteriorat\$3 distort\$3) with lossy same (correct\$3 ameliorat\$3)	US-PGPUB; USPAT	OR	OFF	2014/05/04 18:17
S17	2	signal with (loss deteriorat\$3 distort\$3) with lossy same (switch\$3 with circuit)	US-PGPUB; USPAT	OR	OFF	2014/05/04 18:18
S18	529	signal with (loss deteriorat\$3 distort\$3) with lossy	US-PGPUB; USPAT	OR	OFF	2014/05/04 18:20
S19	11	signal with (loss deteriorat\$3 distort\$3) with lossy same port	US-PGPUB; USPAT	OR	OFF	2014/05/04 18:20
S20	19	signal with (loss deteriorat\$3 distort\$3) with lossy same port	US-PGPUB; USPAT	OR	ON	2014/05/04 18:20
S21	116	signal with (loss deteriorat\$3 distort\$3) with lossy same circuit	US-PGPUB; USPAT	OR	ON	2014/05/04 18:22
S22	8	signal with (loss deteriorat\$3 distort\$3) with lossy same discontin\$6	US-PGPUB; USPAT	OR	ON	2014/05/04 18:23
S23	66	signal with (loss deteriorat\$3 distort\$3) with lossy same (repeater amplifier)	US-PGPUB; USPAT	OR	ON	2014/05/04 18:36
S24	47	transmission with (loss deteriorat\$3 distort\$3) with lossy same (repeater amplifier)	US-PGPUB; USPAT	OR	ON	2014/05/04 18:39
S25	13	signal with (loss deteriorat\$3 distort\$3) with lossy same port	US-PGPUB; USPAT	OR	OFF	2014/05/23 18:39
S26	20	signal with deteriorat\$3 with lossy	US-PGPUB; USPAT	OR	OFF	2014/05/23 18:41
S27	34	signal with discontinuit\$3 with lossy	US-PGPUB; USPAT	OR	OFF	2014/05/23 18:43
S28	3171	365/191.ccls. 365/196.ccls. 711/149.ccls.	US-PGPUB; USPAT	OR	OFF	2014/05/23 18:48
S32	1	"20030070052".pn.	US-PGPUB; USPAT	OR	OFF	2015/09/30 14:47
S33	22	(isolat\$3 mux select\$3) with rand with memory	US-PGPUB; USPAT	OR	OFF	2015/09/30 14:58
S34	1589	(isolat\$3 mux select\$3) with rank with memory	US-PGPUB;	OR	OFF	2015/09/30 14:58

			USPAT			
S35	551	signal with waveform with reshap\$3	US-PGPUB; USPAT	OR	OFF	2015/09/30 15:11
S36	6	signal with waveform with reshap\$3 with loss	US-PGPUB; USPAT	OR	OFF	2015/09/30 15:12
S37	0	signal with waveform with reshap\$3 with rank	US-PGPUB; USPAT	OR	OFF	2015/09/30 15:14
S38	0	signal with waveform with reshap\$3 same rank	US-PGPUB; USPAT	OR	OFF	2015/09/30 15:14
S39	43	signal with waveform with reshap\$3 same memory	US-PGPUB; USPAT	OR	OFF	2015/09/30 15:15
S40	1	"6959369".pn.	US-PGPUB; USPAT	OR	OFF	2015/10/02 12:27
S41	1	"6732244".pn.	US-PGPUB; USPAT	OR	OFF	2015/10/02 12:29
S42	0	impedence near3 match\$3 same rank with memory	US-PGPUB; USPAT	OR	OFF	2017/06/09 00:49
S43	668	impedence near3 match\$3	US-PGPUB; USPAT	OR	OFF	2017/06/09 00:56
S44	0	impedence near3 match\$3 and rank with memory	US-PGPUB; USPAT	OR	OFF	2017/06/09 00:56
S45	9	impedence near3 match\$3 same memory	US-PGPUB; USPAT	OR	OFF	2017/06/09 00:56
S46	0	impedence near3 match\$3 same port\$1 same3 memory	US-PGPUB; USPAT	OR	OFF	2017/06/09 01:10
S47	0	impedence with match\$3 same port\$1 same3 memory	US-PGPUB; USPAT	OR	OFF	2017/06/09 01:10
S48	19	impedence with match\$3 and memory with port\$1	US-PGPUB; USPAT	OR	OFF	2017/06/09 01:52
S49	0	select\$3 with rank and impedence with match\$3	US-PGPUB; USPAT	OR	OFF	2017/06/09 10:54
S50	0	select\$3 with rank and impedence with match\$3	US-PGPUB; USPAT	OR	OFF	2017/06/09 10:54
S51	597	select\$3 with rank and signal with correction	US-PGPUB; USPAT	OR	OFF	2017/06/09 11:00
S52	45	select\$3 with rank same signal with correction	US-PGPUB; USPAT	OR	OFF	2017/06/09 11:00
S53	8280	G06F12/1684.cpc. G11C5/02.cpc. G11C5/06.cpc. G11C8/18.cpc.	US-PGPUB;	OR	OFF	2017/06/09 11:08

		G06F13/4027.cpc.	USPAT			
S54	0	select\$3 with rank same load same noise same (cancel\$5 reduct\$3)	US-PGPUB; USPAT	OR	OFF	2017/06/09 11:30
S55	6165	load same noise same (cancel\$5 reduct\$3)	US-PGPUB; USPAT	OR	OFF	2017/06/09 11:35
S56	0	load same noise same (cancel\$5 reduct\$3) same rank with memory	US-PGPUB; USPAT	OR	OFF	2017/06/09 11:35
S57	28	(US-20010008006-\$ US-20030070052-\$ US-20080025137-\$ US-20090248969-\$ US-20100125681-\$ US-4739473-\$ US-5313624-\$ US-5630096-\$ US-5905401-\$ US-6061754-\$ US-6141245-\$ US-6223650-\$ US-6262938-\$ US-6546476-\$ US-6636446-\$ US-6704910-\$ US-7024518-\$ US-7191302-\$ US-7225303-\$ US-7730254-\$ US-7990746-\$ US-7881150-\$ US-8244971-\$ US-8335894-\$ US-8856464-\$ US-8081536-\$ US-8516185-\$ US-8756364-\$).DID.	US-PGPUB; USPAT	OR	OFF	2017/06/09 11:47
S58	4	"6381140".pn. "7254036".pn. "20090248969".pn. "20100125681".pn.	US-PGPUB; USPAT	OR	OFF	2017/06/09 11:47
S65	21	impedence with match\$3 and memory with port\$1	US-PGPUB; USPAT	OR	OFF	2018/01/21 11:53
S66	16	signal with (loss deteriorat\$3 distort\$3) with lossy same (correct\$3 ameliorat\$3)	US-PGPUB; USPAT	OR	OFF	2018/01/21 12:01
S67	24	signal with deteriorat\$3 with lossy	US-PGPUB; USPAT	OR	OFF	2018/01/21 12:03
S68	47	select\$3 with rank same signal with correction	US-PGPUB; USPAT	OR	OFF	2018/01/21 12:36
S69	9	signal with waveform with reshap\$3 with loss	US-PGPUB; USPAT	OR	OFF	2018/02/02 20:05
S70	16	signal with (loss deteriorat\$3 distort\$3) with lossy same (correct\$3 ameliorat\$3)	US-PGPUB; USPAT	OR	OFF	2018/02/02 20:17
S71	93	bidirectional with ports with (switch\$3 with circuit)	US-PGPUB; USPAT	OR	OFF	2018/02/02 20:41

EAST Search History (Interference)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L5	4842	G06F12/1684.cpc. G11C5/02.cpc. G11C5/06.cpc. G11C8/18.cpc. G06F13/4027.cpc.	USPAT	OR	OFF	2018/02/03 12:39
L6	5049	correction.clm. adj circuit.clm.	USPAT	OR	OFF	2018/02/03 12:39
L7	0	L5 and L6 and impedance.clm.	USPAT	OR	OFF	2018/02/03 12:39

S29	2421	365/191.ccls. 365/196.ccls. 711/149.ccls.	USPAT	OR	OFF	2014/05/23 18:51
S30	1498	lossy.clm.	USPAT	OR	OFF	2014/05/23 18:52
S31	0	S29 and S30	USPAT	OR	OFF	2014/05/23 18:52
S59	4615	G06F12/1684.cpc. G11C5/02.cpc. G11C5/06.cpc. G11C8/18.cpc. G06F13/4027.cpc.	USPAT	OR	OFF	2017/06/09 11:37
S60	4903	correction.clm. adj circuit.clm.	USPAT	OR	OFF	2017/06/09 11:38
S61	15	S59 and S60	USPAT	OR	OFF	2017/06/09 11:38
S62	0	S59 and S60 and impedance.clm.	USPAT	OR	OFF	2017/06/09 11:38
S63	2	"6381140".pn. "7254036".pn. "20090248969".pn. "20100125681".pn.	USPAT	OR	OFF	2017/06/09 11:47
S64	23	"Term Removed"	USPAT	OR	OFF	2017/06/09 11:47

2/3/2018 12:39:43 PM
C:\Users\gbansal\Documents\EAST\Workspaces\14715491.wsp



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PART B - FEE(S) TRANSMITTAL

Electronically filed June 1, 2018

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P.O. Box 1450
Alexandria, Virginia 22313-1450
(571) 273-2885

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CURRENT CORRESPONDENCE ADDRESS

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1400 Page Mill Road
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(Depositor's name)

(Signature)

(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTY'S DOCKET NO.	CONFIRMATION NO.
14/715,491	05/18/2015	Hyun Lee	129980-5044-US	8143

TITLE OF INVENTION **MEMORY MODULE AND CIRCUIT PROVIDING LOAD ISOLATION ISOLATION AND NOISE REDUCTION (AS AMENDED)**

APPLN. TYPE	ENTITY STATUS	ISSUE FEE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	1000	960	40	06/05/2018

EXAMINER	ART UNIT	CLASS-SUB CLASS
BANSAL, Gurtej	2139	711-149000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.

☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47) attached. Use of a Customer Number is required.

2. For printing on the patent front page, list (1) the names of up to 3 registered patent attorneys or agents OR, alternatively, (2) the name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed

1. Morgan, Lewis & Bockius LLP

2. _____

3. _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

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(A) NAME OF ASSIGNEE:

NETLIST, INC.

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

SEOCHO-GU, KOREA

Please check the appropriate assignee category or categories (will not be printed on the patent) ☐ Individual ☒ Corporation or other private group entity ☐ Government

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5. Change in Small Entity Status (from status indicated above)

☐ a. Applicant certifying micro entity status. See 37 CFR 1.29.

NOTE: Absent a valid certification of Micro Entity (see form PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

☐ Applicant asserting small entity status. See 37 CFR 1.27.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

☐ Applicant changing to regular undiscounted fee status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Adjustment date: 06/04/2018 HVUONG2

06/10/2017 INTFSW 00005620 14715491

Authorized Signature: / Jamie J. Zheng /

Date: June 1, 2018

-960.00 OP

Printed Name: Jamie J. Zheng

Registration No. 51167

06/04/2018 HVUONG2 00000002 500310 14715491

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Application of:	Hyun Lee	Confirmation No.:	8143
Serial No.:	14/715,491	Art Unit:	2139
Filed:	May 18, 2015	Examiner:	BANSAL, GURTEJ
For:	<i>MEMORY MODULE AND CIRCUIT PROVIDING LOAD ISOLATION ISOLATION AND NOISE REDUCTION (as amended)</i>	Attorney Docket No.:	129980-5044-US

AMENDMENT UNDER 37 CFR 1.312

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Sir:

This application was allowed on March 5, 2018. Please amend the application as set forth on the following pages.

The Commissioner is hereby authorized to charge any required fee(s) to Morgan, Lewis & Bockius LLP Deposit Account No. 50-0310 (order no. 129980-5044-US).

Amendments to the Specification

Please revise the specification as follows:

Please revise the title as follows:

MEMORY MODULE AND CIRCUIT PROVIDING LOAD ~~ISOLATION~~ ISOLATION AND
NOISE REDUCTION

REMARKS

This amendment is filed after the Notice of Allowance dated March 5, 2018.

Amendments to the Title

The title has been amended to correct typographical errors.

Conclusion

The Examiner is invited to call the undersigned attorney at (650) 843-4000, if a telephone call could help resolve any remaining items.

Respectfully submitted,

Date: June 1, 2018

/ Jamie J. Zheng /

71,397

Jamie J. Zheng

(Reg. No.)

MORGAN, LEWIS & BOCKIUS LLP

1400 Page Mill Road

Palo Alto, CA 94304

(650) 843-4000

PART B - FEE(S) TRANSMITTAL

Electronically filed June 1, 2018

Complete and send this form with applicable fee(s), to: Mail

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 Commissioner of Patents
 P.O. Box 1450
 Alexandria, Virginia 22313-1450
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(Depositor's name)

(Signature)

(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTY'S DOCKET NO.	CONFIRMATION NO.
14/715,491	05/18/2015	Hyun Lee	129980-5044-US	8143

TITLE OF INVENTION **MEMORY MODULE AND CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION (AS AMENDED)**

APPLN. TYPE	ENTITY STATUS	ISSUE FEE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	1000	960	40	06/05/2018

EXAMINER	ART UNIT	CLASS-SUB CLASS
BANSAL, GURTEJ	2139	711-149000

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☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47) attached. Use of a Customer Number is required.

2. For printing on the patent front page, list
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1. **Morgan, Lewis & Bockius LLP**

2. _____

3. _____

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NETLIST, INC.

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

SEOCHO-GU, KOREA

Please check the appropriate assignee category or categories (will not be printed on the patent) ☐ Individual ☒ Corporation or other private group entity ☐ Government

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NOTE: This form must be sign in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature / Jamie J. Zheng /

Date: June 1, 2018

Printed Name: Jamie J. Zheng

Registration No. 51167

Electronic Acknowledgement Receipt

EFS ID:	32787967
Application Number:	14715491
International Application Number:	
Confirmation Number:	8143
Title of Invention:	MEMORY MODULE AND CIRCUIT PROVIDING LOAD ISOLTION AND NOISE REDUCTION
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	129980-5044US
Receipt Date:	01-JUN-2018
Filing Date:	18-MAY-2015
Time Stamp:	20:24:06
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		129980_5044US_AMEND_ISSU E_FEE.pdf	163070	yes	4
			5178edc61190394d904897e422c77cf8c3a 52215		

	Multipart Description/PDF files in .zip description		
	Document Description	Start	End
	Amendment after Notice of Allowance (Rule 312)	1	1
	Specification	2	2
	Applicant Arguments/Remarks Made in an Amendment	3	3
	Issue Fee Payment (PTO-85B)	4	4

Warnings:

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If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



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APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
14/715,491	05/18/2015	2139	800	129980-5044US	20	2

CONFIRMATION NO. 8143

CORRECTED FILING RECEIPT

79141

Morgan Lewis Bockius LLP / Jamie Zheng Ph.D.
1400 Page Mill Road
Palo Alto, CA 94304



Date Mailed: 06/18/2018

Receipt is acknowledged of this non-provisional patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Inventor(s)

Hyun Lee, Ladera Ranch, CA;
Jayesh R. Bhakta, Cerritos, CA;
Jeffrey C. Solomon, Irvine, CA;
Mario Jesus Martinez, Laguna Niguel, CA;
Chi-She Chen, Walnut, CA;

Applicant(s)

Netlist, Inc., Irvine, CA;

Power of Attorney: None

Domestic Priority data as claimed by applicant

This application is a CON of 14/324,990 07/07/2014 PAT 9037809
which is a CON of 13/412,243 03/05/2012 PAT 8782350
which is a CON of 12/422,853 04/13/2009 PAT 8154901
which claims benefit of 61/044,839 04/14/2008
and claims benefit of 61/044,825 04/14/2008
and claims benefit of 61/044,801 04/14/2008

Foreign Applications for which priority is claimed (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.) - None.

Foreign application information must be provided in an Application Data Sheet in order to constitute a claim to foreign priority. See 37 CFR 1.55 and 1.76.

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Permission to Access Search Results: No

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The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 14/715,491**

Projected Publication Date: Request for Non-Publication Acknowledged

Non-Publication Request: Yes

Early Publication Request: No

Title

MEMORY MODULE AND CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION

Preliminary Class

711

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

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Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

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No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).

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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/715,491	05/18/2015	Hyun Lee	129980-5044US	8143
79141	7590	06/18/2018	EXAMINER	
Morgan Lewis Bockius LLP / Jamie Zheng Ph.D.			BANSAL, GURTEJ	
1400 Page Mill Road				
Palo Alto, CALIFORNIA 94304				
UNITED STATES OF AMERICA				
			ART UNIT	PAPER NUMBER
			2139	
			NOTIFICATION DATE	DELIVERY MODE
			06/18/2018	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

padocketingdepartment@morganlewis.com
vskliba@morganlewis.com

Response to Rule 312 Communication	Application No.	Applicant(s)	
	14/715,491	Lee et al.	
	Examiner	Art Unit	AIA Status
	GURTEJ BANSAL	2139	No

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

1. ☒ The amendment filed on 01 June 2018 under 37 CFR 1.312 has been considered, and has been:

a) ☐ entered.

b) ☒ entered as directed to matters of form not affecting the scope of the invention.

c) ☐ disapproved because the amendment was filed after the payment of the issue fee.
Any amendment filed after the date the issue fee is paid must be accompanied by a petition under 37 CFR 1.313(c)(1) and the required fee to withdraw the application from issue.

d) ☐ disapproved. See explanation below.

e) ☐ entered in part. See explanation below.

/GURTEJ BANSAL/ Primary Examiner, Art Unit 2139	
--	--

Electronically filed June 1, 2018

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Application of:	Hyun Lee	Confirmation No.:	8143
Serial No.:	14/715,491	Art Unit:	2139
Filed:	May 18, 2015	Examiner:	BANSAL, GURTEJ
For:	<i>MEMORY MODULE AND CIRCUIT PROVIDING LOAD ISOLATION ISOLATION AND NOISE REDUCTION (as amended)</i>	Attorney Docket No.:	129980-5044-US

AMENDMENT UNDER 37 CFR 1.312

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Sir:

This application was allowed on March 5, 2018. Please amend the application as set forth on the following pages.

The Commissioner is hereby authorized to charge any required fee(s) to Morgan, Lewis & Bockius LLP Deposit Account No. 50-0310 (order no. 129980-5044-US).



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APPLICATION NO.	ISSUE DATE	PATENT NO.	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/715,491	07/17/2018	10025731	129980-5044US	8143

79141 7590 06/27/2018
Morgan Lewis Bockius LLP / Jamie Zheng Ph.D.
1400 Page Mill Road
Palo Alto, CA 94304

ISSUE NOTIFICATION

The projected patent number and issue date are specified above.

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment is 108 day(s). Any patent to issue from the above-identified application will include an indication of the adjustment on the front page.

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Application Information Retrieval (PAIR) WEB site (<http://pair.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Application Assistance Unit (AAU) of the Office of Data Management (ODM) at (571)-272-4200.

APPLICANT(s) (Please see PAIR WEB site <http://pair.uspto.gov> for additional applicants):

Hyun Lee, Ladera Ranch, CA;
Netlist, Inc., Irvine, CA;
Jayesh R. Bhakta, Cerritos, CA;
Jeffrey C. Solomon, Irvine, CA;
Mario Jesus Martinez, Laguna Niguel, CA;
Chi-She Chen, Walnut, CA;

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IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re Patent:	Lee et al.	Confirmation No.:	8143
Patent No.:	10,025,731 B1	Serial No.:	14/715,491
Issued:	July 17, 2018	Filed:	May 18, 2015
For:	MEMORY MODULE AND CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION	Attorney Docket No:	129980-5044-US

REQUEST FOR CERTIFICATE OF CORRECTION

Certificate of Corrections Branch
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Sir:

Upon review of the subject patent, Patentee's attorney discovered 1 error on the part of the Patent Office. Patentee's attorney requests correction of the issued patent. A Form PTO 1050 Certificate of Correction is enclosed.

The Commissioner is authorized to charge required fees to our Deposit Account No. 50-0310 (order no. 129980-5044-US).

Respectfully submitted,

Date: August 13, 2018



Jamie Zheng
MORGAN, LEWIS & BOCKIUS LLP
1400 Page Mill Road
Palo Alto, California 94304
(650) 843-4000

51,167
(Reg. No.)

**UNITED STATES PATENT AND TRADEMARK OFFICE
CERTIFICATE OF CORRECTION**

Page 1 of 1

PATENT NO. : 10,025,731 B1
APPLICATION NO. : 14/715,491
ISSUE DATE : July 17, 2018
INVENTOR(S) : Lee et al.

It is certified that an error appears or errors appear in the above-identified patent and that said Letters Patent is hereby corrected as shown below:

In the claims, claim 16, column 18, line 42, please delete "claim **1**, the" and insert --claim **1**, wherein the--.

MAILING ADDRESS OF SENDER:
MORGAN, LEWIS & BOCKIUS LLP
1400 Page Mill Road
Palo Alto, CA 94304
(650) 843-4000

Electronic Acknowledgement Receipt

EFS ID:	33454551
Application Number:	14715491
International Application Number:	
Confirmation Number:	8143
Title of Invention:	MEMORY MODULE AND CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng/deborah carney
Filer Authorized By:	Jamie Jie Zheng
Attorney Docket Number:	129980-5044US
Receipt Date:	13-AUG-2018
Filing Date:	18-MAY-2015
Time Stamp:	18:16:45
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Request for Certificate of Correction	129980-5044-US_CoC.pdf	64694	no	2
			b41c67c2381d8661e3c822411d446ca96a138336		

Warnings:

Information:	
Total Files Size (in bytes):	64694
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.	



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
14/715,491	05/18/2015	Hyun Lee	129980-5044US	8143
79141	7590	09/11/2018		
Morgan Lewis Bockius LLP / Jamie Zheng Ph.D. 1400 Page Mill Road Palo Alto, CA 94304			EXAMINER BANSAL, GURTEJ	
			ART UNIT	PAPER NUMBER
			2139	
			NOTIFICATION DATE	DELIVERY MODE
			09/11/2018	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

padocketingdepartment@morganlewis.com
vskliba@morganlewis.com



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Patent No.: 10025731
Issue Date: 17 July 2018
Appl. No.: 14/715,491
Filed: 18 May 2015

PART (A) RESPONSE FOR CERTIFICATES OF CORRECTION

This is a decision on the Certificate of Correction request filed 13 August 2018.

The request for issuance of Certificate of Correction for the above-identified correction(s) under the provisions of 37 CFR 1.322 and/or 1.323 is hereby:

(Check one)

☒ Approved ☐ Approved in Part ☐ Denied

Comments: _____

PART (B) PETITION UNDER 37 CFR 1.324 OR 37 CFR 1.48

☐ This is a decision on the petition filed _____ to correct inventorship under 37 CFR 1.324.

☐ This is a decision on the request under 37 CFR 1.48, petition filed _____. In view of the fact that the patent has already issued, the request under 37 CFR 1.48 has been treated as a petition to correct inventorship under 37 CFR 1.324.

The petition is hereby: ☐ Granted ☐ Dismissed

Comment: _____

The patented filed is being forwarded to Certificate of Corrections Branch for issuance of a certificate naming only the actual inventor or inventors.

/REGINALD G BRAGDON/
Technology Center 2100
Phone: (571)272-4204

UNITED STATES PATENT AND TRADEMARK OFFICE
CERTIFICATE OF CORRECTION

PATENT NO. : 10,025,731 B1
APPLICATION NO. : 14/715491
DATED : July 17, 2018
INVENTOR(S) : Lee et al.

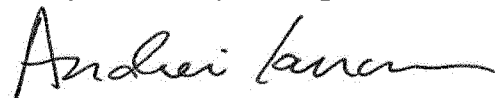
Page 1 of 1

It is certified that error appears in the above-identified patent and that said Letters Patent is hereby corrected as shown below:

In the Claims

Claim 16, Column 18, Line 42, please delete "claim 1, the" and insert --claim 1, wherein the--.

Signed and Sealed this
Twenty-fifth Day of September, 2018

A handwritten signature in black ink, appearing to read "Andrei Iancu".

Andrei Iancu
Director of the United States Patent and Trademark Office

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE OFFICE OF THE UNDER SECRETARY OF COMMERCE
FOR INTELLECTUAL PROPERTY AND DIRECTOR OF THE
UNITED STATES PATENT AND TRADEMARK OFFICE

NOTICE OF DECISIONS ON INSTITUTION

Before JOHN A. SQUIRES, *Under Secretary of Commerce for Intellectual
Property and Director of the United States Patent and Trademark Office.*

NOTICE

Pursuant to 35 U.S.C. § 314(a), after review of discretionary and non-merits considerations, institution of *inter partes* review is denied in the following proceedings:

IPR2025-01458 ¹	IPR2025-01486 ¹	IPR2025-01579 ³
IPR2025-01459 ¹	IPR2025-01488 ¹	IPR2025-01581
IPR2025-01471 ¹	IPR2025-01493 ²	IPR2026-00058
IPR2025-01472 ¹	IPR2025-01525	

¹ Although previously referred, this case is now discretionarily denied in view of *Revvo Technologies, Inc. v. Cerebrum Sensor Technologies, Inc.*, IPR2025-00632, Paper 20 (Director Nov. 3, 2025) (precedential).

² Although previously referred, this case is now discretionarily denied. It is not an efficient use of Office resources to maintain the proceeding at the Board in view of the discretionary denials of IPR2025-01486 and IPR2025-01488, because intrinsically similar issues, where the involved patents are in the same family, will be addressed by the district court.

³ Opinion forthcoming.

Pursuant to 35 U.S.C. § 314(a) and § 324(a), after review of discretionary considerations, the following proceedings will be reviewed for merits and non-discretionary considerations:

IPR2026-00013	IPR2026-00072	PGR2026-00003
IPR2026-00071		

Pursuant to 35 U.S.C. § 314(a), after review of the merits, the petitioner has failed to show a reasonable likelihood of prevailing with respect to at least one of the claims challenged in the petition. Accordingly, institution of *inter partes* review is denied in the following proceedings:

IPR2025-01402	IPR2025-01441 ⁴	IPR2025-01473
---------------	----------------------------	---------------

Pursuant to 35 U.S.C. § 314(a) and § 324(a), after review of the merits, the petitioner has shown a reasonable likelihood of prevailing with respect to at least one of the claims challenged in the petition, or that it is more likely than not that at least one of the claims challenged in the petition is unpatentable, as applicable, and no other non-discretionary considerations warrant denial of institution. Accordingly, institution of *inter partes* review or post-grant review is granted in the following proceedings:

IPR2025-01267	IPR2025-01461	PGR2025-00071
IPR2025-01364	IPR2025-01487 ⁴	PGR2025-00081
IPR2025-01431	IPR2025-01516	

⁴ Opinion forthcoming.